

Convex relaxation gaps and spatial certificates in nonlinear optimization

Abstract

Separate convexification of nonlinear terms can lose compatibility across shared variables. We formulate this loss through probability laws on box vertices and compare it with the resources needed for spatial optimality certificates. For signed bilinear functions, the worst coefficient gap on each interaction graph has the order of the square root of its largest induced edge density, consistent with earlier Schur-multiplier theory. For positive multilinear functions, we prove sharp leading growth $\log d / \log \log d$ in degree, exact and asymptotically sharp structural bounds, and leading growth equal to the physical box aspect ratio. Finite cubic certificates, common-law rounding, and exact scalar-envelope complexity delimit these comparisons. Separately, fractional-cardinality and signed cubic XOR families require exponentially many certified regions under specified spatial moment oracles. The latter bounds survive exact full-box quadratic moment information and bounded monomial reformulations. For the cardinality family, coordinatewise graph lifts admit a degree-loss-free transfer. Matching finite exponential upper certificates clarify the oracle scope. Supporting geometric and complexity comparisons distinguish coordinate dependence, global conic lift size, integer precision, and primitive bound propagation from pointwise envelope gaps and spatial region counts.

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1 Introduction

A convex relaxation can be exact for every individual term and still lose information about their shared variables. Replacing each term by its graph hull permits a separate convex representation for that term. The graph hull of the sum requires a single representation that works simultaneously. This paper studies the resulting width comparison and a complementary question: how many regions must a specified spatial bounding rule certify before it proves a desired optimization tolerance?

For a scalar function f on a box B , write $H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x)$. For a specified factorization $f = \sum_e f_e$, write $T_B(f; x) = \sum_e [\text{cav}_B f_e(x) - \text{vex}_B f_e(x)]$. Always $T_B \geq H_B \geq 0$, but only T_B depends on the factorization. Ratios are formed where $H_B > 0$. A ratio of these pointwise widths is not an approximation ratio for a constrained problem, and supplies no spatial tree lower bound on its own. A spatial theorem must also specify its node oracle, allowed branching coordinates, and how discarded regions are certified.

McCormick’s recursive convexification is the classical starting point [33]. The vertex-law description of multilinear envelopes and earlier gap comparisons appear in Luedtke et al. [32]; their positive-coefficient question motivates the universal laws studied here. For signed bilinear functions, Boland et al. [10] relate the gap to cut ranges and establish square-root-dimensional growth. We give an elementary per-graph density proof, while an explicit transfer from Davidson and Donsig [14] identifies its established asymptotic content. Classical symmetric envelopes, matching theory and balanced-matrix results supply further ingredients, with the exact hypotheses recorded where used.

The positive multilinear results exploit a common upper-attaining law. The remaining task is a simultaneous lower-envelope rounding guarantee that preserves every mean, independently of the objective coefficients. The universal degree and dimension suprema have sharp leading growth $\log d / \log \log d$ and $\log n / \log \log n$, respectively. Explicit finite bounds and cubic witnesses accompany these asymptotics. Incidence orientations, marginal floors, feedback variables, and frequency two give more precise laws; incidence treewidth two has exact constant two. Positive physical boxes require separate coefficient arguments: their aspect-ratio supremum lies between $\max\{2, \rho_B\}$ and $\rho_B + 2$, with a finite-dimensional improvement. A single product already has NP-complete exact midpoint-envelope evaluation on arbitrarily narrow unequal rational boxes, although its factorwise ratio is one. Thus relaxation strength and exact evaluation complexity are distinct.

The spatial results use different examples and proof objects. A fractional-cardinality quadratic supplies an elementary chord obstruction, then explicit SDP–RLT and higher-order pseudomoments extend it. These proofs build on established fractional-knapsack moment lower bounds [22, 35], and are compared with earlier spatial lower bounds [26]. They extend to arbitrary coordinate domains, fixed relative tolerance and coordinatewise graph lifts. A known coupled quadratic cut closes that first family at the root. Signed cubic XOR objectives, using the classical signed-character construction of Schoenebeck [36], give the stronger obstruction: exponentially many regions remain necessary even with exact quadratic moment information for the full coordinate box. The transfer survives bounded signed-monomial lifts, with explicit degree and parity-rank accounting. These signed objectives on $[-1, 1]^n$ are separate from the positive-gap examples on nonnegative boxes.

Table 1 gives a route through the proofs. Readers concerned primarily with spatial methods can read the gap and oracle definitions in Section 2, then start with Section 12; the spatial proofs do not require the positive-coupling chapters. Appendices retain distinct predecessor arguments, finite certificates, and supporting comparisons. The exact check programs and their replay instructions accompany the manuscript; finite checks supplement the general proofs.

Question	Main conclusion	Route
Simultaneous envelopes	Signed graph-density order; sharp positive degree growth and finite cubic bounds	Sections 2–6
Structure and physical boxes	Incidence and marginal laws; exact width two; aspect-ratio bounds and exact evaluation complexity	Sections 7–11
Spatial certificates	Cardinality and XOR cover lower bounds; graph-lift transfer and finite upper certificates	Sections 12–18
Other resources	Coordinate and scale dependence, global lift size, integer precision and primitive propagation	Section 19, Appendices F–K

Table 1: Reader’s roadmap. Each comparison retains its own domain, representation and computational resource.

2 Envelope gaps as probability problems

2.1 The scalar graph and the specified termwise relaxation

Let $B = \prod_{i=1}^n [\ell_i, u_i]$ be a finite box. Fixed coordinates are removed when necessary. A function is *multiaffine* if it is affine in each coordinate separately; equivalently, it is a multilinear polynomial with no repeated variable in a monomial. For a continuous scalar function f on B , let $\text{vex}_B f$ be its greatest convex underestimator and $\text{cav}_B f$ its least concave overestimator. The vertical section of the convex hull of its graph is

$$\{y : (x, y) \in \text{conv}\{(z, f(z)) : z \in B\}\} = [\text{vex}_B f(x), \text{cav}_B f(x)].$$

Define the *scalar graph-hull gap* and, for a specified decomposition $f = \sum_{e \in E} f_e$, the *termwise gap* by

$$H_B(f; x) = \text{cav}_B f(x) - \text{vex}_B f(x), \quad (2.1)$$

$$T_B(f; x) = \sum_{e \in E} [\text{cav}_B f_e(x) - \text{vex}_B f_e(x)]. \quad (2.2)$$

The notation T_B always refers to the displayed decomposition. It is the width obtained by giving each scalar factor its exact graph hull and imposing $y = \sum_e y_e$. It can change when a factor is split into smaller pieces, whereas H_B does not. Always,

$$0 \leq H_B(f; x) \leq T_B(f; x), \quad (2.3)$$

because the sum of convex underestimators is a convex underestimator of the sum, and similarly for concave overestimators. Ratios are taken only where $H_B(f; x) > 0$; gap inequalities apply at every point.

For a positive multilinear polynomial, the factors are its displayed monomials $a_e \prod_{i \in e} x_i$, with $a_e \geq 0$. For a signed bilinear function, they are the signed products $a_{ij} x_i x_j$. A bilinear factor has the familiar four-inequality McCormick hull. For a higher-degree factor, its exact hull and a recursive McCormick relaxation are different objects. A recursive product relaxation is exact for one monomial on a unit box, but need not be exact on a general positive box [32, Sections 2–3]. A bound for (2.2) on that box therefore does not automatically bound the gap of a recursive bilinear factorization.

Adding an affine function to a factor, or adding a separate affine factor, changes neither gap. An affine bijection between boxes preserves both envelopes after composition and hence preserves the gaps for the transformed factors. These facts follow from the graph-hull representation.

2.2 Vertex distributions and compatible marginals

Let $\mathcal{V}(B)$ denote the vertices of B , and let $\mathcal{P}_B(x)$ be the distributions on $\mathcal{V}(B)$ with mean x . On the unit cube these are joint Bernoulli distributions with the prescribed singleton means; coordinates need not be independent.

Lemma 2.1 (Vertex representation). *For a multiaffine f on B ,*

$$\text{vex}_B f(x) = \min_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X), \quad (2.4)$$

$$\text{cav}_B f(x) = \max_{P \in \mathcal{P}_B(x)} \mathbb{E}_P f(X). \quad (2.5)$$

Both extrema are attained. The envelopes are continuous piecewise-affine functions on B .

Proof. Write $x_i = \ell_i + (u_i - \ell_i)p_i$. Independently choose $X_i = u_i$ with probability p_i and $X_i = \ell_i$ otherwise. Multiaffinity gives $\mathbb{E}X = x$ and $\mathbb{E}f(X) = f(x)$. Every graph point therefore belongs to the convex hull of vertex graph points, and the reverse inclusion in the full graph hull is immediate. The vertical sections give the formulas. The feasible distributions form a nonempty compact polytope, so both extrema are attained. The lower and upper surfaces of the vertex graph polytope are respectively maxima and minima of finitely many affine functions on B , which also proves continuity at its boundary. $\square$

These representations are classical [32, Section 3]. They expose the compatibility issue: different factors may use different minimizing distributions in (2.2), whereas (2.4) requires one distribution for their sum.

For $m_e(x) = \prod_{i \in e} x_i$ with $e \neq \emptyset$, the unit-cube formulas are

$$\text{cav } m_e(x) = \min_{i \in e} x_i, \quad \text{vex } m_e(x) = \max\{0, \sum_{i \in e} x_i - |e| + 1\}. \quad (2.6)$$

To prove them, interpret $\mathbb{E}m_e(X)$ as the probability of joint success. The upper bound is attained by $X_i = \mathbf{1}\{U \leq x_i\}$ for one uniform $U \in [0, 1]$. For the lower bound, place failure intervals of lengths $1 - x_i$ consecutively on the unit circle. If their total length is below one they are disjoint, and if it is at least one they cover the circle. Their union therefore has measure $\min\{1, \sum_{i \in e} (1 - x_i)\}$. Each interval has its prescribed measure, so the complement gives the claimed minimum joint-success probability. Unused coordinates can be sampled independently to complete the mean vector.

Proposition 2.2 (Common upper attainment and deficiencies). *Let $f(x) = \sum_e a_e m_e(x)$ on $[0, 1]^n$, with $a_e \geq 0$. Discard supports of size at most one. For each remaining support choose an anchor i_e of smallest mean and put*

$$u_e = x_{i_e}, \quad S_e = \sum_{j \in e \setminus \{i_e\}} (1 - x_j), \quad t_e = \min\{u_e, S_e\}, \quad D_e(X) = X_{i_e} - \prod_{i \in e} X_i.$$

Then D_e is the nonnegative indicator that the anchor succeeds and some other support coordinate fails. Moreover,

$$\text{cav } f(x) = \sum_e a_e u_e, \quad (2.7)$$

$$T(f; x) = \sum_e a_e t_e, \quad (2.8)$$

$$H(f; x) = \max_{P \in \mathcal{P}_{[0,1]^n}(x)} \sum_e a_e \mathbb{E}_P D_e(X). \quad (2.9)$$

Proof. The common-threshold law attains every single-product upper envelope simultaneously. Positivity gives (2.7), and subtracting the two single-product envelopes gives (2.8). Finally, every admissible law has $\mathbb{E}X_{i_e} = u_e$, so minimizing the expected polynomial value is equivalent to maximizing the weighted sum of deficiencies. $\square$

The common-upper statement is classical [32, Theorems 4–5 in the authors’ manuscript]. A distribution used to prove a gap bound must work for all terms together; independently optimizing the distribution for each term merely recovers the termwise relaxation. Mixtures of admissible laws preserve the means, and nonnegative deficiencies allow the guarantees of different laws to be added.

Lemma 2.3 (Two classes handled by independence). *Put $c_0 = 1 - e^{-1}$. Independent Bernoulli rounding gives $\mathbb{E}D_e \geq c_0 t_e/2$ if either every support mean exceeds $1/2$, or at least two support means are at most $1/2$.*

Proof. Independence gives $\mathbb{E}D_e = u_e(1 - \prod_{j \neq i_e} x_j)$. In the second case a nonanchor mean is at most $1/2$, giving at least $u_e/2 \geq t_e/2$. In the first case the expression is at least $u_e(1 - e^{-S_e}) \geq c_0 u_e \min(1, S_e)$. If $S_e \leq 1$, this is at least $c_0 S_e/2 \geq c_0 t_e/2$; otherwise it is at least $c_0 u_e = c_0 t_e$. $\square$

Proposition 2.4 (Nonnegative-box transfer). *For a positive multilinear polynomial on B with $0 \leq \ell_i < u_i$, substitute $x_i = \ell_i + (u_i - \ell_i)p_i$ and expand the original monomials. The unit-cube polynomial $\tilde{f}$ has nonnegative coefficients and no larger degree. With its expanded monomials as factors,*

$$H_B(f; x) = H(\tilde{f}; p), \quad T_B(f; x) \leq T(\tilde{f}; p). \quad (2.10)$$

One common-threshold endpoint law attains the upper envelope of every original monomial and their sum.

Proof. Expansion coefficients are products of nonnegative endpoints and interval widths. The full-envelope equality follows from the affine bijection. For an original factor expanded as $\sum_q g_q$, the sum of the lower convex envelopes is a convex underestimator and the sum of the upper concave envelopes is a concave overestimator. Subtract and sum over the original factors to obtain (2.10). Common-threshold rounding attains all expanded upper envelopes together and therefore also those of each original factor and their sum. $\square$

Expansion is a proof device and need not be an efficient representation. It changes the factor incidence structure. Degree bounds transfer through (2.10); bounds based on the original incidence graph require separate arguments. For bilinear functions, arbitrary box endpoints cause no such difficulty: an affine change introduces only scaled bilinear terms and affine summands, regardless of signs.

2.3 What a spatial certificate counts

Let $\mathcal{F}$ be a nonempty compact feasible set, let f be continuous, and put $f^* = \min_{x \in \mathcal{F}} f(x)$. A region oracle assigns a valid lower bound $\text{LB}(S)$ to feasible points in each region S . A *certificate at target T_** is a finite cover $S_1, \dots, S_N$ of $\mathcal{F}$ with $\text{LB}(S_j) \geq T_*$ for all j . Together with a feasible value U and an absolute tolerance $\epsilon \geq 0$, the target $U - \epsilon$ certifies absolute gap at most ϵ . For $U > 0$ and a relative tolerance $0 \leq \theta < 1$, the target $(1 - \theta)U$ certifies $(U - f^*)/U \leq \theta$. A size lower bound may grant the best incumbent $U = f^*$, requiring $f^* > 0$ in this relative convention. Under these tolerance ranges, either target is nondecreasing in U , so worse incumbents require weakly higher targets.

A complete spatial split tree supplies a cover by its final regions. A lower bound for arbitrary covers is thus independent of branching order and node selection within the specified region and oracle model. If a procedure discards feasible regions by propagation or additional cuts, those deductions must also be included in the certificate model. A lower bound for one oracle does not automatically survive stronger coupled cuts, different branch coordinates, or analytic solution of separate components.

3 Signed bilinear gaps and interaction density

Let $G = (V, E)$ be a finite simple graph and $b_a(x) = \sum_{ij \in E} a_{ij} x_i x_j$ on $[0, 1]^V$, with arbitrary real coefficients. Zero-coefficient edges may be removed. Let $c^*(a)$ be the least $c \geq 0$ with $T(b_a; x) \leq cH(b_a; x)$ everywhere, and set $c^*(0) = 0$. Define

$$\rho_G(G) = \max_{\emptyset \neq W \subseteq V} \frac{|E(G[W])|}{|W|}, \quad (3.1)$$

with value zero for an empty graph. This is half the maximum subgraph average degree, not fractional arboricity, whose denominator is $|W| - 1$. Let d_G be degeneracy and Δ_G maximum degree. Then

$$d_G/2 \leq \rho_G(G) \leq d_G, \quad \rho_G(G) \leq \Delta_G/2. \quad (3.2)$$

The first lower bound uses a subgraph of minimum degree d_G ; the upper bounds use a degeneracy orientation and the degree sum identity.

Theorem 3.1 (Bilinear upper bounds). *For every real coefficient vector on G ,*

$$c^*(a) \leq \min\{4\sqrt{\rho_G(G)}, 4\sqrt{d_G}, 2\sqrt{\Delta_G}\}. \quad (3.3)$$

If G is bipartite with parts P, Q and has an edge, let Δ_P and Δ_Q be the maximum degrees in the respective parts. Then

$$c^*(a) \leq \sqrt{2 \min\{\Delta_P, \Delta_Q\}} \leq \sqrt{2 \min\{|P|, |Q|\}}. \quad (3.4)$$

The factor $\sqrt{2}$ in the first bipartite bound is sharp. The conclusions transfer to arbitrary finite boxes after removing fixed coordinates.

The proof refines the cut argument of Boland et al. [10]. The density order also follows from older operator theory, as shown in Section 3.6; no new general graph-norm principle is claimed.

3.1 An exact characterization by induced cuts

For $W \subseteq V$, set

$$Q_{a,W}(s) = \sum_{ij \in E(G[W])} a_{ij} s_i s_j, \quad L_W(a) = \sum_{ij \in E(G[W])} |a_{ij}|,$$

and define

$$R_W(a) = \frac{1}{2} \left(\max_{s \in \{-1, 1\}^W} Q_{a,W}(s) - \min_{s \in \{-1, 1\}^W} Q_{a,W}(s) \right). \quad (3.5)$$

A cut encoded by s has weight $\frac{1}{2} \sum_{ij} a_{ij} - \frac{1}{2} Q_{a,W}(s)$, so R_W is exactly the maximum cut weight minus the minimum cut weight. If $L_W > 0$ then $R_W > 0$: distinct degree-two sign characters are orthogonal under uniform signs, and their nonzero combination cannot be constant. Set $L_W/R_W = 0$ when both vanish.

Lemma 3.2 (Induced-cut characterization). *For every coefficient vector,*

$$c^*(a) = \max_{W \subseteq V} \frac{L_W(a)}{R_W(a)}. \quad (3.6)$$

Each nonzero induced ratio is attained at $x_i = 1/2$ on W and $x_i = 0$ outside W .

Proof. At that point any admissible vertex law fixes outside coordinates to zero. Write $X_i = (1+s_i)/2$ on W . Since $\mathbb{E}s_i = 0$,

$$\mathbb{E}b_a(X) = \frac{1}{4} \left(\sum_{ij \in E(G[W])} a_{ij} + \mathbb{E}Q_{a,W}(s) \right).$$

An extremizing sign vector and its negative, each of probability $1/2$, attain either extreme of $Q_{a,W}$ with the prescribed means. Thus $H = R_W/2$, whereas each active product has width $1/2$ and $T = L_W/2$. These identities hold at every point of $\{0, 1/2, 1\}^V$, with W its set of half-valued coordinates: fixing coordinates to one instead of zero adds only affine terms on W .

To extend the comparison, the width of the product $x_i x_j$ is

$$\min(x_i, x_j) - \max(0, x_i + x_j - 1) = \min(x_i, x_j, 1 - x_i, 1 - x_j).$$

Partition the cube by the hyperplanes $x_i = x_j$, $x_i + x_j = 1$ ($i \neq j$), and $x_i = 1/2$. The termwise gap is affine on every closed cell. Every cell vertex is in $\{0, 1/2, 1\}^V$. Indeed, active equations fix a coordinate to 0, $1/2$, or 1, or relate two coordinates by equality or complementation. In a connected component of these relations, one fixed coordinate determines all values. If none is fixed, a cycle with an odd number of complementations forces the value $1/2$. Otherwise a free parameter remains and a small perturbation preserves all active equations, contradicting extremality. This accounts for every cell vertex.

The graph-hull gap is concave, being $\text{cav } b_a - \text{vex } b_a$. Hence $T - cH$ is convex on each cell for $c \geq 0$. Its value at any convex combination of cell vertices is at most the combination of its values there. The grid-point bound therefore proves the upper direction of (3.6), and the attained ratios prove the other. $\square$

The grid identities and upper implication are those of Boland et al. [10, Lemma 1 and Corollary 1], following Luedtke et al. [32]. The cell argument above supplies the boundary details.

3.2 Cut ranges, row norms, and fractional orientations

Let A be the symmetric weighted adjacency matrix on W , with zero diagonal and zero entries at nonedges. Write $a_{i,T} = (A_{ij})_{j \in T}$ for $T \subseteq W$ and $a_i = a_{i,W}$. For a rectangular matrix define $\|M\|_{\infty \rightarrow 1} = \max_u \text{sign} \|Mu\|_1 = \max_{u,v} \text{sign } v^\top Mu$; empty matrices have norm zero.

Lemma 3.3 (Polarization and row estimates). *Writing $R = R_W(a)$,*

$$R = \max_{S \subseteq W} \|A_{S, W \setminus S}\|_{\infty \rightarrow 1}, \quad (3.7)$$

$$R \geq \frac{1}{\sqrt{2}} \max_{S \subseteq W} \sum_{i \in S} \|a_{i, W \setminus S}\|_2, \quad (3.8)$$

$$R \geq \frac{1}{4} \sum_{i \in W} \|a_i\|_2. \quad (3.9)$$

Proof. For signs s, t , put $u = (s + t)/2$, $v = (s - t)/2$. Their supports are complementary, and symmetry gives $Q(s) - Q(t) = 2v^\top Au$. Conversely, arbitrary signs on complementary supports arise from $s = u + v$, $t = u - v$. Maximizing half the difference proves (3.7).

The sharp real Khinchin inequality of Szarek [40] states that independent uniform signs satisfy

$$\mathbb{E} \left| \sum_j c_j \varepsilon_j \right| \geq \frac{1}{\sqrt{2}} \left(\sum_j c_j^2 \right)^{1/2}. \quad (3.10)$$

Apply it to every row of $A_{S,T}$ for a fixed partition S, T , and sum. The best sign vector performs at least as well as its average, proving (3.8). Transposition gives the same estimate for side T .

Choose a partition locally maximizing the cut weight with nonnegative weights a_{ij}^2 . At every vertex at least half its squared incident weight crosses, since otherwise moving the vertex improves the cut. Thus

$$X = \sum_{i \in S} \|a_{i,T}\|_2, \quad Y = \sum_{i \in T} \|a_{i,S}\|_2 \quad \text{satisfy} \quad X + Y \geq \frac{1}{\sqrt{2}} \sum_i \|a_i\|_2.$$

The two row estimates yield $R \geq \max(X, Y)/\sqrt{2} \geq (X + Y)/(2\sqrt{2})$, which proves (3.9). $\square$

Lemma 3.4 (Fractional orientation). *Every finite graph H admits nonnegative fractions $\theta_{ij} + \theta_{ji} = 1$ on each edge such that $\sum_j \theta_{ij} \leq \rho_G(H)$ for every vertex. This is the least possible uniform bound on the vertex loads.*

Proof. For $m = |E(H)| > 0$, form a flow network with a source, edge nodes, vertex nodes, and a sink. Source-to-edge capacities are one; arcs from an edge node to its endpoints have capacity $m + 1$; vertex-to-sink capacities are t . A cut of capacity at most m cannot cut an edge-to-endpoint arc. For a fixed set U of vertex nodes on the source side, placing all edge nodes internal to U on that side minimizes capacity, giving $m - |E(H[U])| + t|U|$. At $t = \rho_G(H)$ every cut has capacity at least m . Max-flow/min-cut supplies a flow saturating the source arcs; outgoing flows at each edge define its fractions. Conversely, the loads over U sum to at least $|E(H[U])|$, proving minimality. For $m = 0$ use zero loads. $\square$

Proof of Theorem 3.1. Fix an induced graph H with nonempty weighted support and abbreviate $L = \sum_{ij} |a_{ij}|$, $R = R_{V(H)}(a)$. Weighted Cauchy–Schwarz gives

$$\sum_j \theta_{ij} |a_{ij}| \leq \left(\sum_j \theta_{ij} \right)^{1/2} \left(\sum_j \theta_{ij} a_{ij}^2 \right)^{1/2} \leq \sqrt{\rho_G(H)} \|a_i\|_2.$$

Summing counts each absolute edge weight once. By (3.9), $L \leq 4\sqrt{\rho_G(H)}R$. Also $\|a_i\|_2 \geq \|a_i\|_1/\sqrt{\Delta_H}$ at every nonisolated vertex, and $\sum_i \|a_i\|_1 = 2L$. The same row bound therefore gives $R \geq L/(2\sqrt{\Delta_H})$. The graph parameters do not increase in induced subgraphs, so Lemma 3.2 proves (3.3).

For bipartite H use its inherited parts P', Q' in (3.8). Every edge crosses, giving

$$R \geq \frac{1}{\sqrt{2}} \sum_{i \in P'} \|a_i\|_2 \geq \frac{L}{\sqrt{2\Delta_P}}.$$

Transpose to obtain the bound with Δ_Q and apply the same induced characterization. On a unit-weight four-cycle with one negative edge, $L = 4$, $R = 2$, and $\Delta_P = \Delta_Q = 2$. Its center ratio attains two, so the stated bipartite constant is sharp. An affine box transformation scales each bilinear coefficient by the two interval widths and introduces only affine summands, proving the final assertion. $\square$

The degeneracy bound has a distinct direct proof. Orient along a degeneracy order with outdegree at most d_G and color vertices in reverse order. Place a vertex opposite the side receiving at least half the squared weight of its outgoing edges. Its crossing row norm is at least its outgoing row norm divided by $\sqrt{2}$. Summing and applying Cauchy–Schwarz to at most d_G outgoing entries gives $X + Y \geq L/\sqrt{2d_G}$. The final step of Lemma 3.3 yields $R \geq L/(4\sqrt{d_G})$. This proof is weaker than the density theorem but uses an ordinary acyclic orientation.

3.3 Sharp order on every interaction graph

Define

$$\Gamma(G) = \sup_{a \in \mathbb{R}^E \setminus \{0\}} c^*(a), \quad \Gamma_{\pm}(G) = \max_{a \in \{-1,1\}^E} c^*(a),$$

with both values zero for edgeless graphs.

Theorem 3.5 (Worst coefficients on a fixed graph). *For every graph with an edge,*

$$\max \left\{ 1, \sqrt{\frac{\rho_G(G)}{2 \log 2}} \right\} \leq \Gamma_{\pm}(G) \leq \Gamma(G) \leq 4\sqrt{\rho_G(G)}. \quad (3.11)$$

Requiring every coefficient to be nonzero does not change $\Gamma(G)$. It also equals

$$\sup_{a \neq 0} L_V(a)/R_V(a). \quad (3.12)$$

The lower bound has a full-support signing and a witness in $\{0, 1/2\}^V$.

Proof. On an h -vertex, m -edge graph with $m > 0$, choose independent edge signs σ_e . For fixed vertex signs s and $\lambda > 0$, $\mathbb{E} e^{\lambda Q_{\sigma}(s)} = \cosh(\lambda)^m \leq e^{m\lambda^2/2}$, and similarly for $-\lambda$. Put $Z = \max_s |Q_{\sigma}(s)|$. Global sign reversal preserves Q_{σ} , so

$$e^{\lambda Z} \leq \sum_{s: s_1=1} (e^{\lambda Q_{\sigma}(s)} + e^{-\lambda Q_{\sigma}(s)}).$$

Taking expectations and using Jensen gives $\mathbb{E} Z \leq h \log 2 / \lambda + m\lambda/2$. At $\lambda = \sqrt{2h \log 2 / m}$, at least one signing satisfies

$$\|Q_{\sigma}\|_{\infty} \leq \sqrt{2mh \log 2}. \quad (3.13)$$

The values at different sign vectors need not be independent.

Apply this to a densest induced graph $H = G[W]$. Since $L_W = m$ and $R_W \leq \|Q_{\sigma}\|_{\infty}$, $L_W/R_W \geq \sqrt{\rho_G(G)/(2 \log 2)}$. Give all remaining edges arbitrary signs; they do not affect the witness face of Lemma 3.2. The lower bound one follows from $R_W \leq L_W$, and the upper bound is Theorem 3.1.

Call the supremum in (3.12) C . The full center shows $\Gamma \geq C$. Conversely, every induced ratio is a whole-graph ratio after zeroing all coefficients outside the induced subgraph. Thus $\Gamma \leq C$. On the compact set $L_V(a) = 1$, R_V is continuous and strictly positive, so the ratio attains a maximum. Perturb zero entries of a maximizer to nonzero values. Continuity of the full-graph ratio shows that the full-support supremum is also C . This does not assume continuity of c^* as edges disappear or attainment under the full-support restriction. $\square$

Uniform boundedness of all signed factors in a graph family is therefore equivalent to uniform boundedness of its induced density. No closure assumption on the family is needed: fixing outside

coordinates exposes a densest subgraph. Global average density is insufficient. Attaching q^2 leaves to one vertex of $K_{q,q}$ produces a connected graph with total edge-to-vertex ratio less than two but induced density at least $q/2$.

The density is unweighted. Scaling all coefficients by a nonzero scalar preserves the gap ratio, whereas a sum of coefficient magnitudes scales. A single edge of arbitrarily small magnitude therefore rules out replacing ρ_G by $\max_W \sum_{e \in E(W)} |a_e|/|W|$. The row estimates remain coefficient-sensitive certificates; a uniform gap bound must use them on every induced subgraph, not only the full graph.

The order may equivalently be expressed using degeneracy or arboricity. If α_G is the minimum number of forests partitioning the edges, then $\rho_G(G) \leq \alpha_G \leq d_G \leq 2\rho_G(G)$. For the middle inequality, label outgoing edges in a degeneracy orientation with distinct labels from 1 to d_G . Each label class is a forest: an undirected cycle would have an earliest vertex with two outgoing edges of that label. The other inequalities follow by counting forest edges and (3.2).

3.4 Positive coefficients, exactness, and finite constants

For positive coefficients a proper coloring with $\chi \geq 2$ colors gives the classical stronger bound

$$c^*(a) \leq \begin{cases} 2 - 2/\chi, & \chi \text{ even,} \\ 2 - 2/(\chi + 1), & \chi \text{ odd.} \end{cases} \quad (3.14)$$

Choose uniformly $\lfloor \chi/2 \rfloor$ colors for one side of a cut. Every edge crosses with probability $p_\chi = 2\lfloor \chi/2 \rfloor \lceil \chi/2 \rceil / [\chi(\chi - 1)]$. Thus the maximum cut is at least $p_\chi L$, and the minimum cut is zero. The same coloring works on every induced subgraph, so Lemma 3.2 gives (3.14). This is the bound of Luedtke et al. [32, Theorem 8 in the authors' manuscript]. In particular, the positive bilinear factor is at most two and is one when its nonzero support is nonempty and bipartite. For the all-positive K_n with $n \geq 2$, the center ratio is $\binom{n}{2} / \lfloor n^2/4 \rfloor$, tending to two. Hence the worst-over-signs quantifier in Theorem 3.5 is essential.

Assume the effective support graph, obtained by deleting zero-coefficient edges, has at least one edge. Then $c^*(a) = 1$ if and only if every cycle of this graph has an even number of positive and an even number of negative edges [10, Theorem 4]. Boland et al. identify this as a consequence of the convex-envelope criterion in Theorem 3.10 of Misener, Smadbeck, and Floudas, *Dynamically generated cutting planes for mixed-integer quadratically constrained quadratic programs and their incorporation into GloMIQO 2*; our attribution to that predecessor is through Boland et al., who give the alternative cut proof used here. Indeed, $R_V = L_V$ exactly when one cut contains all positive edges and no negative edges, and another contains all negative edges and no positive edges. An edge set is a cut exactly when it intersects every cycle evenly. Necessity follows by traversing a cycle. For sufficiency, label vertices by the parity of specified edges on a path from a root in each component; cycle parity makes the label path-independent and supplies the cut. The criterion passes to induced subgraphs, proving sufficiency for $c^* = 1$. Conversely, $c^* = 1$ implies $R_V = L_V$ by Lemma 3.2. In particular, every forest with at least one edge has factor one for every signing. For empty effective support the gaps coincide at zero and $c^* = 0$ by convention.

The best universal constant in $\Gamma(G) \leq K_\rho \sqrt{\rho_G(G)}$ lies in $[2, 4]$: the lower bound is the four-cycle with one negative edge, with density one and factor two. For degeneracy the corresponding interval is $[\sqrt{2}, 4]$. These are finite-parameter questions, distinct from the best constants along sequences with density or degeneracy tending to infinity; the four-cycle does not give a growing-density lower bound.

The bounds imply $c^* \leq 4\sqrt{2}$ for series-parallel graphs by degeneracy, $c^* \leq 4\sqrt{3}$ for planar graphs by density, and $c^* \leq \sqrt{2 \min(p, q)}$ for $K_{p,q}$. For K_n , the maximum-degree bound is $2\sqrt{n-1}$,

compared with $600\sqrt{n}$ in Boland et al. [10, Theorem 2]. These are scalar vertical-gap comparisons, not approximation guarantees for a constrained objective.

3.5 Exact finite full-signing constants

For $n \geq 2$, distinguish the largest full-center ratio among complete-graph signings from the largest ratio over all points:

$$M_n = \max_{a \in \{-1,1\}^{E(K_n)}} \frac{\binom{n}{2}}{R_V(a)}, \quad F_n = \Gamma_{\pm}(K_n).$$

All edges here have magnitude one. These maxima differ from the arbitrary-real-coefficient supremum in (3.12).

Proposition 3.6 (Finite exhaustive computation). *The exact values are*

n	3	4	5	6	7
$\min_a R_V(a)$	2	4	4	5	8
M_n	3/2	3/2	5/2	3	21/8
F_n	3/2	3/2	5/2	3	3

Proof. The first row is established by the exhaustive integer computation in Appendix A, which proves completeness of the enumeration, gives executable code, and supplies attaining signings. Dividing $\binom{n}{2}$ by that row gives M_n . Lemma 3.2 gives

$$F_n = \max_{2 \leq k \leq n} M_k.$$

Indeed, each induced support of a full signing is a full signing of some K_k , giving the upper bound. Conversely, any signing of K_k extends to a full signing of K_n by assigning signs to all remaining edges, and its ratio persists on the face fixing the new coordinates to zero. Since $M_2 = 1$, this proves the last row. $\square$

In particular $M_7 < M_6$ while $F_7 = F_6 = 3$. An explicit witness is the K_6 signing in Table 4, extended by a new vertex with all incident coefficients positive. At the point whose six old coordinates are $1/2$ and whose new coordinate is zero, its ratio is three; at the full center its ratio is only $21/10$. The finite computation optimizes precisely these listed full-signing problems; it makes no claim about larger n or the optimal constants for arbitrary real coefficients.

3.6 The relation to Schur-multiplier theory

We record the older implication to separate the density order from the explicit elementary constant. For a real matrix M , its Schur multiplier acts by entrywise multiplication, with norm $\|M\|_{\text{Schur}}$ measured in the Euclidean operator norm. Define the real projective norm

$$\pi(M) = \inf \left\{ \sum_k \|u_k\|_{\infty} \|v_k\|_{\infty} : M = \sum_k u_k v_k^{\top} \right\}.$$

The established results of Davidson and Donsig [14, Theorems 1.2 and 2.4] give

$$\|M\|_{\text{Schur}} \leq 2\sqrt{\beta(P)}, \quad \pi(M) \leq K_G \|M\|_{\text{Schur}}, \quad (3.15)$$

for $|M_{ij}| \leq 1$ supported on a pattern P , where K_G is the real Grothendieck constant and

$$\beta(P) = \max_{R, C: |R|+|C|>0} \frac{|P \cap (R \times C)|}{|R| + |C|}.$$

The first bound uses the weighted matrix theorem, which permits continuous row and column bounds. Using the integer pattern theorem instead would introduce rounding. These are established norm inequalities, not statements about sizes of convex extended formulations.

For the symmetric adjacency pattern,

$$|P \cap (R \times C)| \leq |E(G[R \cup C])| + |E(G[R \cap C])| \leq \rho_G(G)(|R| + |C|).$$

An edge is counted at most once unless both endpoints lie in $R \cap C$, when both matrix orientations are counted. Taking $R = C$ equal to a densest set proves $\beta(P) = \rho_G(G)$. For weighted adjacency A , take $M = \text{sign}(A)$, with zeros where A vanishes. The definition of π gives

$$2L_V = \langle A, M \rangle \leq \pi(M) \|A\|_{\infty \rightarrow 1}.$$

For sign vectors u, v , put $p = (u+v)/2$, $q = (u-v)/2$. Symmetry yields $u^\top A v = 2[Q_{a,V}(p) - Q_{a,V}(q)]$. Multiaffinity puts these two values between the vertex extrema on $[-1, 1]^V$, so $\|A\|_{\infty \rightarrow 1} \leq 4R_V$. Consequently

$$L_V \leq 4K_G \sqrt{\rho_G(G)} R_V. \quad (3.16)$$

Apply this on every induced graph to recover the density upper order. The direct random-sign argument supplies the matching bilinear lower order.

Equivalently, let $S(G) = \sup_{a \neq 0} L_V(a) / \|Q_{a,V}\|_\infty$ be the real Sidon constant of the edge-supported degree-two sign characters, with $S(G) = 0$ for edgeless graphs. Because Q has mean zero, $\|Q\|_\infty / 2 \leq R_V \leq \|Q\|_\infty$ and hence $S(G) \leq \Gamma(G) \leq 2S(G)$. For bipartite graphs, flipping all signs in one part negates Q , making its range symmetric and giving $S(G) = \Gamma(G)$. The constant four improves the particular transfer (3.16); it is not asserted to be the best constant available in all prior literature.

4 Universal bounds for positive multilinear functions

Throughout this section the factors are the positive monomials, with equal supports collected. Let R_d be the supremum of T/H over all dimensions, positive multilinear polynomials of maximum degree at most d , finite nonnegative boxes, and points with $H > 0$. Let C_n be the corresponding supremum in n variables, with no additional degree restriction. Fixed coordinates and affine summands may be removed. Proposition 2.4 shows that both suprema can equivalently be taken on the unit cube: expansion does not increase dimension or degree and can only increase the termwise gap. All logarithms without a subscript are natural.

Theorem 4.1 (Sharp leading growth). *As the indicated integer parameter tends to infinity,*

$$R_d \sim \frac{\log d}{\log \log d}, \quad C_n \sim \frac{\log n}{\log \log n}. \quad (4.1)$$

For every $d \geq 2$, put $\Lambda = \max\{16, 1 + \log d\}$ and $c_0 = 1 - e^{-1}$. The explicit finite bound

$$T \leq \frac{24\Lambda}{c_0 \log \Lambda} H \quad (4.2)$$

holds on every finite nonnegative box. On the unit cube the upper bounds are witnessed by distributions defined from the complete marginal vector and degree allowance, independently of the supports and coefficients.

We first exhibit the obstruction to a constant bound, then construct a common distribution that matches its order. The lower family disproves Conjecture 1 in the author manuscript of Luedtke et al. [32, p. 22], which asks for a uniform constant for positive multilinear functions on nonnegative boxes. The signed bilinear examples of Section 3.6 concern a different coefficient class.

4.1 A sparse multiscale obstruction

Theorem 4.2 (Exact dyadic hull gap). *For every integer $L \geq 2$, there is a unit-coefficient polynomial in $n_L = 2^L + L$ variables, with $2^{L+1} - 2$ monomials, maximum degree $d_L = 2^{L-1} + 1$, and an interior point such that*

$$T = L, \quad H = s + (L - s)2^{-s}, \quad (4.3)$$

where $s \in \{1, \dots, L - 1\}$ satisfies

$$(L - s + 1)2^{-(s+1)} \leq 1 \leq (L - s + 2)2^{-s}. \quad (4.4)$$

Either adjacent choice at a cutoff equality gives the same value. The total number of variable occurrences is $L2^L + 2^{L+1} - 2$.

Proof. Put $m = 2^L$. Use leaf variables $z_1, \dots, z_m$ and anchors $a_1, \dots, a_L$. For each j , partition the leaves into 2^j blocks of size $k_j = 2^{L-j}$, and define

$$f_L(a, z) = \sum_{j=1}^L \sum_{B \in \mathcal{P}_j} a_j \prod_{i \in B} z_i, \quad a_j = 2^{-j}, \quad z_i = 1 - 1/m. \quad (4.5)$$

The upper and lower envelopes of each monomial at this point are 2^{-j} and $\max(0, 2^{-j} - k_j/m) = 0$. Thus $T = \text{cav } f_L = L$. The support and occurrence counts follow by summing 2^j and $2^j(k_j + 1)$, respectively.

For an arbitrary binary law with these means, let R count failed leaves and N_j count blocks hit by a failure. Then $\mathbb{E}R = 1$ and $N_j \leq \min(2^j, R)$. The deficiency representation gives

$$H = \max_j \mathbb{E} \sum_j A_j N_j \leq \sup_j \mathbb{E} \sum_j A_j \min(2^j, R), \quad \mathbb{E}A_j = 2^{-j}. \quad (4.6)$$

For a fixed law of R , an anchor of mass u maximizes its expectation against any increasing function of R by selecting the upper u -tail, splitting an atom if necessary. Indeed, moving selected mass from a smaller value to a larger unselected value cannot reduce the objective. A common nonincreasing quantile $r(t)$, $0 < t < 1$, therefore makes every anchor $A_j = \mathbf{1}\{t \leq 2^{-j}\}$ optimal simultaneously. Consequently the right side of (4.6) equals the supremum of $\int_0^1 S_{l(t)}(r(t)) dt$, where

$$S_l(r) = \sum_{j=1}^l \min(2^j, r), \quad w_l = \mathbb{P}(l(t) = l) = \begin{cases} 2^{-l-1}, & 0 \leq l < L, \\ 2^{-L}, & l = L. \end{cases}$$

Here r is integer valued, bounded by m , and has integral one.

Define $B_q = (L - q + 2)2^{-q}$. The sequence decreases strictly, with $B_1 \geq 1 \geq B_L$, so an s as in (4.4) exists. For every $r \geq 0$,

$$S_l(r) \leq sr + F_l(s), \quad F_l(s) = \begin{cases} 0, & l \leq s, \\ 2^{l-s+1} - 2, & l > s. \end{cases} \quad (4.7)$$

For $l \leq s$, bound each term by r . For $l > s$, cap the first $l - s$ terms at 2^j and the remaining s terms at r . Integrating yields

$$H \leq s + \sum_l w_l F_l(s) = s + (L - s)2^{-s}.$$

To attain this bound, choose l with probabilities w_l , let $A_j = \mathbf{1}\{j \leq l\}$, and for $q \in \{s, s + 1\}$ put

$$R_l^{(q)} = \begin{cases} 0, & l < q, \\ 2^{l-q+1}, & l \geq q. \end{cases}$$

Each anchor has its required mean and $\mathbb{E}R^{(q)} = B_q$. Mix $q = s$ and $q = s + 1$ with respective probabilities $(1 - B_{s+1})/(B_s - B_{s+1})$ and $(B_s - 1)/(B_s - B_{s+1})$. This gives $\mathbb{E}R = 1$. For $l > s$ the two failure counts are the endpoints of $[2^{l-s}, 2^{l-s+1}]$, where (4.7) is an equality. For $l = s$ they are 0, 2, also equality points, and for $l < s$ both are zero.

It remains to realize the hit counts, a step requiring nested partitions. Label leaves by L -bit strings, with level- j blocks defined by the first j bits. The first R strings in bit-reversal order have exactly $\min(2^j, R)$ different prefixes of length j : their prefixes are the reversed residues $0, \dots, R - 1$ modulo 2^j . Declare these leaves failed and apply a uniform bitwise XOR shift to every label. The shift preserves all hit counts and gives each leaf failure probability R/m , conditional on R . Averaging gives failure mean $1/m$ for every leaf. The constructed law therefore attains the bound in the original polynomial. $\square$

The cutoff conditions imply $s = \log_2 L + O(1)$ and $(L - s)2^{-s} = O(1)$. Thus the ratio in (4.3) is asymptotic to $L/\log_2 L$. The exact formula uses nested dyadic partitions. For arbitrary equal-size partitions the same surrogate proves the weaker but still divergent lower bound

$$\frac{T}{H} \geq \frac{L}{5 + \log_2(1 + (L - 1)\log 2)}; \quad (4.8)$$

Appendix B gives its distinct integral proof and the dense symmetric predecessor.

The construction may be made homogeneous of degree d_L : introduce $d_L - 2$ padding variables and multiply a degree- k term by the first $d_L - k$ of them. At padding means one both gaps agree exactly with those above, since all representing laws fix the padding coordinates. Moving them sufficiently close to one makes all means interior with ratio arbitrarily close to (4.3). This adds $O(n_L)$ variables and keeps $O(n_L)$ monomials, but the occurrence count after padding is $O(n_L^2)$, not the original $O(n_L \log n_L)$.

4.2 Harmonic failure distributions

Call a coordinate low when $x_i \leq 1/2$ and high otherwise. Independence handles the terms with zero or at least two low coordinates by Lemma 2.3. For the remaining terms we require failures at several probability scales simultaneously.

Fix a real $M \geq d - 1$, and put

$$\Lambda = 1 + \log M, \quad \eta = (d - 1)/M \in (0, 1]. \quad (4.9)$$

We use η for this cutoff parameter, to distinguish it from graph density and physical box aspect ratio. Both laws below draw one uniform $U \in (0, 1)$ and set every low coordinate to $\mathbf{1}\{U \leq x_i\}$. In

the *threshold law*, a high coordinate with failure mean $p = 1 - x_i$ fails exactly when $U \leq p$. In the *harmonic law*, a coordinate with $p = 0$ never fails; otherwise set

$$h_p = \min(Mp, 1), \quad A_p = 1 + \log(h_p/p), \quad q_p(t) = \frac{\min(1, p/t)}{A_p} \mathbf{1}\{t \leq h_p\}. \quad (4.10)$$

Conditional on $U = t$, high coordinates fail independently with probabilities $q_p(t)$. Since $M \geq 1$, $1 \leq A_p \leq \Lambda$ and $h_p \geq p$. In particular these probabilities are valid, and

$$\int_0^1 q_p(t) dt = \frac{p + p \log(h_p/p)}{A_p} = p.$$

Thus the two laws preserve the full marginal vector, including boundary means, and require no information about a particular term.

Lemma 4.3 (Complete harmonic guarantee). *Consider a term with one low anchor of mean u , and high failure means $p_1, \dots, p_r$, where $r \leq d - 1$. Put $t_e = \min(u, \sum_j p_j) > 0$ and $z = \min(\max_j p_j/t_e, 1)$. The threshold and harmonic deficiencies are at least zt_e and $J_{\Lambda, \eta}(z)t_e$, respectively, where*

$$F_{\Lambda, \eta}(v) = 1 - \exp\left(-\frac{1 - \eta v}{\Lambda v}\right), \quad J_{\Lambda, \eta}(z) = \int_z^1 F_{\Lambda, \eta}(v) dv, \quad F_{\Lambda, \eta}(0) = 1. \quad (4.11)$$

Proof. Threshold deficiency is $\min(u, \max_j p_j) \geq zt_e$. If $\max_j p_j \geq t_e$, the harmonic assertion is nonnegativity. Otherwise integrate over $\max_j p_j \leq t \leq t_e$. The anchor succeeds there and all $p_j \leq t$. An inactive coordinate has $p_j < t/M$, so the inactive failure mass is at most $rt/M \leq \eta t$. Active mass is consequently at least $\sum_j p_j - \eta t \geq t_e - \eta t$. On active coordinates, $q_{p_j}(t) \geq p_j/(\Lambda t)$. Conditional independence gives union failure probability at least $1 - \exp(-(t_e - \eta t)/(\Lambda t))$. Substitution $v = t/t_e$ proves the claim. Zero term gaps require no division and satisfy every asserted nonnegative bound automatically. $\square$

Theorem 4.4 (Optimized finite harmonic bound). *For the parameters in (4.9), let ζ be the unique solution of $J_{\Lambda, \eta}(\zeta) = \zeta$ in $(0, 1)$. Then*

$$R_d \leq \frac{1}{\zeta} + \frac{2}{c_0}. \quad (4.12)$$

If $w > 0$ solves $we^w = \Lambda e^{-\eta}$ and $w \geq 1$, then

$$\zeta \geq \frac{w - 1/(2w)}{\Lambda}, \quad R_d \leq \frac{\Lambda}{w - 1/(2w)} + \frac{2}{c_0}. \quad (4.13)$$

The parameter M can be optimized over $[d - 1, \infty)$ in either bound. For $N = 1 + \log(d - 1)$, choosing $M = (d - 1)N$ gives, as $d \rightarrow \infty$,

$$R_d \leq \frac{N}{\log N - \log \log N + o(1)} + \frac{2}{c_0}. \quad (4.14)$$

This is an upper certificate, with no matching second-order lower assertion.

Proof. F is continuous and strictly decreasing, positive on $(0, 1)$, and nonnegative at one. Thus J is strictly decreasing and strictly convex, and $J(z) - z$ has opposite signs at zero and one. Its unique root is ζ . Set $a = F(\zeta) \in (0, 1)$. The convex tangent inequality gives

$$J(z) + az \geq J(\zeta) + a\zeta = (1 + a)\zeta.$$

Mix the harmonic and threshold laws with weights $1/(1+a)$ and $a/(1+a)$. Lemma 4.3 gives deficiency at least ζt_e for every one-low term. This mixture is optimal for the two *scalar guarantees* $J(z)$ and z : at $z = \zeta$ every convex combination equals ζ . It does not prove optimality over all actual couplings or imply simultaneous attainability of these lower curves by an extremal polynomial. Mix this law and independence with weights proportional to $1/\zeta$ and $2/c_0$. Each term receives at least $t_e/(1/\zeta + 2/c_0)$ from its appropriate component; the other contributions are nonnegative. Proposition 2.2 and then Proposition 2.4 prove (4.12).

For the finite certificate, use $x - x^2/2 \leq 1 - e^{-x} \leq x$ for $x \geq 0$ and, for $0 < z \leq 1$,

$$\int_z^1 (1/v - \eta)^2 dv = 1/z - 1 + 2\eta \log z + \eta^2(1 - z) \leq 1/z.$$

The last inequality follows from $2\eta \log z \leq 0$ and $-1 + \eta^2(1 - z) \leq 0$. For $0 < A \leq \Lambda$ this gives

$$\Lambda J(A/\Lambda) \geq \log \Lambda - \log A - \eta + \eta A/\Lambda - 1/(2A) \geq \log \Lambda - \log A - \eta - 1/(2A). \quad (4.15)$$

Take $A = w - 1/(2w)$. Since $w \geq 1$, $A \geq 1/2$ and $A < \Lambda$. Using $w + \log w = \log \Lambda - \eta$,

$$A + \log A + \frac{1}{2A} - (w + \log w) = \log \left(1 - \frac{1}{2w^2}\right) + \frac{1}{4Aw^2} \leq 0.$$

The inequality follows from $\log(1 - t) \leq -t$ and $A \geq 1/2$. Hence $J(A/\Lambda) \geq A/\Lambda$; monotonicity of $J(z) - z$ proves (4.13).

For the chosen cutoff, $\Lambda = N + \log N$ and $\eta = 1/N$. The defining identity for w first gives $\log \Lambda - \eta - \log \log \Lambda \leq w \leq \log \Lambda$ for all sufficiently large N . Thus $w/\log N \rightarrow 1$, and substitution back gives $w = \log N - \log \log N + o(1)$. Since $(\log N)^2/N \rightarrow 0$,

$$\frac{N}{\Lambda} \left(w - \frac{1}{2w} \right) = \log N - \log \log N + o(1).$$

This proves (4.14). □

Proof of Theorem 4.1. For its original explicit finite bound, take $\Lambda = \max\{16, 1 + \log d\}$ and $M = e^{\Lambda-1} \geq d$. For a one-low term, put $t_e = \min(u, \sum_j p_j) > 0$. If $\max_j p_j \geq t_e/\sqrt{\Lambda}$, threshold deficiency is at least $t_e/\sqrt{\Lambda} \geq t_e \log \Lambda/\Lambda$; here $\log \Lambda \leq \sqrt{\Lambda}$ follows by maximizing $(\log t)/\sqrt{t}$. Otherwise integrate the harmonic law over $[t_e/\sqrt{\Lambda}, t_e/2]$. Inactive failure mass is at most t , so active mass is at least $t_e/2$. The conditional union probability is at least $c_0 t_e/(2\Lambda t)$, using $1 - e^{-z} \geq c_0 \min(1, z)$. The harmonic deficiency is at least

$$\frac{c_0 t_e}{4\Lambda} (\log \Lambda - 2 \log 2) \geq \frac{c_0 t_e \log \Lambda}{8\Lambda}.$$

An equal mixture of independence, threshold and harmonic laws now gives (4.2), since independence supplies at least $c_0 t_e/2$ on the other two classes. Every distribution is global.

Theorem 4.4 implies $R_d \leq (1 + o(1)) \log d / \log \log d$. For the reverse inequality choose the largest L with $2^{L-1} + 1 \leq d$ in Theorem 4.2. Its degree differs from d by a factor at most two, asymptotically, and its ratio is asymptotic to $L/\log_2 L$. For dimension, $C_n \leq R_n$; choose the largest L with $2^L + L \leq n$ and add unused coordinates. Again the dimensions differ by at most a constant factor. These comparisons prove both equivalences in (4.1) for all integers tending to infinity, not merely for a dyadic subsequence. □

For comparison, the earlier leading-constant mixture has a distinct explicit parameterization. Set $\Lambda = \max\{e^6, 1 + \log d\}$, $M = e^{\Lambda-1}$, $b = \log \Lambda$, $a = \Lambda/b^2$, $\beta = 1/b$, and

$$h = \frac{(1 - 1/b)(1 - 1/(2b^2))(b - 3 \log b)}{\Lambda}. \quad (4.16)$$

Here $h > 0$ because $b \geq 6$ and $b - 3 \log b > 0$. If $\max p_j \geq t_e/a$, threshold deficiency is at least t_e/a . Otherwise, on $[t_e/a, \beta t_e]$, active mass is at least $(1 - \beta)t_e$ and the lower exponent $(1 - \beta)t_e/(\Lambda t)$ is at most $1/b^2$. Integrating $1 - e^{-z} \geq z(1 - 1/(2b^2))$ gives harmonic deficiency at least ht_e . Weights proportional to $1/h, a, 2/c_0$ on harmonic, threshold, and independence give $R_d \leq 1/h + a + 2/c_0$, also with leading constant one. The full-curve theorem retains more of the same distributions' information and improves this certificate's lower-order loss.

A useful interpretation is simultaneous interpolation between Fréchet bounds. The global law with $Z = 1/\zeta + 2/c_0$ satisfies, for every $e \subseteq [n]$ with $2 \leq |e| \leq d$,

$$\mathbb{E} \prod_{i \in e} X_i \leq (1 - Z^{-1}) \min_{i \in e} x_i + Z^{-1} \max(0, \sum_{i \in e} x_i - |e| + 1). \quad (4.17)$$

The best fraction guaranteed for every marginal vector is asymptotic to $\log \log d / \log d$: the theorem gives the lower guarantee, and summing any proposed guarantee over the dyadic example gives the reverse order and constant. No convex-envelope evaluation algorithm is inferred.

4.3 Removing coefficients without changing the degree supremum

Theorem 4.5 (Unit coefficients and interior homogeneous examples). *For each fixed $d \geq 2$, the supremum R_d is unchanged by requiring unit coefficients, degree exactly d for every term, and strictly interior unit-cube evaluation points simultaneously. Dimension and support size are unrestricted in this assertion.*

Proof. By positive expansion it suffices to approximate any unit-cube example. Delete affine terms and homogenize with $d - 2$ padding variables as above. At padding means one the gaps are unchanged. Continuity of the finite polyhedral envelopes (Lemma 2.1) permits an interior perturbation preserving any strict target-ratio inequality. Normalize by the largest coefficient. We now have one fixed homogeneous polynomial $f = \sum_{e \in E} a_e \prod_{i \in e} x_i$, $0 < a_e \leq 1$, with n variables, s distinct supports, interior means x , and $H(f; x) > 0$.

Replace coordinate i by m clones $y_{i1}, \dots, y_{im}$, and set

$$\bar{y}_i = m^{-1} \sum_{r=1}^m y_{ir}, \quad F_m(y) = m^d f(\bar{y}), \quad y_{ir} = x_i.$$

There are sm^d distinct candidate clone monomials: each uses one clone from each of its d distinct original groups. At the repeated point,

$$\text{vex } F_m = m^d \text{vex } f, \quad \text{cav } F_m = m^d \text{cav } f, \quad T(F_m) = m^d T(f), \quad H(F_m) = m^d H(f). \quad (4.18)$$

Indeed, any binary clone law induces random group averages $q \in [0, 1]^n$ with $\mathbb{E}q = x$. Conditional independent rounding of the original variables with means q_i has expected polynomial $f(q)$, so every normalized clone expectation is an original feasible expectation. Conversely, setting all clones in a group equal to its original binary coordinate embeds every original law. This proves equality of the entire feasible expectation intervals, and hence of both envelopes. Each clone term has its original list of means, proving the termwise identity.

Independently retain each candidate term arising from e with probability a_e and give it coefficient one; call the result G_m . For N_0 independent random variables with range lengths at most one, the bounded-sum inequality of Hoeffding [25, Theorem 2] gives

$$\mathbb{P}(|\sum_j (Y_j - \mathbb{E}Y_j)| > t) \leq 2e^{-2t^2/N_0}. \quad (4.19)$$

For completeness, if $Y \in [a, b]$, the second derivative of $\log \mathbb{E}e^{\lambda Y}$ is its variance under exponential tilting, at most $(b-a)^2/4$: its squared distance from the interval midpoint is at most that quantity. Integrating twice gives $\mathbb{E}e^{\lambda(Y-\mathbb{E}Y)} \leq e^{\lambda^2(b-a)^2/8}$. Multiplication over independent variables, Markov's inequality, and optimization at $\lambda = 4t/N_0$ prove each tail in (4.19). Thus no concentration result is needed without its hypotheses or proof.

Apply this inequality with $N_0 = sm^d$ at each of the 2^{nm} clone vertices, and once to the termwise gap at the repeated point. The latter is a sum of the same retention indicators with fixed weights in $[0, 1]$. A union bound gives failure probability at most

$$2(2^{nm} + 1) \exp(-2t^2/(sm^d)) \quad (4.20)$$

for the two simultaneous assertions

$$\max_{v \in \{0,1\}^{nm}} |G_m(v) - F_m(v)| \leq t, \quad |T(G_m) - T(F_m)| \leq t.$$

Take $t = Km^{(d+1)/2}$ with $K^2 > sn \log 2/2$. The failure probability tends to zero as $m \rightarrow \infty$, with f, n, s, d fixed. Uniform vertex error bounds the error of each envelope at every marginal vector by t , by comparing expectations under all feasible laws. Thus the hull-gap error is at most $2t$. Since $t/m^d \rightarrow 0$ for $d \geq 2$, successful samples' normalized gaps converge to those of f , and their ratios converge because its hull gap is positive. This proves equality of suprema, not preservation at a fixed dimension or existence of a small sampled support. $\square$

5 Cubic gaps and the role of unequal means

The divergence in Theorem 4.1 requires increasing degree. Degree three already distinguishes positive multilinear functions from the positive bilinear factor-two theorem, however. We prove

$$\frac{1610000}{743033} \leq R_3 \leq \frac{31}{12}. \quad (5.1)$$

The lower endpoint is a bound on a supremum supplied by a sequence of certificates, not a claimed attained finite optimum. We also give exact finite hull values and a homogeneous unit-coefficient example with only 52 variables. The exact value of R_3 remains open.

5.1 Endpoint orientation and a cubic mixture

One particularly simple global law applies in every degree. Draw $U \in [0, 1]$ uniformly and independently orient each coordinate left or right with equal probability. A left coordinate is the indicator of $[0, x_i]$ and a right coordinate of $[1 - x_i, 1]$. Call this law O . Every coordinate has mean x_i .

Proposition 5.1 (Endpoint-orientation bound). *For maximum degree $d \geq 2$, on every nonnegative box,*

$$T \leq \frac{d-1}{1-2^{-(d-1)}} H. \quad (5.2)$$

In particular the constant is $8/3$ for cubics and two for bilinear terms.

Proof. On the unit cube, fix a degree- $(k+1)$ term with minimum mean u at its anchor. Conditional on the anchor orientation, a same-oriented coordinate imposes no further restriction on its success interval. An opposite coordinate excludes an interval of length $a_j = \min(u, 1 - x_j)$ at the opposite end. These excluded intervals are nested. If $a_{(1)} \geq \dots \geq a_{(k)}$, the first selected index has probability 2^{-j} of being j , and hence

$$\mathbb{E}_O D_e = \sum_{j=1}^k 2^{-j} a_{(j)} \geq \frac{1 - 2^{-k}}{k} \sum_j a_j \geq \frac{1 - 2^{-k}}{k} \min(u, \sum_j (1 - x_j)).$$

The first inequality follows by summing $(2^{-i} - 2^{-j})(a_{(i)} - a_{(j)}) \geq 0$ over pairs, and the second by $\sum_j \min(u, p_j) \geq \min(u, \sum_j p_j)$. The fraction $(1 - 2^{-k})/k$ decreases with k , since it is the average of the first k entries of $1/2, 1/4, \dots$. Sum the simultaneous guarantees and use Proposition 2.4. $\square$

Theorem 5.2 (Three-law cubic bound). *Every positive multilinear polynomial of degree at most three on a finite nonnegative box satisfies $T \leq (31/12)H$. On the unit cube, the global law*

$$P = \frac{18}{31}O + \frac{6}{31}I + \frac{7}{31}B \tag{5.3}$$

guarantees deficiency at least $12t_e/31$ for every quadratic or cubic term. Here I is independent rounding. For B , draw one uniform U , set each low coordinate to $\mathbf{1}\{U \leq x_i\}$, and, independently conditional on U , let each high coordinate fail with probability $1/2$ when $U \leq 2(1 - x_i)$ and never otherwise. The weights are optimal among fixed mixtures of these three laws for a uniform termwise guarantee.

Proof. The definition of B gives each high coordinate failure mean $1 - x_i$; thus all three laws have the prescribed means. Consider sorted cubic means $u \leq v \leq w$, and set $t = \min(u, 2 - v - w)$. Terms with $t = 0$ need no estimate. Write D_O, D_I, D_B for their expected deficiencies.

If $u \leq v \leq 1/2$, then $t = u$. Opposite anchor and second-coordinate orientations give $D_O \geq u/2$, while $D_I = u(1 - vw) \geq u/2$. These two contributions alone give $12t/31$.

If exactly one coordinate is low, set $a = 1 - v \geq b = 1 - w$. Directly integrating the nested exclusion and failure intervals gives

$$D_O = \frac{1}{2} \min(u, a) + \frac{1}{4} \min(u, b), \quad D_B = \frac{1}{2} \min(u, 2a) + \frac{1}{4} \min(u, 2b).$$

We verify $18D_O + 7D_B \geq 12 \min(u, a + b)$ in all cases. If $a \leq u/2$, the left side is $16a + 8b \geq 12(a + b)$. If $b \geq u/2$, it is at least $18(3u/8) + 7(3u/4) = 12u$. If $b \leq u/2 \leq a \leq u$, it is $9a + 8b + 7u/2$. When $a + b \leq u$, use $3a + 4b \leq 4u - a \leq 7u/2$; when $a + b \geq u$, use $9a + 8b \geq 8u + a \geq 17u/2$. Finally, if $b \leq u/2$ and $a \geq u$, the expression is $25u/2 + 8b \geq 12u$. These cases exhaust $a \geq b \geq 0$, including their boundaries. The I contribution is nonnegative.

If all coordinates are high, put $c = 1 - u \geq a = 1 - v \geq b = 1 - w$. Integrating the B union probabilities $7/8, 3/4, 1/2$ over intervals of lengths $2b, 2(a - b), 2(c - a)$ gives total failure probability $c + a/2 + b/4$. Subtracting the anchor failure probability c yields

$$D_B = D_O = a/2 + b/4 \geq 3(a + b)/8 \geq 3t/8.$$

Independence gives $D_I = u(a + b - ab) \geq 7t/16$. Indeed, set $s = a + b \leq 1$ and use $ab \leq s^2/4$. For $s \leq u$ the normalized expression is at least $u(1 - s/4) \geq u(1 - u/4) \geq 7/16$. For $s \geq u$ it is at least $s - s^2/4 \geq u - u^2/4 \geq 7/16$. Both comparisons use $u > 1/2$ and monotonicity on the unit interval. Thus the mixture fraction is at least $(25/31)(3/8) + (6/31)(7/16) = 12/31$.

For sorted quadratic means $u \leq v$, set $t = \min(u, 1 - v)$; always $D_O = t/2$. If both means are low, $D_I \geq t/2$. If exactly one is low, $D_B = \min(u, 2(1 - v))/2 \geq t/2$. If both are high, $D_B = t/2$ and $D_I = ut \geq t/2$. The resulting mixture fractions are at least 12/31, 25/62, and 1/2. This proves the simultaneous guarantee. Summation and positive expansion give the claimed box bound.

To check optimality in the stated family, let the arbitrary weights of (O, I, B) be (w, r, v) with $w + r + v = 1$. The following test means give the listed exact or limiting normalized deficiencies:

means	D_O/t	D_I/t	D_B/t
$(u, 1/2), 0 < u \leq 1/2$	1/2	1/2	0
$(\varepsilon, 1 - \varepsilon^2, 1 - \varepsilon^2), \varepsilon \downarrow 0$	3/8	0	3/4
$(u, 1 - u/2, 1 - u/2), u \downarrow 1/2 \text{ from above}$	3/8	7/16	3/8

In the second row $t = 2\varepsilon^2$ for small ε and the independent fraction is $\varepsilon - \varepsilon^3/2$. In the last row $1/2 < u < 2/3$ ensures $t = u$ and the independent fraction is $u - u^2/4$. Therefore a universal fraction α obeys

$$\alpha \leq \frac{1}{2} - \frac{v}{2}, \quad \alpha \leq \frac{3}{8} + \frac{3v}{8} - \frac{3r}{8}, \quad \alpha \leq \frac{3}{8} + \frac{r}{16}.$$

Average these inequalities with weights 3/31, 4/31, 24/31. The variable coefficients cancel, giving $\alpha \leq 12/31$. This conclusion concerns fixed-mixture termwise guarantees; it does not identify R_3 , or exclude a stronger coefficient-dependent analysis. $\square$

5.2 A scalar certificate and the strongest limiting lower bound

For a set Q of variables, write $E_k(Q) = \sum_{S \subseteq Q, |S|=k} \prod_{i \in S} x_i$. At a binary vertex this is $\binom{K}{k}$ when K counts successes in Q . Count-polynomial identities below are vertex identities, not substitutions defining the multilinear functions at continuous points.

Lemma 5.3 (Cubic scalar minorant with positive slack). *On $[0, 1]^3$, let*

$$\begin{aligned} F(a, b, c) &= 6c^3 + 27bc^2 + 18b^2 + 30ac + 20ab + 9a^2, \\ \ell(a, b, c) &= 37a + \frac{79}{2}b + 38c - \frac{103}{3}. \end{aligned}$$

Then $F \geq \ell + \delta_$, where $\delta_* = 901/120000$.*

Proof. For fixed c , the Hessian of $h = F - \ell$ in (a, b) is $\begin{pmatrix} 18 & 20 \\ 20 & 36 \end{pmatrix}$, positive definite with determinant 248. For $0 \leq c \leq 3/10$, set $a = 1$ and $b_*(c) = 13/24 - 3c^2/4 \in (0, 1)$. There $\partial_b h = 0$ and $\partial_a h = -49/6 + 30c - 15c^2 \leq -31/60 < 0$. The convex first-order inequality proves that this is the constrained minimum on the (a, b) square: $(a - 1)\partial_a h \geq 0$ for $a \leq 1$. Its value is

$$p(c) = \frac{-972c^4 + 576c^3 + 1404c^2 - 768c + 101}{96}.$$

For $3/10 \leq c \leq 1$, minimizing over unrestricted real a, b is a valid lower bound. Completing the square gives

$$r(c) = \frac{-78732c^4 + 212256c^3 - 203796c^2 + 82032c - 11275}{2976}.$$

For example, with $v = 30c - 37$, $w = 27c^2 - 79/2$ and $q = 6c^3 - 38c + 103/3$, this value is $q - (18v^2 - 20vw + 9w^2)/248$, which verifies the elimination directly.

Table 2 supplies exact degree-four Bernstein representations. A row with interval $[l, u]$, denominator D , and numerators $v_0, \dots, v_4$ means

$$p(c) \text{ or } r(c) = D^{-1} \sum_{i=0}^4 v_i \binom{4}{i} t^i (1-t)^{4-i}, \quad t = (c-l)/(u-l).$$

These are rational polynomial identities, verified by expanding in t . The five intervals cover the required ranges. The Bernstein basis is nonnegative and sums to one, and the minimum of all 25 coefficients v_i/D is $1802/240000 = \delta_*$. Thus both lower polynomials are at least δ_* wherever used, proving the claim. $\square$

Polynomial	Interval	D	$(v_0, v_1, v_2, v_3, v_4)$
p	$[0, 1/5]$	60000	(63125, 39125, 20975, 9395, 4133)
p	$[1/5, 3/10]$	240000	(16532, 6008, 1802, 3788, 11597)
r	$[3/10, 1/2]$	7440000	(215357, 1285415, 1434125, 1250375, 1008125)
r	$[1/2, 3/4]$	190464	(25808, 18056, 7964, 9230, 15869)
r	$[3/4, 1]$	190464	(15869, 22508, 34520, 45920, 31040)

Table 2: Exact scalar positivity certificate for Lemma 5.3.

Theorem 5.4 (Analytic cubic lower family). *For each multiple $m \geq 9$ of nine, take three groups U, V, W of m variables and write A, B, C for their sums. Define*

$$f_m = 2E_3(W) + 3BE_2(W) + 2mE_2(V) + \frac{5m}{3}AC + \frac{10m}{9}AB + mE_2(U). \quad (5.4)$$

At group means $1/4, 1/2, 3/4$, respectively,

$$\frac{T(f_m)}{H(f_m)} \geq \frac{161/4 - 135/(4m) + 6/m^2}{223/12 - \delta_* + 135/(4m) + 9/m^2}. \quad (5.5)$$

Consequently $R_3 \geq 1610000/743033$.

Proof. All coefficients in (5.4) are positive integers. At binary counts put $a = A/m, b = B/m, c = C/m$. Expansion of binomial coefficients gives the exact identity

$$\frac{18f_m}{m^3} = F(a, b, c) - \frac{18c^2 + 27bc + 18b + 9a}{m} + \frac{12c}{m^2} \geq F(a, b, c) - 72/m. \quad (5.6)$$

Every feasible binary law has expected normalized counts $(1/4, 1/2, 3/4)$. Lemma 5.3 therefore gives

$$\frac{18}{m^3} \text{vex } f_m \geq 139/6 + \delta_* - 72/m.$$

Counting supports and using (2.6) yields

$$\frac{18}{m^3} \text{cav } f_m = 167/4 - 153/(4m) + 9/m^2, \quad (5.7)$$

$$\frac{18}{m^3} T(f_m) = 161/4 - 135/(4m) + 6/m^2. \quad (5.8)$$

For clarity, the six scaled upper contributions in the displayed family order are

$$\frac{9}{2} - \frac{27}{2m} + \frac{9}{m^2}, \quad \frac{27}{2} - \frac{27}{2m}, \quad 9 - \frac{9}{m}, \quad \frac{15}{2}, \quad 5, \quad \frac{9}{4} - \frac{9}{4m}.$$

Only $2E_3(W)$ has a positive termwise lower envelope; its scaled value is $3/2 - 9/(2m) + 3/m^2$. This proves the two exact formulas. Subtraction bounds the scaled hull gap by the denominator in (5.5). It is positive for $m \geq 9$, and the actual hull gap is positive as well: independence gives strictly

smaller expectations than common-threshold rounding for every positive nonlinear term at interior means. Division is therefore valid. Taking arbitrarily large multiples of nine yields

$$R_3 \geq \frac{161/4}{223/12 - 901/120000} = \frac{4830000}{2229099} = \frac{1610000}{743033}.$$

Dropping the positive slack recovers the weaker $483/223$; the same certificate proves the stronger reduced fraction above. Neither minorant is asserted to be attained in expectation, and neither limiting bound asserts an attained finite gap ratio or convergence of the actual family ratios to that lower endpoint. At $m = 36$, even dropping δ_* from (5.5) gives $T/H \geq 16985/8436 > 2$; retaining the positive slack makes this finite bound strictly stronger. $\square$

5.3 Exact finite cubic witnesses

Finite examples permit a stronger type of statement: matching primal and dual certificates determine both envelopes exactly. The three-group polynomials in Table 3 have group sizes m and means $(1/4, 1/2, 3/4)$, and take the form

$$f = c_1 E_3(W) + c_2 B E_2(W) + c_3 E_2(V) + c_4 AC + c_5 AB + c_6 E_2(U). \quad (5.9)$$

Proposition 5.5 (Exact finite values). *The envelope and ratio values in Table 3 are exact. The 18- and 24-variable polynomials have 212 and 464 monomials, respectively. Complete rational primal and dual certificates, and a short executable integer/rational checker, appear in Appendix C.*

Proof. At binary counts $A, B, C \in \{0, \dots, m\}$ the polynomial is

$$\Phi_m(A, B, C) = c_1 \binom{C}{3} + c_2 B \binom{C}{2} + c_3 \binom{B}{2} + c_4 AC + c_5 AB + c_6 \binom{A}{2}.$$

The appendix certifies an affine minorant at every count state and a probability distribution of count states with means $(m/4, m/2, 3m/4)$ attaining it. Conditional uniform subsets in each group realize every individual marginal, so these give matching lower-envelope certificates. The upper and termwise lower values follow directly from the six orbit sizes and monomial bounds in the appendix. Subtracting yields the ratios. $\square$

n	$(c_1, \dots, c_6)$	$\text{cav } f$	$\sum_e \text{vex } f_e$	$\text{vex } f$	T/H
18	(2, 3, 9, 10, 7, 7)	1647/4	10	2750/13	20891/10411
24	(2, 3, 13, 12, 8, 7)	971	28	3572/7	6601/3225
192	(2, 3, 120, 105, 70, 63)	587944	20832	34172072/105	7443345/3445256

Table 3: Exact cubic witnesses, all with ratio above two.

The 24-variable example also gives a quantitative homogeneous interior variant. Write it as $f = P + Q$, where P is cubic and Q quadratic, and put $g(x, z) = P(x) + zQ(x)$ with $z = 999/1000$. All 25 coordinates are interior and all 464 monomials are cubic, with integer coefficients at most 13. Their upper and lower termwise values are unchanged, so $\text{cav } g = 971$ and $T(g) = 943$. The sum of quadratic coefficients is

$$13 \binom{8}{2} + 12 \cdot 8^2 + 8 \cdot 8^2 + 7 \binom{8}{2} = 1840.$$

Thus for any feasible law, $\mathbb{E}g \geq \text{vex } f - 1840/1000$, regardless of dependence involving Z . Consequently

$$H(g) \leq 3225/7 + 46/25 = 80947/175, \quad \frac{T(g)}{H(g)} \geq \frac{165025}{80947} > 2. \quad (5.10)$$

This is a certified bound, rather than an exact new hull value.

For completeness, it yields a finite nonconstructive unit-coefficient example through Theorem 4.5. Normalize by 13, clone each variable $m = 1000$ times, and take error allowance $t = m^3/10$. The failure probability in (4.20) is at most

$$2(2^{25m} + 1) \exp(-m^3/23200) < 1. \quad (5.11)$$

Indeed its logarithm is at most $(25m + 2) \log 2 - m^3/23200 < 0$ at $m = 1000$, even with $\log 2 < 1$. Since $943 - 2(80947/175) = 3131/175$, every successful sample satisfies

$$T(G_m) - 2H(G_m) \geq m^3 \left(\frac{3131}{2275} - \frac{1}{2} \right) > 0. \quad (5.12)$$

It is nonempty and has positive hull gap at its interior point. This proves existence in 25,000 variables with $464m^3$ candidate monomials; no successful sample is claimed to be listed. The explicit construction below is much smaller, while coefficient removal remains useful for approximating an arbitrary fixed-degree ratio.

5.4 Two means and an explicit homogeneous unit-coefficient example

Theorem 5.6 (Two-level cubic family). *For each positive multiple m of four, take two groups U, W of m variables with means $1/2, 3/4$, and define*

$$f_m = AE_2(W) + \frac{5m}{4}E_2(U), \quad A = \sum_{i \in U} x_i. \quad (5.13)$$

Then

$$\frac{243}{115}(1 - 1/m) \leq \frac{T(f_m)}{H(f_m)} \leq \frac{243}{115}. \quad (5.14)$$

For $m = 4, 8, 12, 16$ its exact ratios are, respectively, $27/16, 21/11, 99/50, 135/67$. Among positive multiples of four, this family's ratio exceeds two if and only if $m \geq 16$. There is an explicit polynomial with 52 variables and 4,320 distinct homogeneous cubic monomials, every coefficient one, at an interior point with ratio at least $2700/1343 > 2$.

Proof. For real $a, c \in [0, 1]$, set $F_2(a, c) = ac^2 + 5a^2/4$ and $\ell_2(a, c) = 11a/6 + 20c/27 - 95/108$. The identity

$$F_2 - \ell_2 = \frac{5}{4} \left(a - \frac{11 - 6c^2}{15} \right)^2 + \frac{(1 - c)(3c - 2)^2(3c + 7)}{135} \quad (5.15)$$

proves $F_2 \geq \ell_2$ on the square. Equality holds at $(1/3, 1)$ and $(5/9, 2/3)$; their mixture with weights $1/4, 3/4$ has mean $(1/2, 3/4)$ and expected value $16/27$. Thus the scalar envelope value at that mean is exactly $16/27$.

At binary counts $A = ma, C = mc$,

$$\frac{2f_m}{m^3} = F_2(a, c) - \frac{ac + (5/4)a}{m}.$$

Every feasible law has $\mathbb{E}a = 1/2$ and $\mathbb{E}c = 3/4$, and $ac \leq a$. It follows that $2 \text{vex } f_m/m^3 \geq 16/27 - 9/(8m)$. Each monomial has upper envelope $1/2$ and lower envelope zero; therefore

$$\frac{2}{m^3} \text{cav } f_m = \frac{2}{m^3} T(f_m) = \frac{9}{8}(1 - 1/m). \quad (5.16)$$

Subtraction gives $2H(f_m)/m^3 \leq 115/216$. For the reverse bound, choose one of the two equality atoms above, then sample all coordinates independently conditional on its two probabilities. Distinct variables in each term have conditional product expectation, so $2\mathbb{E}[f_m \mid a, c]/m^3 = (1 - 1/m)F_2(a, c)$. This is a feasible law and gives $2 \text{vex } f_m/m^3 \leq (1 - 1/m)16/27$. Hence

$$\frac{115}{216}(1 - 1/m) \leq \frac{2H(f_m)}{m^3} \leq \frac{115}{216}.$$

Together with (5.16) this proves (5.14), including positive denominators and convergence of the actual ratios to $243/115$. For example, $m = 20$ gives the lower bound $4617/2300 > 2$ in 40 variables.

At $m = 16$, the exact integer-grid certificate is

$$A \binom{C}{2} + 20 \binom{A}{2} \geq 214A + \frac{177}{2}C - 1686 \quad (A, C \in \{0, \dots, 16\}). \quad (5.17)$$

Appendix C supplies its integer verification. Uniform success subsets of sizes 8, 12 give equality, with value 1088. The concave envelope and termwise gap are 2160, so $H = 1072$ and $T/H = 135/67$.

For $m = 4, 8, 12$, Table 7 in Appendix C gives matching affine minorants and uniform fixed-count laws, proving the three exact ratios stated above. All three are below two. For every $m \geq 20$, the lower bound in (5.14) is at least $4617/2300 > 2$, since $1 - 1/m$ is increasing. Together with $m = 16$, these facts prove the if-and-only-if assertion among positive multiples of four in (5.13); they make no minimum-dimension claim over other families.

Introduce 20 distinct coordinates z_k and define explicitly

$$g = AE_2(W) + \left(\sum_{k=1}^{20} z_k \right) E_2(U), \quad x_U = 1/2, \quad x_W = 3/4, \quad z_k = 999/1000. \quad (5.18)$$

There are $16 \binom{16}{2} + 20 \binom{16}{2} = 4320$ distinct cubic supports, all with coefficient one. Every term still has upper envelope $1/2$ and lower envelope zero, so $\text{cav } g = T(g) = 2160$. At binary vertices,

$$g = f_{16} - \left(\sum_k (1 - z_k) \right) E_2(U), \quad 0 \leq E_2(U) \leq 120.$$

Every feasible law projects to the original means, giving $\text{vex } g \geq 1088 - 120(20/1000) = 1088 - 12/5$. Thus $0 < H(g) \leq 5372/5$ and $T(g)/H(g) \geq 2700/1343$. Positivity of H follows by comparing independent and common-threshold laws at these interior means. $\square$

The two-level examples show that means in the upper half of the unit cube do not force factor two. Small upward perturbations of the $1/2$ means preserve a strict ratio above two by continuity, so all means can be strictly above $1/2$. No quantitative perturbation radius or minimal dimension claim is implied. The next result explains why at least two distinct means are needed on the unit cube.

6 Equal means and classical symmetric envelopes

The envelope of the complete elementary symmetric polynomial is classical. Sherali's equation (13) and Theorem 3 [37, pp. 252–253] give, on the entire unit cube,

$$\text{vex } E_d(x) = \max \left(0, \max_{k=d-1, \dots, n-1} \left\{ \binom{k}{d-1} \sum_i x_i - (d-1) \binom{k+1}{d} \right\} \right). \quad (6.1)$$

The following extremal gap formulation is a consequence of that symmetric-envelope machinery. We give a direct proof at equal means to make the common-law guarantee explicit.

Theorem 6.1 (Exact equal-mean formula). *Fix $n \geq 2$ and $0 < u < 1$. Put $b = \lfloor nu \rfloor$, $\theta = nu - b$, and, for $2 \leq d \leq n$,*

$$t_d(u) = \min\{u, (d-1)(1-u)\}, \quad q_{n,d}(u) = \frac{(1-\theta)\binom{b}{d} + \theta\binom{b+1}{d}}{\binom{n}{d}},$$

with $\binom{k}{d} = 0$ for integers $0 \leq k < d$. The maximum of T/H over positive multilinear polynomials with a nonzero nonlinear part at $x = (u, \dots, u)$ is

$$R_n^{\text{eq}}(u) = \max_{2 \leq d \leq n} \frac{t_d(u)}{u - q_{n,d}(u)}. \quad (6.2)$$

A maximum-degree restriction $2 \leq D \leq n$ restricts this maximum to $d \leq D$. Each candidate ratio is attained by E_d .

Proof. Draw $K = b$ or $b+1$ with probabilities $1-\theta, \theta$, then choose a uniformly random K -element success subset. Each mean is u and each degree- d product expectation is $q_{n,d}(u)$. This single law gives

$$H(f; u\mathbf{1}) \geq \sum_e a_e(u - q_{n,|e|}(u)), \quad T(f; u\mathbf{1}) = \sum_e a_e t_{|e|}(u).$$

The quotient is bounded by the largest term ratio, provided its denominators are positive. To see positivity and attainment, the binary value of E_d is $\binom{K}{d}$, whose first forward difference is $\binom{K}{d-1}$ and is nondecreasing. Its linear interpolation on integer counts is convex. Jensen's inequality shows that any integer-valued K with mean nu has expected value at least the adjacent-count interpolation. Our uniform-subset law attains this value with all individual means correct, establishing $\text{vex } E_d = \binom{n}{d} q_{n,d}$. Its upper envelope is $u\binom{n}{d}$. Comparing the minimum with the independent law, whose product expectation is u^d , gives $q_{n,d} \leq u^d < u$. Thus the denominators are positive and every claimed candidate is attained. The maximum is attained by a maximizing degree. $\square$

Corollary 6.2 (Dimension-free equal-mean supremum). *For $0 < u < 1$ the exact supremum over dimensions, and the limit of $R_n^{\text{eq}}(u)$ as $n \rightarrow \infty$, is*

$$R_\infty^{\text{eq}}(u) = \max_{k \geq 1} \frac{\min\{1, k(1-u)/u\}}{1-u^k} \leq 2. \quad (6.3)$$

Writing $r_u = u/(1-u)$, a maximizing k is one of $\max(1, \lfloor r_u \rfloor)$ and $\max(1, \lceil r_u \rceil)$. For $u \leq 1/2$ the value is $1/(1-u)$. The uniform constant two is sharp.

Proof. The comparison $q_{n,d} \leq u^d$ gives the upper bound by the displayed maximum after writing $k = d - 1$. For fixed d , $q_{n,d}(u) \rightarrow u^d$ as $n \rightarrow \infty$, since both adjacent counts divided by n tend to u . Thus any fixed-degree candidate is approached by E_d in increasing dimension.

For $k \geq r_u$ the expression is $1/(1-u^k)$, decreasing with k . For $k \leq r_u$ it equals $k/(u+u^2+\dots+u^k)$, increasing with k because the average of that decreasing sequence decreases. Hence a finite maximizer occurs at one of the stated two integers. Its fixed-degree limit proves both the supremum and convergence assertions.

Set $t = k(1-u)/u$. Bernoulli's inequality gives $u^{-k} = (1 + (1-u)/u)^k \geq 1 + t$, and therefore

$$\frac{\min(1, t)}{1 - u^k} \leq \min(1, t) \frac{1 + t}{t} \leq 2.$$

When $u \leq 1/2$ the maximizing integer is $k = 1$, giving $1/(1-u)$. At $u = 1/2$, the complete bilinear polynomial for even n has ratio $2(n-1)/n$, tending to two. This is the classical positive bilinear sharpness example, now viewed within the equal-mean formula. $\square$

These exact formulas concern equal normalized means on the unit cube. Positive expansion transfers the degree upper bounds to nonnegative boxes, but does not identify an original positive-box factorization with the elementary symmetric unit-cube model. In particular, the precise finite equal-mean optimization above makes no such additional box claim.

7 Incidence structure and marginal restrictions

The *incidence graph* of a specified factorization has variable vertices i , factor vertices e , and one edge ie for each $i \in e$. It is a simple bipartite graph even when two factors have identical supports. This is different from the primal variable graph, which joins two variables occurring together in a factor. All widths below concern the incidence graph. Its treewidth is the minimum, over tree decompositions, of the largest bag size minus one; a tree decomposition covers every edge and has connected bags for each vertex. Its degeneracy is the largest minimum degree of a nonempty subgraph. Its orientation parameter is the minimum possible maximum outdegree, with no acyclicity requirement. Its variable frequency ν is the largest number of factors containing one variable. A feedback variable set is a set of variable vertices whose deletion leaves a forest; write f_F for its size. Affine factors are discarded before these parameters are measured, unless a factorization is explicitly retained.

7.1 Ownership and asymmetric orientations

Write $\kappa = 2/(1 - e^{-1})$. Let $B(d)$ be any valid universal degree- d upper bound from Section 4, with $B(1) = 0$. In particular one can choose $B(d) \sim \log d / \log \log d$.

Theorem 7.1 (Incidence orientation bound). *For a positive multilinear polynomial on a unit cube, suppose an incidence orientation has variable outdegrees at most r and factor outdegrees at most s , where r, s are nonnegative integers. Then*

$$T \leq [r + \min\{s, B(s+1)\} + \kappa]H.$$

In particular frequency ν gives $\nu + \kappa$, and an orientation with all outdegrees at most k gives $k + \min\{k, B(k+1)\} + \kappa$. These conclusions transfer to boxes with zero lower bounds by scaling.

Proof. Call a coordinate low if its mean is at most $1/2$. A term with exactly one low coordinate will be called a one-low term. All other terms receive at least t_e/κ deficiency from independence by Lemma 2.3. In a one-low term the low coordinate is its anchor, of mean u_e . Write $q_i = 1 - x_i$ for the high-coordinate failure means. Partition its high variables into I_e , whose incidence edges point into the factor, and O_e , whose edges point out. Put

$$t_e^I = \min\{u_e, \sum_{i \in I_e} q_i\}, \quad t_e^O = \min\{u_e, \sum_{i \in O_e} q_i\}.$$

Capped sums are subadditive, so $t_e \leq t_e^I + t_e^O$.

For $r > 0$, independently at each high variable label its incidences into one-low factors by distinct labels in $\{1, \dots, r\}$. In round h a variable is owned by its factor with label h , if any. Thus the owned sets $J_{e,h}$ are disjoint over factors and partition I_e over rounds. Take one uniform U and give every low coordinate its success interval $[0, x_i]$. For the high variables owned by e , arrange failure sets so that their union inside $[0, u_e]$ has length $\min\{u_e, \sum_{i \in J_{e,h}} q_i\}$. This can be done by consecutive arcs of lengths $\min(q_i, u_e)$ on the circle of circumference u_e ; when $q_i > u_e$, add measure $q_i - u_e$ outside that interval. For $u_e = 0$ use arbitrary failure sets of the required measures. Unowned variables also receive arbitrary marginal-correct failure sets. Disjoint ownership makes these assignments compatible, including when several terms share one low anchor. Their full deficiencies dominate the owned failure events. A uniform mixture of rounds therefore gives

$$\mathbb{E}D_e \geq \frac{1}{r} \sum_h \min\{u_e, \sum_{i \in J_{e,h}} q_i\} \geq t_e^I/r.$$

If $r = 0$, the incoming gaps all vanish.

A different global law uses the same low success intervals and high failure intervals $[0, q_i]$. Since $|O_e| \leq s$, it gives

$$\mathbb{E}D_e \geq \min\{u_e, \max_{i \in O_e} q_i\} \geq t_e^O/s$$

when $s > 0$; empty outgoing sets have zero gap. Mixing these two laws and independence with weights proportional to r, s, κ proves the bound $r + s + \kappa$, including zero parameters by omitting their components.

For the degree alternative, form the positive polynomial with terms $a_e X_{i_e} \prod_{i \in O_e} X_i$ for one-low factors. Its degree is at most $s + 1$, its termwise gap is $\sum_e a_e t_e^O$, and its deficiency under *every* common-marginal law is at most the original total deficiency. Its hull gap is consequently at most the original hull gap. Thus the outgoing sum is at most $B(s + 1)H$, as well as at most sH . The incoming sum is at most rH by the ownership law, and the other terms' sum at most κH by independence. Adding proves the refinement. This argument needs only a polynomial-level degree bound, not an unproved per-term guarantee from that bound. Orienting all edges from variables to factors gives the frequency specialization. $\square$

Corollary 7.2 (Sharp ownership constant). *At points where every term has exactly one low coordinate, suppose each high variable occurs in at most r factors, with unrestricted low-variable frequency. The sharp supremum of T/H is r for every integer $r \geq 1$.*

The ownership proof gives the upper bound using only its r rounds. The lower bound for $r \geq 2$ follows from the next construction with $L = r$; for $r = 1$ a single nonaffine product attains one. This parameter concerns high variables at the stated point and is distinct from ordinary frequency.

7.2 Variable radix and exact width

Theorem 7.3 (Variable-radix family). *Let $L \geq 2$ and $b \geq L$ be integers, and put $m = b^L$. Index m leaves by length- L base- b words. Let $\mathcal{P}_j$ partition them by their first j digits. With separate anchors $a_1, \dots, a_L$, define*

$$f_{L,b}(a, z) = \sum_{j=1}^L \sum_{Q \in \mathcal{P}_j} a_j \prod_{i \in Q} z_i, \quad a_j = b^{-j}, \quad z_i = 1 - m^{-1}.$$

Then

$$T = L, \quad H = 1 + (L - 1)/b, \quad \text{tw}(G_{L,b}) = L. \quad (7.1)$$

Proof. A level- j block has m/b^j leaves, whose failure means sum to b^{-j} . Its local lower value is zero and upper value b^{-j} ; b^j blocks give one unit of termwise gap at each level. For an arbitrary binary law let R count failed leaves and N_j count hit level- j blocks. Then $\mathbb{E}R = 1$, $N_j \leq \min(b^j, R)$, and total deficiency is $\sum_j A_j N_j$. For a fixed distribution of R , a prescribed-mean indicator has largest correlation with the nondecreasing function $\min(b^j, R)$ by selecting its upper tail: exchanging selected probability at a smaller count with unselected probability at a larger count proves this assertion, with fractional selection at a tied count allowed. All anchors can use these nested tails simultaneously. Represent the count by its nonincreasing quantile $R(U)$ and select $A_j = \mathbf{1}\{U \leq b^{-j}\}$. If exactly l anchors are selected, then

$$\sum_{j=1}^l \min(b^j, R) \leq R + \sum_{j=1}^{l-1} b^j.$$

For $l = 0$ use $0 \leq R$. Integrating gives $H \leq 1 + \sum_{j=1}^{L-1} b^j b^{-j-1} = 1 + (L - 1)/b$.

The probability of level l is $w_l = (b - 1)b^{-l-1}$ for $0 \leq l < L$ and $w_L = b^{-L}$. Two count profiles are

$$R_l^{(1)} = b^l \mathbf{1}\{l \geq 1\}, \quad R_l^{(2)} = b^{l-1} \mathbf{1}\{l \geq 2\}.$$

Their means are $M_1 = [(L - 1)(b - 1) + b]/b$ and $M_2 = [(L - 2)(b - 1) + b]/b^2$. Our parameter range ensures $M_2 \leq 1 \leq M_1$, so mixing the profiles with first-profile weight $(1 - M_2)/(M_1 - M_2)$ gives mean count one. Each profile attains the preceding pointwise inequality at every level, including $l = 0, 1$. For an integer count R , fail the first R leaves in digit-reversal order. Their first j digits are the reversals of the last j digits of $0, \dots, R - 1$, so they hit exactly $\min(b^j, R)$ level- j blocks. A uniform coordinatewise digit shift modulo b preserves every hit count and gives each leaf conditional failure probability R/m . The mixture therefore has all the required marginals and attains the upper bound. This argument works for composite radices as well as prime ones.

For treewidth, eliminate leaves first, filling their L factor neighbors into cliques. These neighbors form nested chains. Next eliminate factor vertices from level L down to level one. A level- j factor's remaining neighbors lie among its $j - 1$ ancestor factors and anchors of levels $j, \dots, L$, at most L vertices. Induction verifies this description: eliminating a descendant joins its ancestor chain and deeper anchors, never two unrelated factors. The final L anchors also have elimination width at most L . For the reverse bound choose a level- $(L - 1)$ block with b leaves and contract each deepest bilinear factor into its leaf. The resulting minor contains $K_{L,b}$, with the $L - 1$ ancestor factors and a_L opposite those leaves. For $b \geq L$, this graph has a clique minor of order $L + 1$: pair $L - 1$ vertices on opposite sides into branch sets, then keep one unused vertex from each side. All branch sets are connected and pairwise adjacent. Minor monotonicity proves the lower bound. $\square$

The restriction $b \geq L$ is part of the exact formula just proved. In particular it cannot be replaced silently by $b = 2$ for arbitrarily many levels; Theorem 4.2 uses a different cutoff. More generally, for integers $b \geq 2, L \geq 2$, the same radix proof gives $H = s + (L - s)b^{-s}$ whenever $1 \leq s < L$ and

$$M_{s+1} \leq 1 \leq M_s, \quad M_q = \frac{(L - q)(b - 1) + b}{b^q}.$$

Indeed, cap the first $l - s$ summands and bound the last s by R , giving $\sum_{j \leq l} \min(b^j, R) \leq sR + \sum_{j \leq l-s} b^j$. Its integrated constant is $(L - s)b^{-s}$. The profiles $R_l^{(q)} = b^{l-q+1} \mathbf{1}\{l \geq q\}$ for $q = s, s + 1$ have means M_q and attain equality; mix them to mean one and use digit reversal as above.

Corollary 7.4 (Sharp leading structural growth). *Let W_k, D_k, P_k be the supremal unit-cube positive-monomial gap ratios under, respectively, incidence treewidth, degeneracy, or orientation parameter at most k . For every integer $k \geq 2$,*

$$W_k \geq k, \quad D_k \geq k + 1, \quad P_k \geq k + 1,$$

whereas

$$W_k \leq D_k \leq P_k \leq k + \min\{k, B(k + 1)\} + \kappa.$$

In particular $W_k \sim D_k \sim P_k \sim k$ as $k \rightarrow \infty$.

Proof. Treewidth at most k implies degeneracy at most k : in a reduced tree decomposition take a vertex exclusive to a leaf bag and delete it, then continue on induced subgraphs. A degeneracy order oriented toward later vertices has outdegrees at most k . This proves the upper chain. Use $L = k$ in Theorem 7.3 and let $b \rightarrow \infty$ for $W_k \geq k$. For the stronger other bounds use $L = k + 1$: orient each deepest factor toward both its leaf and anchor, all other leaf incidences toward their factor, and the remaining factor incidences toward their anchor. Outdegrees are at most k , since deepest factors have outdegree two. For degeneracy delete deepest factors of degree two, their isolated anchor, then leaves of degree $L - 1 = k$, leaving stars. Letting b grow proves both lower bounds. This family still has treewidth $k + 1$. For example $L = 3, b = 5$ gives $15/7 > 2$ for degeneracy and orientation parameter two, with treewidth three. Finally $B(k + 1) = o(k)$. $\square$

For ordinary frequency, the example in Proposition 8.3 below gives a lower bound $2 - 1/\nu$ for every $\nu \geq 2$. The dyadic family has maximum frequency 2^L (at its deepest anchor), so choosing the largest power of two at most ν also gives the lower asymptotic $(1 - o(1)) \log \nu / \log \log \nu$. Frequency two has the sharper exact value $3/2$, proved in Section 8.

7.3 A lower bound on the normalized means

Let C_δ be the supremal positive-monomial gap ratio on unit cubes at points whose coordinates are all at least δ , with dimension and degree unrestricted. This restricts the *evaluation point*; the envelopes remain those on the full unit cube.

Theorem 7.5 (Marginal-floor law). *For $0 < \delta \leq 1/2$, set*

$$M = \max\{2, \log(1/\delta)\}, \quad \tau = \delta/M^2, \quad \Lambda = \log \frac{1 + \tau}{\tau}, \quad I_\delta = \int_0^1 \left(1 - e^{-1/[\Lambda(z + M^{-2})]}\right) dz.$$

Then $I_\delta > 0$ and $C_\delta \leq I_\delta^{-1} + \kappa$. For $1/2 < \delta < 1$, $C_\delta \leq C_{1/2}$. Moreover

$$C_\delta \sim \frac{\log(1/\delta)}{\log \log(1/\delta)} \quad (\delta \downarrow 0).$$

The same leading asymptotic holds with every coordinate in $[\delta, 1 - \delta]$. The upper bound transfers to nonnegative boxes when all nonfixed normalized means $(x_i - \ell_i)/(u_i - \ell_i)$ are at least δ .

Proof. The density $h(t) = 1/[\Lambda(t + \tau)]$ integrates to one on $(0, 1)$. Use low-coordinate success intervals $[0, x_i]$. For a high coordinate with failure mean $q < 1/2$ define

$$q_0(t) = \min\{1, qh(t)\}, \quad m_q = \int_0^1 q_0(t) dt, \quad q_1(t) = q_0(t) + \frac{q - m_q}{1 - m_q}(1 - q_0(t)).$$

Here $0 \leq m_q \leq q < 1$, so $q_1 \in [q_0, 1]$ and its integral is exactly q , including $q = 0$. Conditional on the common uniform variable $U = t$, sample high failures independently with probabilities $q_1(t)$. For a set of high coordinates with failure sum S , the probability of some failure is at least $1 - e^{-Sh(t)}$. If any $qh(t) \geq 1$, a failure is certain; otherwise $q_1(t) \geq qh(t)$ and the usual product bound proves it. The clipping case is needed for this conclusion.

A one-low term with anchor mean $u \geq \delta$ consequently has

$$\mathbb{E}D_e \geq \int_0^u (1 - e^{-S/[\Lambda(t+\tau)]}) dt.$$

For $S > 0$, substitute $t = uz$, $y = S/u$, and note

$$\frac{1 - e^{-ay}}{\min(1, y)} \geq 1 - e^{-a} \quad (a \geq 0, y > 0).$$

Concavity proves this for $y \leq 1$, monotonicity for $y \geq 1$. Since $\tau/u \leq M^{-2}$, division by $t_e = u \min(1, y)$ now gives $\mathbb{E}D_e/t_e \geq I_\delta$. Zero S requires no division. Mix this law and independence with weights proportional to I_δ^{-1}, κ ; Lemma 2.3 handles all other factors.

As $\delta \downarrow 0$, $\Lambda = M + 2 \log M + o(1)$ and $\Lambda/M^2 \rightarrow 0$. The inequalities $1 - e^{-a} \leq \min(1, a)$ and $1 - e^{-a} \geq a - a^2/2$ give

$$I_\delta \leq \frac{1 + \log \Lambda}{\Lambda}, \quad I_\delta \geq \frac{\log \Lambda}{\Lambda(1 + \Lambda/M^2)} - \frac{\Lambda - 1}{2\Lambda^2}.$$

For the second bound integrate only over $[1/\Lambda, 1]$, using $z + M^{-2} \leq z(1 + \Lambda/M^2)$. Hence $I_\delta \sim \log \Lambda/\Lambda$, proving the upper asymptotic. The dyadic family with $L = \lfloor \log_2(1/\delta) \rfloor$ has all means in $[\delta, 1 - \delta]$ and ratio asymptotic to $L/\log_2 L$, which gives the matching lower bound. Monotonicity gives the finite large- δ claim. Finally positive affine expansion on a nonnegative box preserves the lower restriction on normalized means, so Proposition 2.4 gives its stated extension. $\square$

Theorem 7.6 (Uniform joint asymptotic). *Let $C(n, d, k, \delta)$ be the supremal unit-cube positive-monomial ratio in dimension at most n , degree at most d , incidence treewidth at most k , and with every mean at least δ . Assume integers $n, d \geq 2$, $k \geq 1$, and $0 < \delta \leq 1/2$. Put $q = \min\{n, d, 1/\delta\} \geq e^e$, $\phi(q) = \log q / \log \log q$, and $h = \min\{k, \phi(q)\}$. Along every joint parameter sequence with $h \rightarrow \infty$,*

$$C(n, d, k, \delta) \sim h.$$

The same statement holds for the two-sided strip and with degeneracy or the orientation parameter replacing treewidth.

Proof. The separate dimension, degree, marginal-floor, and incidence estimates give $C \leq \min\{(1 + o(1))\phi(q), k + o(k)\}$. Their relative errors vanish uniformly along the stated sequences, since both q and k then diverge. For a simultaneous lower construction take

$$L = \lfloor h \rfloor, \quad b = \lfloor (q/2)^{1/L} \rfloor, \quad m = b^L.$$

Then $b \geq (1 - o(1)) \log q$, $L/b \leq (1 + o(1))/\log \log q \rightarrow 0$, and eventually $b \geq L \geq 2$. The radix family has dimension $m + L \leq q/2 + \phi(q) \leq q \leq n$, degree $m/b + 1 \leq q \leq d$, and width $L \leq k$. Its smallest anchor mean and leaf failure mean are $1/m \geq 2/q \geq \delta$, and all other means also lie in the two-sided strip. Thus its ratio $L/[1 + (L - 1)/b] \sim h$. Its degeneracy and orientation parameter are at most its treewidth, proving the alternatives. This is not an asymptotic assertion for a fixed small value of k . $\square$

8 Feedback variables and frequency two

8.1 A universal repair law

The next argument works on the original factor scopes, including on boxes with positive lower endpoints.

Lemma 8.1 (Feedback repair and gluing). *Fix singleton Bernoulli means and a feedback variable set F of size f_F . For any nonnegative local functions g_e and any prescribed-mean local laws P_e on their scopes, there is one global prescribed-mean law P' satisfying*

$$\mathbb{E}_{P'} g_e \geq 2^{-f_F} \mathbb{E}_{P_e} g_e \quad \text{for every } e.$$

Proof. Extend each local law to $F \cup e$ by independently sampling any missing feedback coordinates. Put $C = 2^{f_F}$. Construct a global reference law Q with a common uniform U : each feedback coordinate independently chooses, fairly, its success interval $[0, x_j]$ or $[1 - x_j, 1]$; every outside coordinate uses $[0, x_i]$. Write Q_F for its feedback marginal and Q_i for its (F, i) marginal. For a feedback state s and outside state b , every prescribed-mean local law satisfies

$$P(s, b) \leq m(s, b; i) := \min(\{x_j : s_j = 1\}, \{1 - x_j : s_j = 0\}, \{x_i \text{ if } b = 1; 1 - x_i \text{ if } b = 0\}).$$

The minimum includes all feedback coordinates and the outside coordinate. Choose the feedback orientations that place every desired event at the left endpoint if $b = 1$, and at the right endpoint if $b = 0$. This single orientation pattern has probability $1/C$, and its intersection has length $m(s, b; i)$. Therefore $Q_i(s, b) \geq P(s, b)/C$. The same endpoint argument gives $Q_F(s) \geq P_F(s)/C$, also when there are no outside variables. For $F = \emptyset$ its unique state has mass one.

For a fixed extended factor law P_e , define the nonnegative residuals

$$h_s = Q_F(s) - (P_e)_F(s)/C, \quad r_{i,s,b} = Q_i(s, b) - (P_e)_{F,i}(s, b)/C \quad (i \in e \setminus F).$$

For each i , $\sum_b r_{i,s,b} = h_s$. At state s with $h_s > 0$, place residual mass h_s and sample these outside coordinates independently with probabilities $r_{i,s,b}/h_s$. For $h_s = 0$ all these residuals vanish. For an empty outside scope place just the mass h_s . The resulting measure R_e has total mass $1 - 1/C$, and $P'_e = P_e/C + R_e$ has feedback marginal Q_F and (F, i) marginal Q_i , while dominating P_e/C entrywise.

Condition on any s with $Q_F(s) > 0$. The factor laws $P'_e(\cdot \mid s)$ have consistent outside singleton marginals $Q_i(s, b)/Q_F(s)$. Their incidence graph is a forest. Root each nontrivial component at a factor, sample its law, then traverse the tree. Each newly reached factor shares exactly one already sampled variable; sample its new variables using its local conditional law given that variable. Null conditioning states are never encountered. Isolated variables use their stated conditional means; factors with no outside variables need no sampling. This preserves every conditional factor law. Mix over s with weights $Q_F(s)$. The resulting global law has marginals P'_e on $F \cup e$, and nonnegativity gives the claimed domination of expectations. There is one loss C , regardless of the lengths of the forest paths. $\square$

Theorem 8.2 (Feedback-variable gap). *For positive monomials on any finite nonnegative box, deletion of f_F variable vertices to an incidence forest implies $T \leq 2^{f_F} H$.*

Proof. Substitute fixed coordinates first. In normalized endpoint variables Y , an original physical product is $\varphi_e(Y) = \sum_{A \subseteq e} c_A \prod_{i \in A} Y_i$ with $c_A \geq 0$. This expansion is used only inside the original scope. For each nonempty A choose an index k_A of smallest normalized mean, and let

$$\ell_e(Y) = c_\emptyset + \sum_{\emptyset \neq A \subseteq e} c_A Y_{k_A}.$$

On binary vertices $\ell_e \geq \varphi_e$, while every prescribed-mean law has $\mathbb{E}\ell_e = \text{cav } \varphi_e$, by common upper attainment. Thus $g_e = \ell_e - \varphi_e$ is a nonnegative function on the original scope whose local maximum expectation is the original factor's exact gap. Apply Lemma 8.1 to maximizing local laws, then sum with the positive coefficients. The full upper value is the sum of local upper values, proving the result. Replacing the factors by expanded hyperedges would change the incidence graph and would not justify this proof. $\square$

For $f_F = 0$ this recovers classical incidence-forest exactness; exactness of standard linearization on Berge-acyclic hypergraphs is established in Del Pia and Khajavirad [15]. The general domination lemma is sharp for arbitrary nonnegative local payoffs: on f_F feedback bits and one outside bit, all fair, use the 2^{f_F+1} indicators of their states. Every local maximum is $1/2$, while their sum is identically one. Private variables fixed to one can make the scopes distinct. These indicators have zero literals, so they do not prove sharpness for positive monomials when $f_F > 1$.

Proposition 8.3 (One feedback variable is sharp). *For $n \geq 2$, let $f(a, x) = a \sum_{i=1}^n x_i + \prod_{i=1}^n x_i$ and evaluate at $a = 1/n$, $x_i = 1 - 1/n$ on the unit cube. Then $T = 2 - 1/n$ and $H = 1$. Its incidence graph has treewidth two and becomes a tree after deleting a .*

Proof. The bilinear terms contribute one in total and the large product contributes $(n-1)/n$. If R counts failed leaves and A is the binary anchor, the full deficiency is $\mathbb{E}[AR] + \mathbb{P}(R \geq 1) - 1/n$, with $\mathbb{E}R = 1$ and $\mathbb{E}A = 1/n$. Pointwise $AR + \mathbf{1}\{R \geq 1\} \leq R + A$, giving the upper bound one. Exactly one uniformly chosen failed leaf and an independent anchor attain equality. Write e_0 for the large factor and e_i for ax_i . Path bags $\{a, e_0, x_i\}$ with attached bags $\{a, x_i, e_i\}$ give width two. A cycle rules out width one, and deleting a leaves a tree. $\square$

8.2 Half-integral degree slabs

For frequency at most two, form a loopless dual multigraph with factor vertices. A shared variable is an edge between its two factors; a private variable is an edge to its own new dummy vertex. Parallel edges are allowed. Unused variables may be completed independently. Dummy factors have zero weight or identically zero costs. Let g be its odd girth, with $g = \infty$ in the bipartite case.

Lemma 8.4 (Degree-slab vertices). *For a loopless multigraph and $p \in [0, 1]^E$, put $s_v(p) = \sum_{i \in \delta(v)} p_i$. Every vertex of*

$$\mathcal{S}(p) = \{z \in [0, 1]^E : \lfloor s_v(p) \rfloor \leq s_v(z) \leq \lceil s_v(p) \rceil \ (v \in V)\}$$

has fractional edges forming vertex-disjoint odd cycles, all valued $1/2$.

Proof. Let J be the fractional edges at a vertex and R the tight degree rows incident to them, counting a repeated lower/upper equality once. The restricted rows must have full column rank on J : otherwise a small perturbation in a null direction preserves every tight constraint and box bound, contradicting extremality. Hence $|J| \leq |R|$. An integer tight residual degree cannot contain exactly

one fractional edge, so every such row has at least two fractional incidences. Thus $2|R| \leq 2|J|$. Equality forces each fractional edge to have both endpoints in R , and each such vertex to have fractional degree two. These are cycles; an even cycle admits an alternating perturbation and is excluded. On an odd cycle the two incident fractional values sum to the integer one, forcing all values to $1/2$. The argument excludes parallel fractional two-cycles and degree-one dummy vertices automatically. $\square$

This is the usual fractional matching rank argument, here stated for integer degree slabs; compare the fixed-degree formulation in Barrus [6, Theorem 2.1].

Theorem 8.5 (Frequency two and odd girth). *For positive monomials of frequency at most two on a unit cube,*

$$T \leq \frac{g}{g-1}H \quad (g < \infty), \quad T = H \quad (g = \infty).$$

Every finite odd-girth constant is sharp, and the unrestricted frequency-two supremum is $3/2$.

Proof. Use failure probabilities $p_i = 1 - x_i$, and define $b_v(p) = \max_{i \in \delta(v)} p_i$ and $c_v(p) = \min\{1, s_v(p)\}$, with empty values zero. For a failure edge set J , write $I_v(J)$ for its coverage indicator. Exact single-product envelopes give

$$T = \sum_v a_v(c_v - b_v), \quad H = \max_{\mathbb{P}(i \in J) = p_i} \sum_v a_v(\mathbb{E}I_v(J) - b_v). \quad (8.1)$$

In particular an approximation of unshifted coverage alone is insufficient. Decompose $p = \mathbb{E}Z$ into vertices of $\mathcal{S}(p)$. On each allowed slab c_v is affine (linear below one, constant above), so $\mathbb{E}c_v(Z) = c_v(p)$; convexity gives $\mathbb{E}b_v(Z) \geq b_v(p)$.

Given $Z = z$, keep integral edges. On each fractional odd cycle of length l , choose one of its l maximum matchings uniformly, then fairly choose that matching or its complement in the cycle. Each edge has marginal $1/2$. A matching leaves its uniformly selected vertex uncovered; its complement covers every vertex. Thus coverage probability is $1 - 1/(2l)$ at each cycle vertex. A vertex with another selected integral edge is covered surely. Otherwise its $c_v(z) = 1$ and $b_v(z) = 1/2$. All vertices off these cycles have equal integral $b_v(z) = c_v(z)$. With $\alpha = 1 - 1/g$, this proves

$$\mathbb{E}[I_v(J) \mid Z] \geq \alpha c_v(Z) + (1 - \alpha)b_v(Z).$$

Averaging retains the baseline: $\mathbb{E}I_v(J) - b_v(p) \geq \alpha[c_v(p) - b_v(p)]$. Sum this in (8.1). For a bipartite graph there are no fractional cycles, so the slab decomposition itself attains $c_v(p)$ simultaneously, proving exactness.

For sharpness take a unit-weight odd cycle of length l with all edge means $1/2$. Its termwise gap and coverage baseline both equal $l/2$. For every selected edge set J , $\sum_v I_v(J) \leq |J| + (l-1)/2$: if $|J| \leq (l-1)/2$ use $2|J|$, otherwise use l . Expected coverage is therefore at most $l - 1/2$, attained by the matching/complement law. The hull gap is $(l-1)/2$. $\square$

Bipartite equality is a statement about the scalar envelopes. It does not assert compatibility of all separately feasible product-value coordinates in the full lifted multilinear polytope. Zero-lower boxes transfer by scaling; unequal positive aspect ratios need a separate analysis.

8.3 Convex cardinality factors

A discrete sequence $\varphi(0), \dots, \varphi(d)$ is convex if $\varphi(k+2) - 2\varphi(k+1) + \varphi(k) \geq 0$. Its cardinality factor is the unique multiaffine interpolant of the values $\varphi(\sum_i X_i)$ on binary vertices. It is generally not the literal continuous composition $\varphi(\sum_i x_i)$. Sequences may be negative, decreasing, or nonmonotone; only their discrete convexity is needed.

Lemma 8.6 (Cardinality envelopes). *Let f_e be such a factor and $\hat{\varphi}_e$ its piecewise-affine interpolation between integers. Then*

$$\text{vex } f_e(p) = \hat{\varphi}_e\left(\sum_{i \in e} p_i\right).$$

A prescribed-mean law supported on the two adjacent cardinalities attains this minimum. One common-threshold law maximizes every such factor and their sum. If $p_{(1)} \geq \dots \geq p_{(d)}$ are the local means, its upper value is

$$\varphi_e(0) + \sum_{j=1}^d [\varphi_e(j) - \varphi_e(j-1)] p_{(j)}.$$

Proof. Jensen's inequality gives the lower bound for the convex function $\hat{\varphi}_e$. The cube slab between the two adjacent integer sums is integral: two fractional coordinates admit an opposite perturbation; one fractional coordinate cannot make an integer sum bound tight and admits a small perturbation. Decompose p into its binary vertices to attain equality. Discrete convexity implies supermodularity of the cardinality set function. Among laws maximizing a factor, choose one maximizing also $\mathbb{E}|X|^2$. If two incomparable support sets have positive mass, replace equal mass on them by their intersection and union. Singleton means are unchanged; supermodularity cannot decrease the primary value, while the secondary value strictly increases by twice the product of the two nonempty set differences' sizes. This is impossible. The maximizer is chain-supported, and the chain law with prescribed means is precisely the common-threshold law. The same law therefore maximizes each factor individually, and summing establishes simultaneous upper attainment. Integrating its cardinality gives the displayed formula. $\square$

The single-factor principles are classical; symmetric envelope formulas appear in Sherali [37], and the common-aspect product formula in Adams et al. [1, Proposition 4.1].

Theorem 8.7 (Cardinality odd-girth bound). *For a sum of convex cardinality factors of frequency at most two, with each entire factor kept in the relaxation, the sharp constants of Theorem 8.5 hold unchanged.*

Proof. Use success probabilities in the degree slab, and decompose $p = \mathbb{E}Z$ as before. Each local lower envelope is affine on its allowed slab, so

$$\mathbb{E} \text{vex } f_v(Z) = \text{vex } f_v(p), \quad \mathbb{E}T(Z) \leq T(p),$$

where the second assertion also uses concavity of each upper envelope. At a fractional-cycle vertex v , let m count its incident integral ones. Its local lower value is $\varphi_v(m+1)$, and its upper value is $[\varphi_v(m) + \varphi_v(m+2)]/2$, since precisely two coordinates are half-valued. Matching/complement rounding on a length- l cycle selects zero or two of them with probability $1/(2l)$ each, and one otherwise. Therefore the expected local cost is exactly $\text{vex } f_v(Z) + T_v(Z)/l$. Off the cycles the local gap is zero. Averaging gives

$$\text{vex } f(p) \leq \mathbb{E}f(X) \leq \sum_v \text{vex } f_v(p) + T(p)/g.$$

Subtract from the common upper value. Bipartite slabs are integral and need no rounding. Positive bilinear odd cycles prove sharpness. $\square$

Corollary 8.8 (Common-aspect physical products). *For original positive products on boxes $[\ell_i, \rho_e \ell_i]$, $\ell_i > 0$, assume each scope e has a common coordinate aspect ratio $\rho_e > 1$. Then the frequency-two odd-girth theorem applies to the original factors. Its constant is sharp for every fixed common ratio.*

Proof. A normalized binary vertex gives the table $a_e(\prod_{i \in e} \ell_i) \rho_e^K$, which is discrete convex. Shared variables enforce ratio compatibility, so different connected components may have different ratios. No new factors are introduced. For a bilinear odd cycle on $[1, \rho]^n$, expansion gives affine terms plus $(\rho - 1)^2$ times the unit-cube bilinear polynomial. Both gaps scale by the same positive number. Fixed coordinates are substituted first, and ratio one is trivial. $\square$

8.4 Rational scalar-envelope algorithms

Proposition 8.9 (Frequency-two binary oracle). *For a rational positive polynomial of frequency at most two on a unit cube and arbitrary rational λ , one can minimize $f(X) - \lambda^\top X$ over binary X in polynomial bit time.*

Proof. With $Z = 1 - X$, the objective is

$$-\sum_i \lambda_i + \sum_i \lambda_i Z_i + \sum_v a_v \mathbf{1}\{v \text{ uncovered by } Z\}.$$

Dummy vertices have zero penalties. Include every negative-cost edge: adding it strictly lowers cost and cannot increase an uncovered penalty. Add its cost to the constant, remove it from the remaining choices, and set its endpoints' penalties to zero, retaining those vertices. Remaining costs and penalties are nonnegative. Add a hub h , its mate k , their zero-cost edge hk , and an edge vh of cost equal to each vertex v 's penalty. Any selected original edge set extends to an edge cover by paying exactly its uncovered penalties and adding hk . Conversely, retaining original edges of an augmented cover leaves uncovered only vertices whose penalty edges were paid. These two comparisons prove value equality with minimum-weight edge cover. For completeness this reduces to matching: for a nonnegative-cost graph without isolated vertices let $\mu(v)$ be its cheapest incident cost. From a matching M , add a cheapest edge at each unmatched vertex. Its cover cost is at most $\sum_v \mu(v) - \sum_{uv \in M} (\mu(u) + \mu(v) - c_{uv})$. Conversely an inclusion-minimal cover is a union of stars (a path of three edges or a cycle would have a removable edge). Choose one edge from each star as a matching. Every other edge can be charged to its unmatched leaf, with $\mu(v)$ no larger than its cost. This shows the minimum of the displayed expression is no larger than the minimum cover cost. Hence equality holds and maximum-weight matching supplies an optimal cover. Parallel edges can keep their cheapest representative. Weighted matching is polynomial-time [17]. $\square$

Proposition 8.10 (Convex-degree matching oracle). *For explicitly listed rational discrete-convex tables and arbitrary rational edge costs c_i , the binary problem*

$$\min_{J \subseteq E} \sum_v \varphi_v(\deg_J(v)) + \sum_{i \in J} c_i$$

is solvable in polynomial bit time.

Proof. At vertex v create d_v slots with nondecreasing charges $\Delta_v(j) = \varphi_v(j) - \varphi_v(j-1)$. For each edge $i = uv$, create mandatory vertices i_u, i_v , joined by an inactive zero-cost edge. Connect i_u to all slots of u with cost $\Delta_u(j) + c_i$, and i_v to all slots of v with cost $\Delta_v(j)$. A matching covering

all mandatory vertices either uses the inactive edge or uses two slots; mixed activation cannot cover both endpoints. Thus it represents one binary choice per original edge. Every chosen set is representable, and the occupied slots at a vertex can be reassigned to its cheapest k slots because their available gadget neighbors are identical. Their charges telescope to $\varphi_v(k) - \varphi_v(0)$.

To enforce mandatory coverage by ordinary unconstrained matching, let W be the sum of absolute original gadget edge costs and subtract $M = 2W + 1$ per mandatory endpoint covered by each edge. The all-inactive matching covers every mandatory vertex. Missing even one such vertex loses bonus M , exceeding any possible original cost difference. Thus an optimum covers them all, also with negative increments or negative c_i . Adding $\sum_v \varphi_v(0)$ and undoing the fixed bonus recovers the desired value. The gadget has $O(|E|)$ vertices and $O(\sum_v d_v^2)$ edges with polynomial rational encoding length. This is the incremental-slot matching mechanism of convex degree optimization; compare Deza and Onn [16, Theorem 1.2 and Section 3]. $\square$

Corollary 8.11 (Exact scalar envelopes and supporting minorants). *Both preceding factor classes admit exact rational convex-envelope evaluation, a supporting affine minorant, and scalar graph-hull separation in polynomial bit time. This includes rational common-aspect physical products. No polynomial-size explicit description of the full polytope carrying all individual factor values is asserted.*

Proof. The finite vertex-law LP has dual

$$\max_{\eta, \lambda} \{ \eta + \lambda^\top p : \eta + \lambda^\top z \leq f(z) \ (z \in \{0, 1\}^n) \}.$$

Its separation oracle is precisely binary minimization of $f(z) - \lambda^\top z$, with arbitrary signed linear costs. Affine summands are absorbed into these costs and restored in the value; unused coordinates optimize independently. Let $A = \sum_v \max_j |\varphi_v(j)|$, or $A = \sum_v a_v$ for positive monomials, adding the absolute coefficient sum of separate affine terms if present. Then $|f(z)| \leq A$. The dual polyhedron contains no line, since the rows $(1, z)$ span $\mathbb{R}^{n+1}$, and its attained optimal face contains a vertex. Such a vertex solves $n + 1$ independent tight equations with a zero-one matrix. Cramer's rule bounds each coordinate by $(n + 1)!A$; its nonzero integer denominator determinant has magnitude at least one. Clearing input denominators also gives polynomial bit length for each coordinate. Thus the box of radius $\max\{1, (n + 1)!A\}$ retains an optimum and has polynomial encoding size. Rational separation-to-optimization [23, Theorem 6.4.9] gives the exact optimum and minorant, also at boundary means. Every vertex inequality extends to the whole cube by multiaffine interpolation. Sorting gives the common-threshold upper value and a supporting upper affine function; together with cube bounds these separate the scalar graph hull. Rational common-ratio products generate their convex tables by polynomially many rational multiplications. $\square$

9 The exact incidence-treewidth-two constant

Theorem 9.1 (One-sided cycle coloring). *In a finite simple bipartite graph $G = (V, F; I)$ of treewidth at most two, one can color the factor side F with two colors so that every cycle whose factor vertices all have one color contains an even number of factor vertices. Consequently every zero-one matrix with this bipartite support has a partition of its rows into two totally unimodular matrices.*

The all-cycle assertion is stronger than excluding induced odd-factor cycles (balancedness). The stronger statement will supply total unimodularity for arbitrary integer cardinality slabs.

Proof. We prove a two-terminal series-parallel invariant. A network starts with an edge; series composition identifies one terminal of each of two otherwise disjoint networks, and parallel composition identifies both terminal pairs. Intermediate networks are simple and bipartite subgraphs of the eventual graph. A coloring is good if its monochromatic-factor cycles have even factor count. A terminal path is c -monochromatic when all its factor vertices, including factor terminals, have color c . Its parity counts those factor vertices modulo two.

For each network there is a parity $\epsilon \in \{0, 1\}$ with all the following witnesses available. They are possibly different colorings. For V, V terminals and either color c , an *active* good coloring has at least one c -monochromatic terminal path, all such paths have parity ϵ , and no path is monochromatic in the other color. If $\epsilon = 0$, a *blocking* good coloring with neither type of monochromatic terminal path is also available. For mixed terminals, with either prescribed color c of the factor terminal, an active good coloring has paths of color c , all of parity ϵ ; a blocking witness with that prescribed terminal color is available whenever there is no direct terminal edge. For F, F terminals, with either prescribed common color c , the same active witness exists; if $\epsilon = 1$ a blocking witness with those colors also exists. In addition, for either assignment of distinct terminal colors, a good coloring exists. Such colors automatically block monochromatic terminal paths. Finally a network with a direct terminal edge is mixed and uses $\epsilon = 1$.

The base edge satisfies the mixed invariant with parity one. For a series composition let δ be one if the shared vertex is a factor, and zero otherwise. The output parity is $\epsilon = \epsilon_1 \oplus \epsilon_2 \oplus \delta$. Every path concatenates component terminal paths; subtracting the shared factor once accounts for δ . Every cycle stays in a component. Active witnesses concatenate by using the same desired color and matching the shared factor's color when present. The required additional witnesses have the following complete case analysis.

1. For outer types V, V and a shared variable, use opposite active colors in the two V, V components. No monochromatic path can cross both.
2. For outer types V, V and a shared factor, blocking is required only if the component parities differ. The even mixed component has no direct edge and can block with the prescribed shared-factor color; use an active witness of that color in the other component.
3. For mixed outer types and a shared variable, use the active color opposite to the outer factor's color in the V, V component and use that prescribed factor color in the mixed component. This blocks.
4. For mixed outer types and a shared factor, give the shared factor the color opposite to the outer factor. The F, F component uses its distinct-terminal witness, and the mixed component uses a compatible active witness. This again blocks.
5. For outer types F, F of common color and a shared variable, blocking is required only for output parity one, so the two mixed component parities differ. Block the even component, which has no direct edge, and activate the other, respecting both outer colors.
6. For outer types F, F of common color and a shared factor, color the shared factor oppositely. Both F, F components have their required distinct-terminal good witnesses and block.
7. For outer F, F terminals of distinct colors and a shared variable, activate each mixed component in its outer terminal's color. If instead the shared vertex is a factor, give it either color and use in each component the equal-color active or distinct-color witness as appropriate.

These cases include reversed mixed types and exhaust all triples of terminal types. A nontrivial series composition has no direct outer edge, so its mixed blocking requirement has also been checked.

In a parallel composition every new simple cycle consists of one terminal path from each component. If the paths are monochromatic in one color, its factor parity is their parity XOR plus τ , the number of factor terminals modulo two. Thus $\tau = 0$ for equal terminal types and $\tau = 1$ for mixed types. We check the invariant as follows. For V, V terminals with equal component parities, activate both in the same color. If both are even, both may also block. For different parities, activate the odd component and block the even component. Output parity is $\epsilon_1 \vee \epsilon_2$; whenever both components contribute paths their equal parities make new cycles even. For F, F terminals of common prescribed color, activate both when parities agree; if both are odd, both may also block. If parities differ, activate the even component and block the odd one. Output parity is $\epsilon_1 \wedge \epsilon_2$. With distinct prescribed terminal colors, use their good witnesses in both components; a crossing monochromatic cycle would contain both differently colored terminals and is impossible. For mixed terminals with no direct edge, both components can block in the prescribed factor color. Activate either one and block the other for an active witness, choosing its parity as output, or block both for the blocking witness. For mixed terminals with a direct edge, simplicity implies exactly one component contains it. Activate its parity-one witness and block the edge-free component. No blocking output is required. These mixed choices create no crossing monochromatic cycle. The induction is complete.

A treewidth-two graph has no K_4 minor, hence no subdivision of K_4 . The classical block characterization supplies a two-terminal series-parallel representation for each nontrivial block [24, Theorem 3.1]. Apply the invariant to each block. Root the block-cut forest. If a child block attaches at a factor, interchange its two colors if needed to match the articulation color; if it attaches at a variable no matching is needed. Bridges use the base edge and isolated factors receive arbitrary colors. Every cycle lies in one block, proving the graph assertion.

For the matrix assertion use Camion's criterion: a $0, \pm 1$ matrix is totally unimodular exactly when every Eulerian submatrix has entry sum divisible by four [13, Theorem 6.5]. In either color class an Eulerian submatrix has even row and column degrees. Its bipartite support decomposes into edge-disjoint simple cycles. Each has an even number of factor vertices, hence length divisible by four. Its total number of ones is consequently divisible by four, as required. The argument covers rectangular Eulerian submatrices too. $\square$

Theorem 9.2 (Sharp width-two gap). *A sum of discrete-convex cardinality factors with incidence treewidth at most two satisfies $T \leq 2H$. This includes positive monomials on unit cubes and zero-lower boxes, and original positive products with a common aspect ratio within each scope. The sharp supremum is two, including for every fixed common positive aspect ratio $\rho > 1$.*

Proof. Partition factor rows into two TU classes by Theorem 9.1. For a class C and prescribed means p , the polytope

$$0 \leq z \leq 1, \quad \left\lfloor \sum_{i \in e} p_i \right\rfloor \leq \sum_{i \in e} z_i \leq \left\lceil \sum_{i \in e} p_i \right\rceil \quad (e \in C)$$

is integral and contains p . Indeed row sign changes, duplicate rows, and unit rows preserve total unimodularity; each basic solution with integer right-hand side is integral by its determinant ± 1 . A binary-vertex decomposition of p attains all class lower envelopes by Lemma 8.6. Under that law every other factor has nonnegative expected deficiency from its own upper value. The equal mixture of the two class laws therefore attains at least half the sum of local gaps. Their common upper extremizer makes this a lower bound for half the full termwise gap in the scalar hull gap.

For positive unit-cube monomials one can equivalently use failure rows $\sum_{i \in e} z_i \leq 1$ when their mean sum is at most one and $\sum_{i \in e} z_i \geq 1$ otherwise. A classwise integral decomposition attains the failure union bound in every row. The mixed unit-right-hand-side integrality theorem for balanced matrices suffices for this particular argument [13, Theorem 6.13]; it would not justify arbitrary integer cardinality slabs without the stronger TU conclusion. Common-aspect products are cardinality factors by Corollary 8.8. Proposition 8.3 proves the unit-cube lower supremum two.

For sharpness at a fixed positive ratio, put $\epsilon = 1/\rho$ and $\alpha = 1 - \epsilon$. On $[\epsilon, 1]^{n+1}$ use

$$f_n(a, x) = \alpha^{-1} a \sum_{i=1}^n x_i + \prod_{i=1}^n x_i, \quad \mathbb{E}A = 1/n, \quad \mathbb{E}X_i = 1 - 1/n,$$

where $a = \epsilon + \alpha A$ and $x_i = \epsilon + \alpha X_i$ are endpoint variables. The normalized nonlinear bilinear terms are $\alpha A X_i$, so their gaps sum to α . The large product is ϵ^R for failure count R , with $\mathbb{E}R = 1$. Its lower value is ϵ by adjacent-cardinality attainment, and its upper value $1 - (1 - \epsilon^n)/n$. Hence

$$T_n = 2\alpha - \frac{1 - \epsilon^n}{n}, \quad H_n = \max \mathbb{E}[\alpha A R + 1 - \epsilon^R] - \frac{1 - \epsilon^n}{n}.$$

For any $M > 0$, $1 - \epsilon^r \leq \alpha r$ on integer $r \geq 0$ gives

$$\mathbb{E}[1 - \epsilon^R] \leq \alpha(1 - \mathbb{E}[R; R > M]) + 1/M, \quad \alpha \mathbb{E}[AR] \leq \alpha M/n + \alpha \mathbb{E}[R; R > M].$$

Markov's inequality was used in the first bound. Thus $H_n \leq \alpha + \alpha M/n + 1/M$. Exactly one uniformly failed leaf and an independent anchor give $H_n \geq \alpha + \alpha/n - (1 - \epsilon^n)/n$. Taking $M = \sqrt{n}$ yields $H_n \rightarrow \alpha$ and $T_n/H_n \rightarrow 2$. The incidence graph is the same width-two graph as before; all means are interior. Scaling physical coordinates by ρ maps the box to $[1, \rho]^{n+1}$ and preserves the gaps of the correspondingly rescaled positive polynomial. $\square$

9.1 What fails for general nonnegative factors

Example 9.3 (An exact width ratio of three). Use six fair variables in pairs AB, BC, CA , and put

$$E(u, v) = 1 - u - v + 2uv, \quad N(u, v) = u + v - 2uv.$$

These are the multiaffine equality and inequality indicators, both in $[0, 1]$ on the square. Define

$$g_A = E(AB)E(CA), \quad g_B = N(AB)E(BC), \quad g_C = N(BC)N(CA).$$

Each local upper envelope is one at the center: mix a satisfying assignment and its complement. Each local lower envelope is zero, by the analogous mixture of a falsifying assignment. No two factors can equal one on a binary assignment, because adjacent factors demand opposite parities on their common pair. Thus the sum's upper envelope is at most one, attained by the all-zero/all-one mixture, which satisfies g_A . The pair assignment $AB = 01, BC = 01, CA = 00$ and its complement make all factors zero, proving the sum's lower envelope zero. Therefore $T = 3$ and $H = 1$ for these three entire factors.

A central bag $\{A, B, C\}$ and six attached bags $\{A, B, X_{AB,j}\}$, $\{B, C, X_{BC,j}\}$, $\{C, A, X_{CA,j}\}$, $j = 1, 2$, form a width-two decomposition. A four-cycle proves the width is exactly two. Every variable has frequency two. The factors are nonnegative on the full cube, but their monomial coefficients have mixed signs, so neither the positive-monomial nor the convex-cardinality factor theorem applies.

Proposition 9.4 (Independent-set payoff family). *For a simple graph G with nonnegative vertex weights w_v , place two fair bits on each edge and assign its endpoints opposite demanded parities. Give vertex v payoff w_v if all its incident demanded parities hold, and zero otherwise. The sum of separate maximum expected payoffs is $\sum_v w_v$, while the global maximum is $\alpha_w(G)$, the maximum independent-set weight. With all constructed factor scopes retained, including zero-weight factors, a graph with an edge has incidence treewidth $\max\{2, \text{tw}(G)\}$.*

Proof. Each local satisfaction pattern and its complement attain w_v with fair means. Satisfied vertices form an independent set, since the two endpoints of an edge demand opposite parities. Conversely fix a maximum independent set, give each incident edge its selected endpoint's demand, and choose remaining parities arbitrarily. Complement-pair mixing satisfies all selected factors with fair means, proving the maximum $\alpha_w(G)$; any extra satisfied factors still form an independent set and cannot improve that maximum. To a decomposition of G attach two size-three bags $\{u, v, X_{uv,j}\}$ for every edge at a bag containing u, v . This proves the treewidth upper bound. Contracting one length-two path per edge and deleting the other gives G as a minor; each edge also creates a four-cycle, proving both lower bounds. $\square$

Thus K_{k+1} with unit vertex weights gives maximum-payoff ratio $k + 1$ at incidence width k for $k \geq 2$. A general payoff maximum ratio is not an envelope-width ratio: the minima must also be computed. Example 9.3 does compute them and gives the claimed exact width obstruction.

9.2 Limits of the coloring method

A distinct open question at incidence treewidth two asks whether the factors admit two classes whose incidence subgraphs are each chordal bipartite, meaning they contain no induced cycle of length at least six. The proved all-cycle parity property permits induced eight-cycles and does not establish this target. The recorded experiment found such a partition for all 5,000 generated supports; this is finite evidence only.

The all-cycle coloring invariant cannot extend with a bounded number of colors to width three. In $K_{3,m}$, $m \geq 3$, with its three-vertex side interpreted as variables, every three factors lie on a six-cycle. Each permitted color class therefore contains at most two factors, requiring $\lceil m/2 \rceil$ colors, although the graph has treewidth three. Its all-ones matrix is already TU: determinants of order at least two vanish. The six-cycles have chords and do not obstruct balancedness. This separates the stronger coloring mechanism from a TU-row-partition target.

Finite searches found three balanced classes in 3,000 generated width-at-most-three supports (212 required more than two), and three TU classes in 300 further supports. Their accepted widths were certified by particular elimination orders; rejected greedy orders did not certify larger width. The TU tests enumerated column signings by the Ghouila-Houri criterion and used exact coloring backtracking. These searches are finite evidence, not proofs of universal partitions. The exact W_k for $k \geq 3$, k -class balanced or TU partitions, and sharp planar constants remain open here. A planar bipartite incidence graph does admit an orientation of maximum outdegree at most two: Hall's theorem assigns every edge to one of two slots at an endpoint, since every edge set uses at most twice as many edges as incident vertices. Orient from the assigned endpoint. Thus the general orientation theorem gives a finite planar bound, without settling its sharp value. Auxiliary reductions and the simpler forest-coloring obstruction are recorded in Appendix D.

10 Positive boxes and coefficient regularity

Let $C_{\text{box}}(\rho_B)$ be the supremal gap ratio for positive monomial factorizations on strictly positive boxes with every coordinate aspect ratio $u_i/\ell_i \leq \rho_B$, over arbitrary dimensions and evaluation points with positive hull gap. Unlike C_δ , this restricts the box, not the normalized evaluation point. Fixed coordinates are removed.

10.1 A coefficient inequality with one ambient orientation law

Fix an ambient dimension $N \geq 2$. Choose a uniform subset $J \subseteq [N]$ of size $\lfloor N/2 \rfloor$ and an independent uniform $U \in [0, 1]$. Coordinate i of mean p_i succeeds on $[0, p_i]$ if $i \in J$ and on $[1 - p_i, 1]$ otherwise. Call this balanced orientation law O . It has all prescribed means, though for odd N its individual orientation choices are not fair. The probability that a fixed pair has opposite orientations is

$$q_N = \frac{2\lfloor N/2 \rfloor \lceil N/2 \rceil}{N(N-1)}, \quad \beta_N = q_N^{-1} = \begin{cases} 2 - 2/N, & N \text{ even,} \\ 2 - 2/(N+1), & N \text{ odd.} \end{cases}$$

All local supports and smaller-dimensional induction steps below use restrictions of this one law. Deleting a deterministic coordinate neither conditions nor resamples the orientation subset.

For a support S of size $d \leq N$, let $K = \sum_{i \in S} X_i$. Write C_j, P_j, O_j for $\mathbb{E}(\binom{K}{j})$ under common-threshold, independent, and ambient orientation rounding. With $s = \sum_{i \in S} p_i$, $k = \lfloor s \rfloor$, and $\theta = s - k$, put

$$V_j = (1 - \theta) \binom{k}{j} + \theta \binom{k+1}{j}.$$

Degree-zero moments are one; negative orders and orders above support size are zero. By Lemma 8.6, C_j and V_j are the true upper and lower envelope values of the elementary symmetric polynomial of degree j .

Lemma 10.1 (Finite spreading inequality). *For every support $S \subseteq [N]$, every mean vector, and every integer $j \geq 0$,*

$$P_j - V_j \leq \beta_N (C_j - O_j) + (C_{j-1} - P_{j-1}). \quad (10.1)$$

The same statement holds with β_N replaced by 2 and independently fair endpoint orientations in every coordinate.

Proof. Write β for the indicated constant and set $F_{S,j} = \beta C_j + V_j - P_j - \beta O_j + C_{j-1} - P_{j-1}$. Orders zero and one vanish; order $d+1$ is $C_d - P_d \geq 0$, and higher orders vanish. For order two let $L_2 = \sum_{a < b} \max(0, p_a + p_b - 1)$. Equal orientations give a pair's upper Fréchet value; opposite orientations give its lower value. Their opposite probability is $1/\beta$, including under restriction. Thus $O_2 = (1 - 1/\beta)C_2 + L_2/\beta$, and

$$F_{S,2} = (C_2 - P_2) + (V_2 - L_2) \geq 0.$$

The first inequality follows from upper optimality, the second because every law, including an adjacent-cardinality law, respects every pair lower bound.

We induct on support size. A mean-zero coordinate is deterministic and its deletion gives $F_{S,j} = F_{S \setminus \{i\},j}$. A mean-one coordinate and Pascal's identity give $F_{S,j} = F_{S \setminus \{i\},j} + F_{S \setminus \{i\},j-1}$. The adjacent-cardinality values obey the same identity after shifting the sum by one. These are restrictions of the ambient law, not newly sampled balanced laws, so its correlated coins cause no exception. All boundary points follow by induction; dimension one is immediate.

For an interior vector and $3 \leq j \leq d$, select distinct coordinates of global minimum x and global maximum y . Move them to $x - h, y + h$ with $0 \leq h \leq \min(x, 1 - y)$. The sum stays fixed, so $\Delta V_j = 0$. In increasing order of means,

$$C_j = \sum_{i=1}^d p_{(i)} \binom{d-i}{j-1}.$$

Putting the selected minimum first and maximum last also handles ties. Hence, with $B = \binom{d-1}{j-1}$ and $D = \binom{d-1}{j-2}$, $\Delta C_j = -Bh$ and $\Delta C_{j-1} = -Dh$; the last coordinate has zero coefficient since $j \geq 3$. If E_r is the elementary symmetric polynomial in the other $d - 2$ means, the unchanged sum and product change $\Delta(xy) = -h(y - x + h)$ give exactly

$$\Delta(-P_j - P_{j-1}) = h(y - x + h)(E_{j-2} + E_{j-3}) \leq Dh.$$

Here $0 \leq y - x + h \leq 1$ and $E_r \leq \binom{d-2}{r}$. Finally couple both orientation vectors using the same U and coins. Increasing a single mean by h changes only that bit, monotonically, on an event of probability h . The increase in $\binom{K}{j}$ is then $\binom{K_{\text{rest}}}{j-1} \leq B$. Decreasing the minimum first and increasing the maximum second therefore gives $\Delta O_j \geq -Bh$, regardless of dependence among orientation coins. Combining the finite changes yields $\Delta F_{S,j} \leq -\beta Bh - Dh + Dh + \beta Bh = 0$. Take $h = \min(x, 1 - y)$ to reach a boundary point, whose coefficient is nonnegative by induction. The original coefficient is at least that boundary value. No differentiability assumption is used. $\square$

The choice of global extrema in this proof matters. For independently fair orientations, the stronger Schur-concavity assertion is false:

$$\begin{aligned} F_{[4],3}(2/5, 7/10, 24/25, 97/100) &= 6201/12500, \\ F_{[4],3}(2/5, 7/10, 959/1000, 971/1000) &= 4961031/10000000. \end{aligned} \tag{10.2}$$

Spreading the last two coordinates by $1/1000$ increases the coefficient by $231/10000000$. These rational values follow by averaging the sixteen orientation patterns, integrating between their interval endpoints, and evaluating the displayed symmetric moments. They refute arbitrary pair spreading, without affecting the global-minimum/global-maximum comparison.

Theorem 10.2 (Positive-box upper bounds). *For $N \geq 2$ nonfixed coordinates and $\rho_B > 1$, every positive multilinear polynomial on a strictly positive box of aspect ratios at most ρ_B satisfies*

$$T \leq (\rho_B + \beta_N)H \leq (\rho_B + 2)H.$$

The common mixture of independence and balanced orientation with weights $\rho_B/(\rho_B + \beta_N)$ and $\beta_N/(\rho_B + \beta_N)$ captures this fraction of each original monomial's exact gap simultaneously. For zero or one nonfixed coordinate all gaps vanish.

Proof. First consider a support on $[1, \rho_B]^N$ and put $t = \rho_B - 1$. Its binary physical value is $(1 + t)^K$, so its upper, independent, orientation, and lower values are $C(t) = \sum_j C_j t^j$, $P(t)$, $O(t)$, and $V(t) = \sum_j V_j t^j$. The last identity follows from a single adjacent-cardinality law attaining every order together. The coefficient of t^j in $(\beta_N + t)C + V - (1 + t)P - \beta_N O$ is $F_{S,j} \geq 0$. Consequently the true local gap satisfies

$$C - V \leq \rho_B(C - P) + \beta_N(C - O). \tag{10.3}$$

The same mixture works on all supports in the ambient N coordinates. Summing with nonnegative coefficients and using common upper attainment proves the common-aspect conclusion. Fair independent orientation coins and $\beta = 2$ prove $\rho_B + 2$ directly in every dimension.

For unequal endpoints $0 < \ell_i < u_i$ put

$$s_i = \frac{u_i/\ell_i - 1}{\rho_B - 1} \in (0, 1], \quad z_i = \ell_i[(1 - s_i) + s_i y_i], \quad y_i \in [1, \rho_B].$$

This is an affine bijection onto the original box. Each original monomial becomes a positive sum of physical y -monomials. Its true factor gap is at most the sum of their gaps, by (2.3), while the full hull gap is affine invariant. Apply the common-aspect bound to the expansion, always using restrictions of the same ambient orientation law. Moreover the original factor's upper value is the sum of the expanded upper values by common-threshold attainment. Its deficiency under either law is thus the sum of expanded deficiencies, proving the claimed per-original-factor guarantee. The certificate can be sampled without forming the expansion: the endpoint laws pull back to the original endpoints. Box inclusion alone would not prove any of these comparisons. $\square$

Theorem 10.3 (Coefficient-regularity cardinality bound). *In ambient dimension $N \geq 2$, suppose each entire local factor is the multiaffine interpolant of*

$$\psi_e(k) = \sum_{j=0}^{d_e} a_{e,j} \binom{k}{j}, \quad a_{e,j} \geq 0, \quad a_{e,j+1} \leq L a_{e,j} \quad (j \geq 2),$$

where $L \geq 0$ is common to all factors and coefficients beyond the degree are zero. Then

$$T \leq (L + 1 + \beta_N)H.$$

Separate affine terms may be added freely. Fair independent endpoint orientations give the dimension-free bound $L + 3$. These are bounds for the gaps of the entire cardinality factors, not the gaps obtained by separately relaxing their expanded monomials.

Proof. The nonnegative binomial coefficients make each sequence discrete convex: its second difference is $\sum_{j \geq 2} a_{e,j} \binom{k}{j-2} \geq 0$. Lemma 8.6 supplies a common upper extremizer, and within each factor one adjacent-cardinality law attains every V_j together. Denote that factor's four values by $C_\psi, V_\psi, P_\psi, O_\psi$. Multiply (10.1) by its a_j and sum. Since $C_j - P_j \geq 0$ and the order-zero and order-one deficiencies vanish, reindexing gives

$$\begin{aligned} P_\psi - V_\psi &\leq \beta_N(C_\psi - O_\psi) + \sum_{j \geq 2} a_{j+1}(C_j - P_j) \\ &\leq \beta_N(C_\psi - O_\psi) + L(C_\psi - P_\psi). \end{aligned}$$

Add $C_\psi - P_\psi$ to obtain

$$C_\psi - V_\psi \leq (L + 1)(C_\psi - P_\psi) + \beta_N(C_\psi - O_\psi).$$

One global mixture with weights proportional to $L + 1$ and β_N therefore captures the required fraction for all factors. Summing and using common upper attainment proves the theorem. This also handles $L = 0$, deterministic means, and affine factors. With at most one nonfixed coordinate the interpolants are affine and all gaps vanish. The product sequence ρ_B^k has $a_j = (\rho_B - 1)^j$, so $L = \rho_B - 1$ recovers the common-aspect product inequality. $\square$

This refinement is a direct consequence of the coefficient mechanism; no optimality or independent priority is asserted. Distinct earlier low/high and asymmetric proofs, and the precise limitation of fixed independence/orientation mixtures, appear in Appendix E.

10.2 A lower family on each fixed positive box

Theorem 10.4 (Aspect-ratio lower bound). *For every $\rho_B > 1$,*

$$\max\{2, \rho_B\} \leq C_{\text{box}}(\rho_B) \leq \rho_B + 2.$$

Thus $C_{\text{box}}(\rho_B) = \rho_B + O(1)$ as $\rho_B \rightarrow \infty$. The exact finite-aspect-ratio constant is not determined.

Proof. The lower bound two already holds at width two by Theorem 9.2. For the lower bound ρ_B , fix $\epsilon = 1/\rho_B \in (0, 1)$ and put $\alpha = 1 - \epsilon$. Use the radix supports with $L \geq 2$, $b = L^2$, $m = b^L$, now on $[\epsilon, 1]^{m+L}$ with normalized means $\mathbb{E}A_j = b^{-j}$ and $\mathbb{E}Z_i = 1 - m^{-1}$. A level- j term has $k = m/b^j$ leaves and total failure mean $(1 - b^{-j}) + k/m = 1$. Its vertex value is ϵ^{R_e} , so its exact lower value is ϵ , attained by exactly one failure with the prescribed categorical probabilities. Common-threshold rounding gives its upper value

$$C_{j,Q} = \epsilon + \alpha b^{-j} - \frac{\epsilon}{m}(1 - \epsilon^k).$$

Indeed its successive threshold states have product values $1, \epsilon, \epsilon^{k+1}$ with interval lengths $b^{-j}, 1 - m^{-1} - b^{-j}, m^{-1}$. Therefore

$$T_L = \alpha L - D_L, \quad D_L = \epsilon \sum_{t=0}^{L-1} \frac{1 - \epsilon^{b^t}}{b^t}, \quad 0 \leq D_L \leq \frac{\epsilon b}{b-1}. \quad (10.4)$$

For any common endpoint law let R_Q count failed leaves in block Q and R count them in total. Since $\mathbb{E}R = 1$, subtracting expected physical products from their common upper value gives exactly

$$H_L = \max \mathbb{E} \left[\alpha \sum_j A_j \sum_{Q \in \mathcal{P}_j} (1 - \epsilon^{R_Q}) + \epsilon \sum_j \sum_{Q \in \mathcal{P}_j} (1 - \epsilon^{R_Q}) \right] - D_L. \quad (10.5)$$

For its first term use $1 - \epsilon^{R_Q} \leq \mathbf{1}\{R_Q > 0\}$ and the unit-cube radix hull upper bound. For its second use $1 - \epsilon^r \leq \alpha r$ and $\sum_{Q \in \mathcal{P}_j} R_Q = R$. This proves

$$H_L \leq \epsilon \alpha L + \alpha[1 + (L-1)/b] - D_L.$$

Exactly one uniformly failed leaf, with independent anchors of the required means, instead gives

$$H_L \geq \epsilon \alpha L + \alpha^2 \sum_{j=1}^L b^{-j} - D_L.$$

The same geometric inequality gives $D_L \leq \epsilon \alpha L$, so this last bound is strictly positive for every finite L . The bounded correction in (10.4) now implies $T_L/L \rightarrow \alpha$, $H_L/L \rightarrow \epsilon \alpha$, and their ratio tends to ρ_B . Scale coordinates by ρ_B to get positive coefficients on $[1, \rho_B]^{m+L}$; both gaps are preserved. This is a limit in dimension for each fixed aspect ratio, requiring no exchange of those limits. The upper bound is Theorem 10.2. $\square$

10.3 Unequal aspect ratios at frequency two

Example 10.5 (A rational bipartite ratio 7/6). For $f(x, y, z) = xy + xyz$ on $[1, 2] \times [1, 2] \times [1, 3]$, evaluate at $(5/4, 5/4, 5/2)$. Put $x = 1 + a, y = 1 + b, z = 1 + 2c$, so the normalized mean is $(1/4, 1/4, 3/4)$. The following valid affine bounds give exact values:

$$xy \geq 1 + a + b, \quad xyz \geq 2a + 2b + 3c, \quad f \geq 4a + 4b + 4c, \quad f \leq 2 + 8a + 4b + 2c.$$

Validity follows by substitution at all eight binary vertices and multiaffine interpolation. The respective attaining distributions are

xy lower	000, 001, 011, 101 each of mass $1/4$
xyz lower	001 of mass $3/4$; 110 of mass $1/4$
f lower	001 of mass $1/2$; 011, 100 each of mass $1/4$
f upper	000 of mass $1/4$; 001 of mass $1/2$; 111 of mass $1/4$.

Each row has the required means. The separate upper bounds $xy \leq 1+2a+b$ and $xyz \leq 1+6a+3b+2c$ are attained by the last law. Thus $T = 13/2 - 19/4 = 7/4$, while $H = 13/2 - 5 = 3/2$. The dual multigraph is bipartite: its two factor vertices share two parallel edges, with a dummy leaf for z . This disproves unequal-box bipartite scalar exactness, while leaving the common-aspect theorem intact.

Proposition 10.6 (Bipartite positive-box boundary). *For arbitrary nonnegative boxes with a bipartite dual graph, positive monomial gaps satisfy $T \leq 2H$. Their worst ratio over strictly positive boxes is at least $3/2$, already with two factors and three variables. Its exact value between $3/2$ and 2 remains open here, as does a universal $3/2$ bound for unequal-box frequency-two instances.*

Proof. More generally a proper coloring of the term conflict graph into q classes gives $T \leq qH$: terms in one class have disjoint scopes, so join their minimizing laws independently and complete unused means. They attain their entire class gap, and other factors have nonnegative expected deficiency from their upper envelopes. Since those upper values are commonly attained, H is at least each class's total gap. Sum these inequalities. This also proves a bound two for any two positive terms regardless of their boxes or frequency.

For the lower family let $0 < \epsilon \leq 2$ and use $f_\epsilon = (2/\epsilon)xy + xyz$, with $x, y \in [1, 1 + \epsilon]$, $z \in [1, 3]$, at normalized means $(a, b, c) = (1/4, 1/4, 3/4)$. After removing affine parts the two factors are

$$A = 2\epsilon ab, \quad B = \epsilon^2 ab + 2\epsilon ac + 2\epsilon bc + 2\epsilon^2 abc, \quad C = A + B.$$

Their lower bounds at the point are attained at the first three laws of Example 10.5, respectively. The affine certificates are $A \geq 0$, $B \geq \epsilon^2(a + b + c - 1)$, and $C \geq 2\epsilon(a + b + c - 1)$. In binary order 000, 001, 010, 011, 100, 101, 110, 111, the last two residual lists are

$$\begin{aligned} &(\epsilon^2, 0, 0, \epsilon(2 - \epsilon), 0, \epsilon(2 - \epsilon), 0, \epsilon(\epsilon + 4)), \\ &(2\epsilon, 0, 0, 0, 0, 0, \epsilon^2, \epsilon(3\epsilon + 2)), \end{aligned}$$

which are nonnegative throughout the stated range. The local lower values sum to $\epsilon^2/4$ and the full lower value is $\epsilon/2$. The same common upper law as before gives $3\epsilon/2 + 3\epsilon^2/4$. Hence

$$T = \epsilon(3/2 + \epsilon/2), \quad H = \epsilon(1 + 3\epsilon/4), \quad \frac{T}{H} = \frac{2(\epsilon + 3)}{3\epsilon + 4} \longrightarrow 3/2.$$

To obtain unit coefficients divide the polynomial by $2/\epsilon$ and put $z' = (\epsilon/2)z$. It becomes $xy + xyz'$ on $[1, 1 + \epsilon]^2 \times [\epsilon/2, 3\epsilon/2]$, with the same ratio. $\square$

For frequency-two products on unequal positive boxes, normalizing upper endpoints to one gives a binary failure cost $\prod_{i \in J \cap e} \alpha_i = \exp(-\sum_{i \in J \cap e} w_i)$, where $\alpha_i = \ell_i/u_i$ and $w_i = -\log \alpha_i$. Each edge has the *same* weight w_i at both incident factors. This is a weighted convex degree potential, whereas the matching theorem above uses unweighted cardinalities. Finite floating-point searches approaching $3/2$ from below do not supply an upper bound. If the dual multigraph is a forest, conditional gluing actually preserves all local minima and maxima even for signed multiaffine factors on arbitrary boxes; Appendix D gives the elementary argument.

11 Exact complexity of a single physical product

A ratio bound can be small even when evaluating an individual factor's exact envelope is difficult. For one factor the ratio is identically one wherever its gap is positive. The next result concerns computing that factor's envelope, not loss caused by separating interacting terms. All complexity statements use rational binary encodings and bit time.

Theorem 11.1 (Midpoint decision on narrow unequal boxes). *Given rational $u_i > 1$ and q , deciding whether*

$$\text{vex}_B m(\bar{x}) \leq q, \quad m(x) = \prod_{i=1}^n x_i, \quad B = \prod_i [1, u_i], \quad \bar{x}_i = (1 + u_i)/2,$$

is NP-complete. For every fixed rational $\eta > 0$ it remains NP-complete when $u_i \leq 1 + \eta$ for all i .

Proof. Reduce from PARTITION, the classical NP-complete integer partition problem [27]. Let its positive integers be $a_1, \dots, a_n$ with total A . Doubling them permits us to assume A even and at least two. Fix an integer $K \geq 1$ and put

$$\epsilon = (16KA^3)^{-1}, \quad u_i = 1 + \epsilon a_i.$$

These boxes satisfy $1 < u_i \leq 1 + 1/(16KA^2) < 2$. In normalized endpoint variables $X_i = 1 + \epsilon a_i Z_i$, the midpoint means are $\mathbb{E}Z_i = 1/2$. Write $S(z) = \sum_i a_i z_i$. On a binary vertex,

$$\prod_i (1 + \epsilon a_i z_i) = 1 + \epsilon S(z) + \frac{\epsilon^2}{2} \left(S(z)^2 - \sum_i a_i^2 z_i \right) + R(z). \quad (11.1)$$

All remainder coefficients are nonnegative. Its order- j elementary symmetric sum is at most A^j , and $\epsilon A \leq 1/16$, so

$$0 \leq R(z) \leq \frac{(\epsilon A)^3}{1 - \epsilon A} \leq 2\epsilon^3 A^3 = \epsilon^2/(8K) \leq \epsilon^2/8. \quad (11.2)$$

Every admissible law has $\mathbb{E}S = A/2$, whence

$$\mathbb{E}m(X) = b + \frac{\epsilon^2}{2} \text{Var}(S) + \mathbb{E}R(Z), \quad b = 1 + \frac{\epsilon A}{2} + \epsilon^2 \left(\frac{A^2}{8} - \frac{1}{4} \sum_i a_i^2 \right).$$

A partition state and its complement, with equal probabilities, have all means $1/2$ and $S = A/2$ deterministically. Thus a YES instance has lower envelope at most $b + \epsilon^2/8$. In a NO instance every binary state has integer $S \neq A/2$, so $\text{Var}(S) \geq 1$ under every feasible law. Its lower envelope is at least $b + \epsilon^2/2$. The rational threshold $q = b + \epsilon^2/4$ separates the cases. All numbers have polynomial encoding length, since $\log A$ is polynomial in the original input length. For a fixed $\eta > 0$, choose fixed K with $1/(16K) \leq \eta$ to enforce the narrower interval.

For NP membership, the vertex-law LP has equality rows $1, z_1, \dots, z_n$ and right-hand side $(1, 1/2, \dots, 1/2)$. An optimal basic solution has at most $n + 1$ positive probabilities. It provides that many binary states and rational weights of polynomial bit length: a nonsingular subsystem has zero-one entries and half-integer right-hand side, so determinant bounds apply. Each vertex product has polynomial bit length as a product of at most n input rationals. Exact arithmetic verifies nonnegative weights, total mass, all means, and expected product at most q in polynomial time. $\square$

The NO case also has an explicit affine certificate in normalized coordinates:

$$\ell(z) = 1 - \epsilon^2 A^2/8 + \epsilon^2/2 + \sum_i \left[\epsilon a_i + \frac{\epsilon^2}{2} (A a_i - a_i^2) \right] z_i. \quad (11.3)$$

Subtracting it from (11.1) gives $\epsilon^2[(S - A/2)^2 - 1]/2 + R(z) \geq 0$ at every binary vertex of a NO instance, hence everywhere by multiaffine interpolation. Its midpoint value is $b + \epsilon^2/2$. The reduction is from weakly NP-complete PARTITION; it establishes ordinary, not strong, NP-hardness. It concerns varying unequal boxes contained in a narrow common box, not the envelope on that one common box. The separating accuracy can be exponentially small in input length.

Corollary 11.2 (Graph-hull membership). *Deciding whether $(\bar{x}, q)$ belongs to the scalar graph hull of the same single monomial is NP-complete under the same narrow-box restrictions.*

Proof. At the midpoint its common-threshold upper value is $C = (1 + \prod_i u_i)/2$. Retaining the product's nonnegative quadratic terms gives

$$C \geq 1 + \epsilon A/2 + \epsilon^2(A^2 - \sum_i a_i^2)/4, \quad C - b \geq \epsilon^2 A^2/8 \geq \epsilon^2/2 > q - b.$$

Thus the upper-hull condition always holds on the reduced instances, and membership is exactly the lower-envelope threshold question. Membership lies in NP: a basic distribution for the normalization, coordinate, and value equations has at most $n + 2$ positive weights. These equations have rational polynomial-bit entries and determinant bounds give polynomial-bit weights. Exact verification proves membership. $\square$

Proposition 11.3 (A fixed number of ratio types). *For one product on a rational strictly positive box having at most k distinct coordinate aspect ratios, its exact scalar envelope and a supporting affine minorant can be computed in $n^{O(k)}$ times a polynomial in the input bit length. In particular fixed k is tractable.*

Proof. After substituting fixed coordinates and normalizing endpoints, a binary product with count h_j in ratio group j has value $(\prod_i \ell_i) \prod_{j=1}^k r_j^{h_j}$. To minimize this value minus $\sum_i \pi_i z_i$ for arbitrary dual coefficients, sort the coefficients within each group. For fixed counts choose the h_j largest coefficients, and use prefix sums. Enumerate all $\prod_j (n_j + 1) \leq (n + 1)^k$ count tuples. Rational multiplication and sorting have polynomial bit cost; this is an exact vertex oracle. The determinant-bounded dual argument of Corollary 8.11 then computes the lower envelope and its minorant. The common-threshold upper envelope is explicit. Thus mere closeness of the ratios does not substitute for a bound on their number of distinct values. $\square$

11.1 Rank-one transport and logarithmic accuracy dependence

Multimarginal optimal transport with specified finite marginals minimizes expected cost over their joint laws. With n fair binary marginals and factored cost tensor

$$C = \bigotimes_{i=1}^n (1, 1 + \epsilon a_i), \quad C_z = \prod_i (1 + \epsilon a_i z_i),$$

it is exactly the midpoint envelope above. This tensor is strictly positive and rank one, with no sparse correction, and

$$1 \leq C_z \leq \prod_i (1 + \epsilon a_i) \leq \sum_{j \geq 0} (\epsilon A)^j \leq 16/15.$$

Corollary 11.4 (Rank-one value and weighted dual accuracy). *Exact threshold evaluation for these rank-one MOT instances is NP-complete. Unless $P = NP$, neither their values nor their weighted vertex minima admit a deterministic additive- δ approximation algorithm polynomial in rational input length and $\log(C_{\max}/\delta)$. For randomized bounded-error algorithms the corresponding consequence requires $NP \not\subseteq BPP$.*

Proof. Take $\delta = \epsilon^2/32$ in the YES and NO intervals of Theorem 11.1. Their approximation intervals remain disjoint and $\log(1/\delta)$ is polynomial in the PARTITION input length. Such an algorithm would decide PARTITION. The weighted vertex oracle is hard directly: set

$$\pi_i = \epsilon a_i + \frac{\epsilon^2}{2}(Aa_i - a_i^2), \quad d_0 = 1 - \epsilon^2 A^2/8.$$

Equation (11.1) gives

$$C_z - \pi^\top z = d_0 + \frac{\epsilon^2}{2}(S(z) - A/2)^2 + R(z).$$

Its minimum is at most $d_0 + \epsilon^2/8$ in a YES instance and at least $d_0 + \epsilon^2/2$ in a NO instance. The same approximation margin and polynomial encoding bounds apply. Exact MOT decision has the NP certificates already given. If an input convention requires bit lengths polynomial in the number of marginals, append a number of dummy binary factors $(1, 1)$ linear in the original input bit length. Give them fair marginals and zero dual potentials. They change neither rank, cost range, value, nor oracle minimum, and meet that convention. $\square$

The published structured-MOT paper of Altschuler and Boix-Adserà [3, Section 7.2, after Theorem 7.4] asks whether constant-rank approximate dual minimization can have logarithmic inverse-accuracy dependence, leading to exact algorithms. The preceding reduction rules out that dependence in the rational bit model under the stated complexity assumption, already at rank one. Their Corollary 7.5 has polynomial dependence on inverse accuracy and is consistent with it. This comparison identifies the historical question; it is not a claim that no subsequent work addressed it. Nor is it a lower bound in an unrestricted unit-cost arithmetic model. The unweighted smallest entry of this tensor is simply one; arbitrary dual weights are essential. None of these arguments proves hardness at fixed additive error.

11.2 A boundary of multilinearity

Example 11.5 (Two positive rational-power terms). On the fixed interval $[1, 3]$, let $f_\epsilon(x) = x^{1-\epsilon} + x^{1+\epsilon}$, $0 < \epsilon < 1$. The first term is concave and the second convex, while

$$f''_\epsilon(x) = \epsilon x^{-1-\epsilon}[(1+\epsilon)x^{2\epsilon} - (1-\epsilon)] > 0.$$

At $x = 2$ their separate widths and full width are consequently

$$T_\epsilon = 3 \sinh(\epsilon \log 3) - 4 \sinh(\epsilon \log 2), \quad H_\epsilon = 1 + 3 \cosh(\epsilon \log 3) - 4 \cosh(\epsilon \log 2).$$

These follow by subtracting each function from its endpoint secant, with the opposite subtraction for the concave term. Taylor expansion gives

$$T_\epsilon = (3 \log 3 - 4 \log 2)\epsilon + O(\epsilon^3), \quad H_\epsilon = \frac{1}{2}[3(\log 3)^2 - 4(\log 2)^2]\epsilon^2 + O(\epsilon^4).$$

The first leading coefficient is $\log(27/16) > 0$; the second is positive by strict convexity of $x(\log x)^2$ on $[1, 3]$. Thus their ratio diverges like a positive constant times $1/\epsilon$. Taking $\epsilon = 1/k$, $k \geq 2$, keeps

rational exponents in $[1/2, 3/2]$ with only two positive unit coefficients and one variable. Under $x = e^y$, however, both terms become convex exponentials on $[0, \log 3]$. Their lower envelopes are exact and endpoint secants add, so the termwise and full widths coincide in that representation. The example concerns the original coordinate and does not preclude useful logarithmic reformulation.

12 Fractional cardinality and spatial certificates

The following optimization problems have explicit solutions. Their role is to distinguish the size of certificates produced by specified spatial oracles from the difficulty of finding an optimizer. The obstruction uses a concave quadratic penalty and an affine balance; it is a separate construction from the positive multilinear gap examples above.

Fix integers $n \geq 2$ and $0 \leq k \leq n - 1$, and write

$$K = k + \frac{1}{2}, \quad \mathcal{F}_{n,k} = \{x \in [0, 1]^n : \sum_i x_i = K\}, \quad F(x) = \sum_{i=1}^n x_i(1 - x_i). \quad (12.1)$$

The dimension parameter n counts original coordinates. A vertex of $\mathcal{F}_{n,k}$ has exactly k coordinates equal to one, one equal to $1/2$, and the others zero: two strictly fractional coordinates would admit a small opposite perturbation, whereas a point with at most one fractional coordinate is determined by its active bounds and the balance. Concavity and compactness therefore give

$$\min_{\mathcal{F}_{n,k}} F = \frac{1}{4}. \quad (12.2)$$

For each oracle below, a certified region is either infeasible or has oracle bound at least T_* . A certificate is a finite cover of $\mathcal{F}_{n,k}$ by such regions, as in Section 2.3. For absolute tolerance $\epsilon \geq 0$ we grant the best incumbent and set $T_* = 1/4 - \epsilon$. Pruning at $U - \epsilon$, for any feasible incumbent $U \geq 1/4$, implies this condition. Infeasible regions contain no witness and do not help cover the feasible set. All lower bounds also hold for a weaker valid oracle: certifying a region with that oracle certifies it with the stronger oracle used in the proof.

There are close precedents in other certificate models. Jarre’s binary knapsack construction retains an exponential branch-and-bound obstruction with a max-cut semidefinite bound [26, Sections 2–3]. Coniglio’s hyperplane-clustering analysis gives a related spatial obstruction under midpoint branching on symmetric coordinate domains [12, Assumption 1, Propositions 2–4, Appendix C in the review version]. Our arguments concern arbitrary coordinate restrictions and a fixed objective gap under the explicitly defined node oracles. We make no priority claim for exponential spatial lower bounds in general. The fractional-cardinality moments used later are classical; the additional argument places them in spatial regions and counts how many witnesses a certified region can contain.

12.1 Chords and endpoint avoidance

For a box $B = \prod_i [a_i, b_i] \subseteq [0, 1]^n$, including degenerate intervals, define

$$\ell_i(x_i) = a_i b_i + (1 - a_i - b_i)x_i, \quad x_i(1 - x_i) - \ell_i(x_i) = (x_i - a_i)(b_i - x_i). \quad (12.3)$$

Its chord bound is

$$\text{LB}_{\text{ch}}(B) = \min_{x \in B \cap \mathcal{F}_{n,k}} \sum_i \ell_i(x_i),$$

with $+\infty$ for an empty feasible intersection. If g_i is any convex underestimator of $x_i(1 - x_i)$ on $[a_i, b_i]$, convexity between the endpoints gives $g_i \leq \ell_i$. This also holds on a singleton interval. Thus

the infimum for any separable convex underestimator $\sum_i g_i$ is at most $\text{LB}_{\text{ch}}(B)$; an infimum is used because arbitrary convex underestimators need not attain their infima on a closed interval without lower semicontinuity at its endpoints.

For clarity, two common formulations yield exactly these chords. For $y_i = 1 - x_i \in [1 - b_i, 1 - a_i]$, both McCormick lower planes for $w_i = x_i y_i$ become $w_i \geq \ell_i(x_i)$. The upper planes remain compatible: the exact product lies above the chord and below each upper plane. The separable αBB expression $x_i(1 - x_i) - \alpha(x_i - a_i)(b_i - x_i)$ has second derivative $2(\alpha - 1)$, so the smallest convexifying value $\alpha = 1$ gives the chord. Larger valid values give weaker underestimators. These statements concern the specified separable formulations, without additional coupled cuts or nonlinear subproblem solution rules.

We isolate the counting argument so that later oracle changes do not obscure it. Let $n = k + m + z$, where $k, z \geq 0$ and $m \geq 1$ are integers. For each ordered partition (H, M, Z) of $[n]$ with these sizes, put

$$p = \frac{1}{2m}, \quad w = \mathbf{1}_H + p\mathbf{1}_M. \quad (12.4)$$

These $n!/(k!m!z!)$ distinct witnesses belong to $\mathcal{F}_{n,k}$. For an arbitrary product $S = \prod_i S_i$, with nonempty $S_i \subseteq [0, 1]$, define

$$A(S) = \{i : 0 \notin S_i\}, \quad D(S) = \{i : 1 \notin S_i\}, \quad R(S) = A(S) \cup D(S). \quad (12.5)$$

For boxes these are $A = \{i : a_i > 0\}$ and $D = \{i : b_i < 1\}$.

Lemma 12.1 (Endpoint-avoidance count). *Suppose each certified product domain that contains a witness has $|R(S)| \geq q$, where $q \geq 0$ is real. Any certified cover of $\mathcal{F}_{n,k}$ contains at least*

$$\mathcal{N}(q) := \min \left\{ \left(\frac{n}{n-k} \right)^{q/2}, \left(\frac{n}{n-z} \right)^{q/2} \right\} \quad (12.6)$$

regions. The same conclusion follows from the stronger hypothesis $|R(S)| > q$. Both denominators are positive, and $\mathcal{N}(0) = 1$.

Proof. Choose the ordered partition uniformly. A contained witness must have $Z \cap A = \emptyset$ and $H \cap D = \emptyset$. For a fixed set of s excluded positions, a uniformly chosen z -subset avoids it with probability

$$\frac{\binom{n-s}{z}}{\binom{n}{z}} = \prod_{j=0}^{s-1} \frac{n-z-j}{n-j} \leq \left(\frac{n-z}{n} \right)^s \quad (s \leq n-z).$$

For $s > n - z$ the probability is zero, so the upper bound still holds; for $s = 0$ the product is one. The analogous inequality holds for H . No independence between the two avoidance events is required. A fixed region's witness fraction is at most

$$\min \left\{ \left(\frac{n-z}{n} \right)^{|A|}, \left(\frac{n-k}{n} \right)^{|D|} \right\}.$$

Since $|A| + |D| \geq |R| \geq q$, at least one exponent is at least $q/2$. The displayed fraction is consequently at most $1/\mathcal{N}(q)$. Regions containing no witness have fraction zero. The union bound over a cover proves the assertion, whether or not its regions overlap. Bases equal to one when $k = 0$ or $z = 0$ cause no exception. $\square$

Theorem 12.2 (Separable spatial certificates). *Let $n = k + m + z$, with integers $k, z \geq 0$, $m \geq 1$, $n \geq 2$, and $0 \leq \epsilon < 1/4$. For problem (12.1), every cover by boxes certified by the chord oracle at $T_* = 1/4 - \epsilon$ has at least $\mathcal{N}(h)$ members, where*

$$h = m(1/2 - 2\epsilon). \quad (12.7)$$

This includes arbitrary real coordinate split points and all separable convex underestimators dominated by the chord oracle. For $k = m = z = t$, $n = 3t$, the bound is $(3/2)^{t(1/4-\epsilon)}$, hence $(3/2)^{n/24}$ at $\epsilon = 1/8$.

Proof. In a box containing w , the chord values at its zero and unit witness coordinates vanish. On $U = [n] \setminus R$ the interval is $[0, 1]$, whose chord vanishes identically. Consequently

$$T_* \leq \text{LB}_{\text{ch}}(B) \leq \sum_{i \in M \cap R} \ell_i(p) \leq |M \cap R|p(1-p) < \frac{|M \cap R|}{2m}.$$

The last strict inequality is applicable because $T_* > 0$ forces $M \cap R \neq \emptyset$. Thus $|R| > 2mT_* = h$, and Lemma 12.1 applies. A complete coordinate split tree provides such a cover, regardless of split locations or branching order. $\square$

12.2 Upper certificates, tightening, and tolerances

Proposition 12.3 (Explicit upper certificates). *For $0 < \epsilon < 1/4$, a chord certificate for (12.1) has at most $2^{n+2} - 3$ nodes and $2^{n+1} - 1$ leaves. For $\epsilon = 0$, a midpoint certificate has exactly $\binom{n+1}{k+1}$ leaves if only the stopping rules in the proof are used. In particular its node count is $O(n^{\min\{k+1, n-k\}})$ when that exponent is fixed.*

Proof. Choose $0 < \alpha < 1/2$ with $\alpha(1-\alpha) = 1/4 - \epsilon$. Process coordinates in order. At each surviving node split the next coordinate at α , then split its upper child at $1 - \alpha$. The middle interval has constant chord $\alpha(1 - \alpha)$, and all other chords are nonnegative, so the middle child is certified. Continue on the low and high intervals. At a final corner box the chord sum is $(1 - \alpha)\|x - v\|_1$, where $v \in \{0, 1\}^n$ is its corner. On the demand slice, $\|x - v\|_1 \geq |K - \sum_i v_i| \geq 1/2$. This gives a bound $(1 - \alpha)/2 > 1/4 > T_*$. Processing coordinate i adds four nodes per surviving prefix, of which there are 2^{i-1} . The stated counts follow. For a strict pruning rule choose a slightly larger middle-chord target still below $1/4$.

For exact certification split at $1/2$ instead. If a prefix has c high intervals and d low intervals, its chord bound is at least each of

$$\frac{c - K}{2}, \quad \frac{K - (n - d)}{2}.$$

Indeed, the sum on high coordinates is at most K , whereas the sum on low coordinates is at least $K - (n - d)$; all other chord contributions are nonnegative. Stop at $c = k + 1$ or $d = n - k$. No path survives depth n . With remaining high and low stopping counts a, b , the number of leaves satisfies $L(a, b) = L(a - 1, b) + L(a, b - 1)$ and $L(0, b) = L(a, 0) = 1$, hence $L(a, b) = \binom{a+b}{a}$. Start at $(a, b) = (k + 1, n - k)$. A full binary tree has twice as many leaves minus one nodes. This proves the exact count and polynomial assertion. $\square$

The balanced fixed-tolerance family therefore has certificate size $2^{\Theta(n)}$ under the chord model. If both k and $n - k$ are proportional to n , choose m, z proportional to $n - k$ to obtain the same exponential order. Fixed demand level, or fixed complementary demand level, admits the polynomial

certificate just proved. At $\epsilon \geq 1/4$, the root chord bound zero already meets the target; no positive exponent is asserted there.

The chord error is genuinely quadratic in interval width:

$$0 \leq F(x) - \sum_i \ell_i(x_i) \leq \frac{1}{4} \sum_i (b_i - a_i)^2 \leq \frac{n}{4} \max_i (b_i - a_i)^2. \quad (12.8)$$

In particular, the difference between the exact node optimum and the chord bound is at most this quantity, by evaluating F at a chord minimizer. Thus the fixed-tolerance lower bound persists despite this second-order error estimate; it does not require a first-order local approximation error.

Proposition 12.4 (Charged reductions and equality tolerance). *Suppose a run keeps feasible points covered by its final boxes together with discarded boxes certified by the same oracle at T_* . If it processes N nodes and discards at most a such boxes per node, a cover lower bound L implies $N \geq L/(a+1)$. Pure feasibility reductions need not be charged when they discard no feasible point.*

For a demand feasibility tolerance $0 \leq \delta < 1/2$, replace $\mathcal{F}_{n,k}$ by $\mathcal{F}_{n,k}^\delta = \{x \in [0, 1]^n : |\sum_i x_i - K| \leq \delta\}$. Its exact optimum is $1/4 - \delta^2$. If $\epsilon + \delta^2 < 1/4$, a chord certificate at absolute tolerance ϵ for this enlarged set satisfies Theorem 12.2 with ϵ replaced by $\epsilon + \delta^2$.

Proof. The augmented cover has at most $(a+1)N$ members. A discarded open coordinate slab may be charged by its closed box only when that box is certified; this condition is part of the assertion, not an unproved closure assumption. Subsequent product-domain results can charge the removed coordinate set itself.

For a fixed sum $s \in [k + 1/2 - \delta, k + 1/2 + \delta] \subset (k, k+1)$, the same vertex argument as above gives minimum penalty $(s - k)(1 - s + k)$. Its minimum on this interval is $1/4 - \delta^2$, attained by k ones and a remaining coordinate $1/2 - \delta$ or $1/2 + \delta$. A tolerance-feasible incumbent is therefore at least this value. The exact-balance witnesses remain feasible for the enlarged problem, and the chord bound over the enlarged feasible set is at most its value at each witness. The proof of Theorem 12.2 applies at the positive target $1/4 - (\epsilon + \delta^2)$. $\square$

The restriction $\delta < 1/2$ is material: once the tolerance interval contains an integer sum, the optimum is zero. An unrestricted formula $1/4 - \delta^2$ would then be false. Relative pruning in the convention $\text{LB} \geq (1 - \theta)U$, $0 \leq \theta < 1$, on the original problem implies the absolute target with $\epsilon = \theta/4$.

12.3 Full second moments and box RLT

For $B = \prod_i [a_i, b_i] \subseteq [0, 1]^n$, let $\text{LB}_{\text{SDP}}(B)$ be the infimum of $\sum_i (\mu_i - X_{ii})$ over symmetric X and $\mu \in \mathbb{R}^n$ satisfying

$$a \leq \mu \leq b, \quad \sum_i \mu_i = K, \quad \begin{pmatrix} 1 & \mu^T \\ \mu & X \end{pmatrix} \succeq 0, \quad \sum_j X_{ij} = K\mu_i \quad (i \in [n]), \quad (12.9)$$

and all pairwise box inequalities

$$\begin{aligned} X_{ij} - a_i\mu_j - a_j\mu_i + a_ia_j &\geq 0, \\ b_j\mu_i - X_{ij} - a_ib_j + a_i\mu_j &\geq 0, \\ b_i\mu_j - X_{ij} - b_ia_j + a_j\mu_i &\geq 0, \\ b_ib_j - b_i\mu_j - b_j\mu_i + X_{ij} &\geq 0 \quad (i, j \in [n]). \end{aligned} \quad (12.10)$$

Indices may coincide. These are the linearizations of all products of $x_i - a_i$ and $b_i - x_i$. Exact evaluation $(\mu, X) = (x, xx^T)$ shows validity. As before the infimum of an empty relaxation is $+\infty$, and a region is certified when its bound is at least $T_* = 1/4 - \epsilon$ (or it is infeasible).

Lemma 12.5 (Restricted fractional second moments). *Let $k, m, z \geq 1$, and let B contain a witness (12.4). If $|R(B)| < \min\{k, z\}$, the model (12.9)–(12.10) has a feasible point of value $|M \cap R|p(1 - p)$.*

Proof. Fix $x_i = w_i$ deterministically on R and put

$$U = [n] \setminus R, \quad s = |U|, \quad t = K - \sum_{i \in R} w_i = |H \cap U| + p|M \cap U|.$$

At least one unit and one zero witness coordinate remain, so $s \geq 2$ and $1 \leq t \leq s - 1$. On U set

$$c = t/s, \quad d = t(t - 1)/(s(s - 1)), \quad \mu_i = c, \quad X_{ii} = c, \quad X_{ij} = d \quad (i \neq j).$$

On R set $\mu_i = w_i$, and set $X_{ij} = \mu_i \mu_j$ whenever at least one index lies in R . First moments meet the box bounds because the intervals on U are $[0, 1]$ and $w \in B$. Their sum is K . The covariance is zero on rows and columns in R , and on U it is

$$X_{UU} - \mu_U \mu_U^T = (c - d)(I - J/s), \quad c - d = \frac{t(s - t)}{s(s - 1)} \geq 0.$$

It is positive semidefinite, proving augmented PSD by the Schur complement of the entry one. For $i \in U$, $c + (s - 1)d = tc$ gives $\sum_j X_{ij} = Kc$; for $i \in R$, the equality is $\sum_j X_{ij} = w_i \sum_j \mu_j = Kw_i$.

For distinct $i, j \in U$, the four slack-product expectations are

$$d, \quad c - d, \quad c - d, \quad 1 - 2c + d = \frac{(s - t)(s - t - 1)}{s(s - 1)},$$

all nonnegative. For $i = j \in U$ they are $c, 0, 0, 1 - c$. If at least one index is in R , the expectation factors into a nonnegative deterministic slack and a nonnegative expected slack; this also covers repeated restricted indices and degenerate intervals. Finally $\mu_i - X_{ii} = 0$ on U , and on R it is $w_i(1 - w_i)$. $\square$

Theorem 12.6 (SDP–RLT cover bound). *Let $n = k + m + z$, where $k, m, z \geq 1$ are integers, and let $0 \leq \epsilon < 1/4$. Every cover for (12.1) certified by (12.9)–(12.10) at $T_* = 1/4 - \epsilon$ has at least $\mathcal{N}(q)$ members, where $q = \min\{k, z, m(1/2 - 2\epsilon)\}$. In the balanced case $k = m = z = t$, this is the chord lower bound in Theorem 12.2.*

Proof. If a containing box has $|R| < q$, Lemma 12.5 gives a feasible objective at most $|R|p(1 - p) < q/(2m) \leq T_*$, where the strict comparison holds also when $R = \emptyset$ because $q > 0$. The box cannot be certified. Therefore $|R| \geq q$ on every certified box containing a witness, and Lemma 12.1 applies. The diagonal mixed inequality gives $X_{ii} \leq (a_i + b_i)\mu_i - a_i b_i$, so the SDP bound dominates the chord bound. The upper certificates of Proposition 12.3 remain valid. $\square$

This resolves the behavior under spatial branching for this SDP–RLT model; it is not merely a calculation of its root value.

One can also grant all products of two affine inequalities valid on $B \cap \mathcal{F}_{n,k}$ without strengthening this oracle. Indeed, linear-programming duality represents any such nonnegative affine function, on a nonempty polytope, as a nonnegative constant plus a nonnegative combination of box slacks plus a multiple of $e(x) = \sum_i x_i - K$. The representation is an identity of affine polynomials, including for degenerate boxes. Expanding the product of two representations leaves only existing nonnegative slack moments and terms $L[ev] = 0$ with v affine. This assertion concerns affine inequalities in the original x coordinates. It does not grant all valid affine cuts in the lifted (μ, X) coordinates.

13 Full preorderings and fractional-cardinality moments

An order- r oracle below acts on polynomials of total degree at most $2r$, where $r \geq 1$ is an integer. On a box $B = \prod_i [a_i, b_i] \subseteq [0, 1]^n$, it consists of linear functionals $L : \mathbb{R}[x]_{\leq 2r} \rightarrow \mathbb{R}$ satisfying

$$L[1] = 1, \tag{13.1}$$

$$L[ev] = 0 \quad (\deg v \leq 2r - 1), \quad e = \sum_i x_i - K, \tag{13.2}$$

$$L[gP^2] \geq 0 \quad \left(\sum_{j=1}^a \deg g_j + 2 \deg P \leq 2r, \quad g = \prod_{j=1}^a g_j \right), \tag{13.3}$$

where each g_j is a coordinate-bound slack $x_i - a_i$ or $b_i - x_i$. The empty product and repeated slack factors are included. The square polynomial P may couple every coordinate. Zero polynomials impose no condition; constants have degree zero. The node bound $\text{LB}_r(B)$ is the infimum of $L[F]$ over these functionals. This is the full truncated box preordering, including all allowed products of slacks, not just the separate single-generator localizers. Evaluation at a feasible point proves validity.

At $r = 1$ this model is exactly (12.9)–(12.10): positivity on affine squares is augmented PSD, products of at most two slacks give box RLT, single slacks give first-moment bounds, and the balance times affine polynomials gives its first and second moment equations. At every order $r \geq 1$ the model dominates that degree-two model and hence the chord oracle. Certification again means an infeasible region or $\text{LB}_r(B) \geq T_* = 1/4 - \epsilon$ for problem (12.1). Original dimension remains n .

13.1 A self-contained positivity proof

The moments below are the classical fractional-cardinality moments associated with Grigoriev's knapsack obstruction [22]. Potechin gives the explicit construction and its classical provenance [35, Theorem 1, Example 18, Theorem 44]. We prove the precise sufficient range required for the present full preordering; no external positivity theorem is needed.

For real v and integer $a \geq 0$, write $(v)_a = v(v-1)\cdots(v-a+1)$, with $(v)_0 = 1$. In s formal Boolean variables $u_1, \dots, u_s$, let red replace every positive power of u_i by u_i . This is the canonical multilinear representative modulo $u_i^2 - u_i$; it respects sums and products modulo these identities and cannot increase degree. For $u_S = \prod_{i \in S} u_i$ define

$$E_{s,t}[u_S] = \frac{(t)_{|S|}}{(s)_{|S|}}, \quad E_{s,t}[P] = E_{s,t}[\text{red } P]. \tag{13.4}$$

All subset sizes are at most s , so denominators are positive. The functional is defined on the Boolean quotient and is used only through the degree stated in each assertion; no positivity on all polynomials is assumed. In particular $E_{s,t}[1] = 1$. For $s = 0$, deterministic evaluation of constants will be used instead of a fractional-cardinality functional.

Lemma 13.1 (Cardinality identities and moment positivity). *Let $d \geq 1$ be an integer, $s \geq 2d$ an integer, and $t, s - t \geq 2d - 1$. Then*

$$E_{s,t} \left[\left(\sum_i u_i - t \right) v \right] = 0 \quad (\deg v \leq 2d - 1), \tag{13.5}$$

and $E_{s,t}[P^2] \geq 0$ for every $\deg P \leq d$.

Proof. For squarefree u_S of degree $a \leq 2d - 1 \leq s - 1$,

$$E_{s,t} \left[\left(\sum_i u_i \right) u_S \right] = a \frac{(t)_a}{(s)_a} + (s - a) \frac{(t)_{a+1}}{(s)_{a+1}} = t \frac{(t)_a}{(s)_a}.$$

Boolean reduction and linearity prove (13.5) for every allowed multiplier.

To reduce positivity to homogeneous polynomials, for $|S| = a \leq d$ put

$$H_S(u) = \frac{\sum_{T \supseteq S, |T|=d} u_T}{\binom{t-a}{d-a}}. \quad (13.6)$$

The denominator is positive under $t \geq 2d - 1$ (and is one for $a = d$). In the Boolean quotient the numerator equals $u_S \binom{\sum_i u_i - a}{d-a}$: at a Boolean vector it counts the d -subsets containing S when all coordinates of S equal one, and both sides vanish otherwise. The polynomial $\binom{z-a}{d-a} - \binom{t-a}{d-a}$ is divisible by $z - t$ with quotient degree at most $d - a - 1$ when $a < d$. Consequently

$$u_S - H_S = \left(\sum_i u_i - t \right) Q_S \quad \text{in the Boolean quotient,} \quad \deg Q_S \leq d - 1.$$

Replace each monomial of red P by its H_S , obtaining a homogeneous squarefree polynomial H of degree d . Then $P - H = (\sum_i u_i - t)Q$ modulo Boolean identities with $\deg Q \leq d - 1$. Since $\deg(Q(P + H)) \leq 2d - 1$, the cardinality identity gives $E_{s,t}[P^2] = E_{s,t}[H^2]$. Every expression used here has degree at most $2d$ before Boolean reduction.

Index the homogeneous moment matrix N by d -subsets I, J of $[s]$:

$$N(I, J) = \frac{(t)_{2d-|I \cap J|}}{(s)_{2d-|I \cap J|}}.$$

Let $P_j(I, J) = \binom{|I \cap J|}{j}$ for $0 \leq j \leq d$. This is a Gram matrix: associate with I the vector indexed by j -subsets A , with entry $\mathbf{1}\{A \subseteq I\}$. The exact decomposition is

$$N = \sum_{j=0}^d \frac{(t)_{2d-j}(s-t)_j}{(s)_{2d}} P_j. \quad (13.7)$$

Indeed, in an entry with $|I \cap J| = \ell$, the numerator on the right is

$$\begin{aligned} \sum_{j=0}^{\ell} \binom{\ell}{j} (t)_{2d-j}(s-t)_j &= (t)_{2d-\ell} \sum_{j=0}^{\ell} \binom{\ell}{j} (t-2d+\ell)_{\ell-j}(s-t)_j \\ &= (t)_{2d-\ell}(s-2d+\ell)_{\ell}. \end{aligned}$$

The last equality is the falling-factorial Vandermonde identity, obtained by counting ordered choices from two disjoint finite sets for nonnegative integer arguments and then by polynomial identity for real arguments. Division by $(s)_{2d}$ gives the claimed entry. No division by $(t)_{2d-\ell}$ was used, so zero coefficients cause no exception. All coefficients in (13.7) are nonnegative because $t, s - t \geq 2d - 1$, and $(s)_{2d} > 0$. Thus $N \succeq 0$ and $E_{s,t}[H^2] \geq 0$. $\square$

Lemma 13.2 (Assignment indicators and every slack product). *Suppose $r \geq 1$, $s \geq 2r$, and $t, s - t \geq 2r - 1$. For every product g of u_i and $1 - u_i$, including repetitions, and every polynomial P with $\deg g + 2 \deg P \leq 2r$, one has $E_{s,t}[gP^2] \geq 0$.*

Proof. Boolean reduction makes g zero if both u_i and $1 - u_i$ occur for some i . Otherwise it becomes an assignment indicator

$$I_{A,B} = \prod_{i \in A} u_i \prod_{j \in B} (1 - u_j), \quad A \cap B = \emptyset.$$

Put $a = |A|$, $b = |B|$, and $v = a + b \leq \deg g$. For a squarefree monomial u_C on the remaining coordinates, direct expansion gives

$$E_{s,t}[I_{A,B}u_C] = \frac{(t)_{a+|C|}(s-t)_b}{(s)_{v+|C|}} \quad (v + |C| \leq s). \quad (13.8)$$

One verification is induction on b : inserting an additional factor $1 - u_j$ subtracts the expression with j also fixed to one, and putting the two terms over their common denominator supplies the next factor $s - t - b$. The case $b = 0$ is (13.4). In particular the indicator weight is

$$\pi = E_{s,t}[I_{A,B}] = \frac{(t)_a(s-t)_b}{(s)_v} \geq 0.$$

If P is constant, the desired inequality is $\pi P^2 \geq 0$. If P is nonconstant, $v \leq \deg g \leq 2r - 2$, so all factors in this weight are strictly positive. Write $\hat{P} = P(1_A, 0_B, u_{[s] \setminus (A \cup B)})$. For $v < s$, cancellation in (13.8) proves

$$E_{s,t}[I_{A,B}P^2] = \pi E_{s-v,t-a}[\hat{P}^2]. \quad (13.9)$$

This identity is used only when its remaining moments are defined. If $\hat{P}$ is constant, its value is simply $\pi \hat{P}^2$; the same direct evaluation applies when $v = s$, without defining a zero-variable conditional functional. Otherwise let $d = \deg \hat{P} \geq 1$. Then

$$v + 2d \leq 2r, \quad s - v \geq 2d, \quad t - a \geq 2r - 1 - a \geq 2d - 1, \quad s - t - b \geq 2d - 1.$$

Lemma 13.1 applies to $E_{s-v,t-a}$. This proves positivity, including for arbitrary squares that couple all unassigned variables and for repeated original slacks. $\square$

The sufficient degree range in Lemma 13.1 is not the sharp classical range for the moment matrix alone. For the *full preordering* and this particular functional, the slack products explain the two-sided restriction: if $0 < t < s$ is noninteger, $s \geq 2r$, and $t < 2r - 1$, a product of $\lfloor t \rfloor + 2 \leq 2r$ distinct lower slacks has a negative falling-factorial moment. Applying the same argument to upper slacks treats $s - t < 2r - 1$. Integer t is different: (13.4) is then expectation under the uniform distribution of t -subsets. Sharper moment-matrix positivity alone therefore does not extend this full-preordering construction past the noninteger slack-product obstruction.

13.2 The order versus region tradeoff

Theorem 13.3 (Full-preordering spatial cover bound). *For problem (12.1), let $n = k + m + z$ with positive integers k, m, z , let $r \geq 1$ be an integer, and let $0 \leq \epsilon < 1/4$. Put*

$$q_r = \min\{k - 2r + 2, z - 2r + 2, m(1/2 - 2\epsilon)\}. \quad (13.10)$$

If $q_r > 0$, every box cover certified by (13.1)–(13.3) at $T_ = 1/4 - \epsilon$ has at least $\mathcal{N}(q_r)$ members. If $q_r \leq 0$, only the trivial bound of one region is asserted.*

Proof. Take a box containing a witness (12.4) and suppose $|R| < q_r$. Since $|R|$ and $k - 2r + 2$ are integers, $|R| < k - 2r + 2$ implies $|H \cap U| \geq k - |R| \geq 2r - 1$; likewise $|Z \cap U| \geq 2r - 1$. Set

$$s = |U|, \quad t = |H \cap U| + p|M \cap U|.$$

Then $t, s - t \geq 2r - 1$. Also $s \geq 2r$: for $r = 1$ a unit and a zero coordinate remain, and for $r \geq 2$ we have $s \geq 4r - 2 \geq 2r$. Define L by fixing $x_R = w_R$ and applying $E_{s,t}$ to $x_U = u$. This degree-preserving substitution is normalized. It changes e into $\sum_{i \in U} u_i - t$, whose products with every substituted multiplier of degree at most $2r - 1$ vanish by Lemma 13.1. Bound slacks on R are nonnegative constants, and those on U are Boolean slacks. Removing constant factors cannot increase degrees, so Lemma 13.2 proves every required positivity constraint.

The Boolean penalty on U vanishes, leaving

$$L[F] = |M \cap R|p(1 - p) \leq |R|p(1 - p) < \frac{q_r}{2m} \leq \frac{1}{4} - \epsilon. \quad (13.11)$$

The strict inequality follows from $|R| < q_r$ and $q_r > 0$, including $R = \emptyset$. This feasible functional precludes certification. Thus every certified box containing a witness has $|R| \geq q_r$, and Lemma 12.1 proves the bound. In particular non-strict pruning at the target is covered: the constructed value is strictly below it. No integer rounding of $q_r/2$ is necessary. $\square$

For balanced parts $k = m = z = t$, positivity of q_r is precisely $r < (t + 2)/2$ together with $\epsilon < 1/4$. A uniform exponential bound as $n = 3t \rightarrow \infty$ follows whenever $\epsilon < 1/4$ is fixed and $r \leq ct$ for a fixed $c < 1/2$. At $\epsilon = 1/8$, the exponent $n/24$ survives whenever

$$t - 2r + 2 \geq t/4, \quad \text{equivalently} \quad r \leq 3t/8 + 1.$$

An order approaching the positivity boundary can instead give only a constant q_r ; positivity alone is not an exponential-size assertion. When $\epsilon = 1/4$, the root has an available certificate because the mixed slacks give $L[F] \geq 0$. We do not divide by T_* or by a vanishing q_r in any of these cases. Increasing r may itself make the oracle costly; the theorem grants that oracle and counts regions only.

The affine-inequality closure following Theorem 12.6 extends to this degree. Expand a product of valid affine inequalities in their nonnegative box-slack and balance representations and multiply by an allowed square. Terms with a balance factor vanish within degree $2r$, and all remaining terms are existing preordering inequalities. This still grants no arbitrary coupled nonlinear valid cut, no localizer of an objective cutoff, and no analytic solution of the coupled problem. The charged-cover convention of Proposition 12.4 applies to objective-based spatial reductions.

13.3 Unique minimizers and increasing concave costs

Corollary 13.4 (Distinct perturbations). *Let $0 < d_1 < \dots < d_n$ and $\sum_i d_i \leq \eta$, where $\eta > 0$. On $\mathcal{F}_{n,k}$ replace F by $F_d = F + \sum_i d_i x_i$. Its unique optimizer has*

$$x_i^* = 1 \ (i \leq k), \quad x_{k+1}^* = 1/2, \quad x_i^* = 0 \ (i \geq k + 2), \quad (13.12)$$

with value $F_d^ = 1/4 + \sum_{i \leq k} d_i + d_{k+1}/2$. For the cover assertion take $n = k + m + z$ with positive integers k, m, z and integer $r \geq 1$, as in Theorem 13.3. Under the order- r box oracle, a cover at $T_* = F_d^* - \epsilon$ has at least $\mathcal{N}(q_{r,\eta})$ members provided $0 \leq \epsilon, \epsilon + \eta < 1/4$, and*

$$q_{r,\eta} = \min\{k - 2r + 2, z - 2r + 2, m(1/2 - 2\epsilon - 2\eta)\} > 0. \quad (13.13)$$

There is no nonidentity coordinate permutation preserving both feasible set and objective, even when objective equality is required only on the demand slice.

Proof. The Hessian of F_d is $-2I$, so a nonvertex, lying strictly between two distinct feasible points, cannot minimize this strictly concave function. All vertices have base value $1/4$. Among their linear costs, an exchange puts the largest coordinate value at the smallest coefficient: if $i < j$ and $x_i < x_j$, exchanging them reduces cost by $(d_j - d_i)(x_j - x_i) > 0$. Hence the unique minimizing vertex is (13.12). It also uniquely minimizes the linear cost over the whole slice: if $i < j$, $x_i < 1$, and $x_j > 0$, transferring $\min\{1 - x_i, x_j\} > 0$ from coordinate j to coordinate i strictly decreases that cost. At a linear-cost minimizer no such pair remains, so the balance forces exactly (13.12).

The node constraints themselves impose $0 \leq L[x_i] \leq 1$, since x_i and $1 - x_i$ are nonnegative combinations of node slacks and constants. The constructed functional therefore gives linear cost at most η . If $|R| < q_{r,\eta}$, its total value is strictly below

$$\frac{m(1/2 - 2\epsilon - 2\eta)}{2m} + \eta = 1/4 - \epsilon \leq F_d^* - \epsilon,$$

so the previous count applies.

The slice has an interior point relative to its balance hyperplane, namely $x_i = K/n \in (0, 1)$. A linear form constant on this slice has coefficient vector proportional to the all-ones vector. If a coordinate permutation preserves the objective there, the permuted coefficient vector differs from d by a constant vector. Comparing sums forces this constant to be zero, and distinctness forces the identity permutation. $\square$

An explicit choice is $d_i = 2\eta i / [n(n+1)]$, whose sum is η . For $n = 3t$, $\epsilon = 1/8$, and $\eta = 1/32$, the lower bound is $(3/2)^{n/32}$ whenever $r \leq 13t/32 + 1$. If $\eta < \epsilon$, the chord certificate at tolerance $\epsilon - \eta$ has base bound at least $1/4 + \eta - \epsilon \geq F_d^* - \epsilon$; the nonnegative linear term preserves it. Thus this explicit perturbed family again has $2^{\Theta(n)}$ certificate size under the specified oracles.

Adding the conserved flow gives increasing strictly concave coordinate costs:

$$C_d(x) = \sum_i [x_i(2 - x_i) + d_i x_i] = F_d(x) + K \quad (x \in \mathcal{F}_{n,k}). \quad (13.14)$$

Each derivative is $2 - 2x_i + d_i > 0$ on $[0, 1]$ and each second derivative is -2 . Every equality-preserving functional adds exactly K , so the optimizer and all absolute-gap conclusions are unchanged. When k is proportional to n , this raises the optimum to order n , while the unshifted obstruction has constant size. A fixed relative tolerance cannot be inferred from that absolute bound. Section 15 uses fixed-dimensional blocks to address precisely this distinction.

14 Arbitrary coordinate domains and local graph lifts

The endpoint argument uses membership, not interval geometry. We first strengthen the information available about each original coordinate, then show that adding arbitrarily many auxiliaries depending on that coordinate does not cost degree in this particular moment construction.

14.1 All valid local polynomial information

Let $S = \prod_i S_i$, where each S_i is an arbitrary nonempty subset of $[0, 1]$. Closedness, connectedness, semialgebraicity, and a finite description are not required. Define the order- r product-domain oracle by normalized functionals on $\mathbb{R}[x]_{\leq 2r}$, all balance products (13.2), and all inequalities

$$L \left[\left(\prod_{j=1}^a g_j \right) P^2 \right] \geq 0, \quad \sum_j \deg g_j + 2 \deg P \leq 2r, \quad (14.1)$$

where each g_j is any univariate polynomial nonnegative on its coordinate set S_i . The empty product is allowed, factors may repeat, and P may couple every coordinate. We also grant

$$L[hv] = 0 \quad (\deg h + \deg v \leq 2r) \quad (14.2)$$

for every univariate polynomial h vanishing on its S_i and every polynomial multiplier v in the original coordinates. In particular this model grants all local information through degree, not merely a chosen list of input generators. Its representation need not be finite or computationally efficient. The node bound is the infimum of $L[F]$ or $L[F_d]$, as appropriate. A certified cover has original feasible set $\mathcal{F}_{n,k}$, original dimension n , and target $1/4 - \epsilon$ or $F_d^* - \epsilon$ respectively.

Theorem 14.1 (Product-domain certificates). *Under the respective parameter assumptions of Theorem 13.3 and Corollary 13.4, the cover bounds $\mathcal{N}(q_r)$ and $\mathcal{N}(q_{r,\eta})$ hold for the product-domain oracle (14.1)–(14.2).*

Proof. For a domain containing a witness, use the exclusions (12.5). If $|R| < q_r$, or $|R| < q_{r,\eta}$ in the perturbed case, fix $x_R = w_R$ and use $E_{s,t}$ on U exactly as in Theorem 13.3. Its parameter and balance calculations are unchanged.

A local inequality on a restricted coordinate becomes a nonnegative constant because $w_i \in S_i$. For $i \in U$, both endpoints belong to S_i , and its Boolean reduction is

$$\text{red } g(u_i) = g(0)(1 - u_i) + g(1)u_i, \quad g(0), g(1) \geq 0. \quad (14.3)$$

Group factors belonging to the same coordinate and expand their product in endpoint indicators. After removing constant factors, let J be the coordinates with nonconstant reduced factors. The expansion has the form

$$\sum_{\sigma \in \{0,1\}^J} c_\sigma \prod_{i \in J: \sigma_i=1} u_i \prod_{i \in J: \sigma_i=0} (1 - u_i), \quad c_\sigma \geq 0.$$

Every such coordinate requires at least one original degree, so $|J| \leq \sum_j \deg g_j$. If $\hat{P}$ is the substituted square polynomial, $\deg \hat{P} \leq \deg P$ and $|J| + 2 \deg \hat{P} \leq 2r$. Lemma 13.2 gives nonnegativity of each indicator times $\hat{P}^2$, hence every inequality in (14.1). This includes repeated factors and squares coupling restricted and unrestricted coordinates.

A locally vanishing h becomes zero at w_i on R , or has zero values at both endpoints on U . Its Boolean reduction is zero. This remains true after multiplication by every allowed, possibly globally coupled, v , proving (14.2). The objective value and perturbation estimate are exactly (13.11) and Corollary 13.4. Thus certification forces the same restriction count, and Lemma 12.1 finishes the proof. $\square$

The constructed first moment need not itself belong to a disconnected S_i ; it is a moment satisfying the stipulated oracle. The proof uses only endpoint and witness membership. It invokes no compact-domain hull theorem and takes no closure of S_i . A branch on $\psi_i(x_i) \leq \beta$ versus $\psi_i(x_i) \geq \beta$ changes only S_i , for any real-valued univariate function ψ_i . In particular, branching on $x_i(1 - x_i)$, with its disconnected endpoint-side region, is covered. For an objective-based reduction $S_i \rightarrow T_i \subset S_i$, the removed product region has coordinate set $S_i \setminus T_i$. If nonempty and certified at the target, it can be charged directly under Proposition 12.4, without a closure convention.

14.2 Endpoint interpolation in lifted coordinates

For each original coordinate i , introduce finitely many auxiliaries

$$y_{i,j} = \psi_{i,j}(x_i), \quad j = 1, \dots, a_i, \quad \psi_{i,j} : [0, 1] \rightarrow \mathbb{R}. \quad (14.4)$$

Every function is defined and finite on the entire interval. Neither continuity nor polynomiality is required. Write $\Gamma_i(v) = (v, \psi_{i,1}(v), \dots, \psi_{i,a_i}(v))$. A node restricts each local graph independently to $\Gamma_i(S_i)$ for arbitrary nonempty $S_i \subseteq [0, 1]$. Thus restrictions in the lifted variables are measured by their exact preimages. Original coordinates x_i are retained and the original balances remain affine. The number of auxiliaries is finite for each instance but otherwise unrestricted.

The lifted order- r oracle acts on polynomials in all retained and auxiliary coordinates of total *lifted* degree at most $2r$. It imposes normalization; all multiples of the original affine balance through that degree; every local polynomial equality vanishing on $\Gamma_i(S_i)$ times every allowed global polynomial multiplier; and

$$L \left[\left(\prod_j g_j \right) P^2 \right] \geq 0, \quad \sum_j \deg g_j + 2 \deg P \leq 2r, \quad (14.5)$$

where each g_j is a polynomial in one local graph block, nonnegative on that block's restricted graph. Degrees in this display are measured before substitution. Squares can couple every lifted block, and factors may repeat. Graph equations are included whenever their lifted degrees allow them. Arbitrary coupled valid inequalities are not supplied.

Let $\tilde{F}(x, y)$ be a polynomial of lifted degree at most $2r$ that agrees exactly with $F(x)$, or with $F_d(x)$, on the *full* product graph $\prod_i \Gamma_i([0, 1])$. Agreement merely on the balance-feasible graph is not the stated assumption. The node bound is $\inf L[\tilde{F}]$. A graph-region cover means that its preimages cover the original compact set $\mathcal{F}_{n,k}$. Full-graph objective agreement preserves its optimum and an attaining optimizer, even if the lifted graph itself is noncompact because some $\psi_{i,j}$ is noncontinuous or unbounded near a point. Targets are still $T_* = 1/4 - \epsilon$ or $T_* = F_d^* - \epsilon$.

Theorem 14.2 (No order loss for coordinatewise graph lifts). *For the lifted model just defined, under the respective assumptions of Theorem 13.3 and Corollary 13.4, every certified cover has at least $\mathcal{N}(q_r)$ or $\mathcal{N}(q_{r,\eta})$ regions. The parameter is the lifted order r itself, independent of the number or polynomial degrees of the univariate auxiliaries. The dimension in the bound is the number n of original coordinates.*

Proof. Fix a node containing a graph witness $\Gamma(w)$ and form A, D, R, U from its preimage sets. Suppose $|R|$ is below the appropriate threshold. The cardinality parameters s, t satisfy the same hypotheses as before. Define an algebraic substitution Φ by

$$\begin{aligned} \Phi(x_i) &= w_i, & \Phi(y_{i,j}) &= \psi_{i,j}(w_i) & (i \in R), \\ \Phi(x_i) &= u_i, & \Phi(y_{i,j}) &= \psi_{i,j}(0) + (\psi_{i,j}(1) - \psi_{i,j}(0))u_i & (i \in U). \end{aligned} \quad (14.6)$$

Every image is affine or constant, so $\deg \Phi(P) \leq \deg P$ for every lifted polynomial. Define

$$L[P] = E_{s,t}[\Phi(P)] \quad (\deg P \leq 2r).$$

This is well-defined and normalized. The original affine balance maps to $\sum_{i \in U} u_i - t$, and any allowed multiplier maps to degree at most $2r - 1$. Lemma 13.1 proves all balance products.

A local valid inequality g maps to a nonnegative constant on R . For $i \in U$, the two values of $\Phi(g)$ are $g(\Gamma_i(0))$ and $g(\Gamma_i(1))$. Both are nonnegative because both endpoint graph tuples belong to

the restricted local graph. Thus its Boolean reduction has the nonnegative endpoint form (14.3). Grouping and expanding all local factors gives nonnegative combinations of assignment indicators on at most $\sum_j \deg g_j$ distinct Boolean coordinates. Moreover $\deg \Phi(P) \leq \deg P$, so each indicator times $\Phi(P)^2$ lies in the degree range of Lemma 13.2. This proves every inequality (14.5), including squares coupling all original and auxiliary variables.

For a local equality h , $\Phi(h)$ vanishes at the deterministic witness tuple on R or at both endpoint graph tuples on U . Its Boolean reduction is zero. For any allowed multiplier v , $\text{red}(\Phi(h)\Phi(v)) = 0$, and the degree bound ensures $L[hv]$ is defined. Thus every granted local equality product holds. The same reasoning would also respect any polynomial identity valid on the full product graph, even a coupled identity, and its allowed multiples: at every Boolean assignment the image is an actual graph point. This observation does not grant coupled inequalities that use the balances or objective cutoff.

At each Boolean assignment of u_U , (14.6) is an actual full-graph tuple with $x_R = w_R$. Therefore $\Phi(\tilde{F})$ agrees at every Boolean point with the polynomial obtained by substituting $x_R = w_R, x_U = u$ in F or F_d . Two polynomials agreeing on the Boolean cube have the same multilinear reduction: successively evaluating at zero and one in each coordinate proves uniqueness of that representation. Both polynomials have degree at most $2r$. Their functional values are consequently identical. The penalty is $|M \cap R|p(1-p)$ and any perturbation contributes at most η . The strict below-target estimate and the endpoint count from the preceding proofs finish the theorem. $\square$

The substitution Φ constructs a moment functional. It is not an identity $\psi_i(x_i) = \psi_i(0) + (\psi_i(1) - \psi_i(0))x_i$ for continuous x_i . The proof checks true graph identities only at deterministic or Boolean endpoint tuples. For example, it covers a square-root auxiliary $y_i = \sqrt{x_i}$ with its polynomial identity $y_i^2 = x_i$, and a penalty auxiliary $y_i = x_i(1 - x_i)$ with linear objective $\sum_i y_i$. Both maps are finite on all of $[0, 1]$ and their objective agreement, when used to encode the intended objective, is checked on the full graph. It also covers finite-valued nonpolynomial maps that have few polynomial identities; the oracle still grants every local polynomial inequality valid on the restricted graph.

The theorem requires independent local restrictions and retained original coordinates. Eliminating x_i through a nonlinear inverse, branching on functions of several original coordinates, granting arbitrary coupled valid cuts, or replacing the oracle by the exact hull of the coupled feasible graph changes the model. Arbitrarily many auxiliaries also mean that an exponential bound in original dimension need not be exponential in lifted dimension.

Proposition 14.3 (Literal polynomial substitution as a coarser bound). *If all $\psi_{i,j}$ are polynomials of degree at most an integer $D \geq 1$, with the originals counted as degree one, literal graph substitution also proves the lifted cover bounds with effective order rD in place of r . This is a valid weaker conclusion of the same type.*

Proof. Let Ψ replace each auxiliary by its defining polynomial and fix no coordinates yet. Then $\deg \Psi(P) \leq D \deg P$. Take the product-domain functional of order rD from Theorem 14.1 and set $\tilde{L}[P] = L[\Psi(P)]$. A lifted local inequality pulls back to a univariate polynomial valid on S_i , and

$$\sum_j \deg \Psi(g_j) + 2 \deg \Psi(P) \leq D \left(\sum_j \deg g_j + 2 \deg P \right) \leq 2rD.$$

Thus every localizer holds; local equalities pull back to equalities vanishing on S_i , and graph identities vanish identically. The original balance has degree one after substitution, so for a lifted multiplier v of degree at most $2r - 1$,

$$\deg(e\Psi(v)) \leq 1 + D(2r - 1) \leq 2rD.$$

All equality products are therefore available. Objective agreement on the full graph means the substituted objective equals F or F_d on $[0, 1]^n$. Since both are polynomials and the cube has interior, this is a polynomial identity. Hence its evaluated objective is the required one. The product-domain count at order rD applies. $\square$

For quadratic penalty auxiliaries the literal argument uses $\min\{k - 4r + 2, z - 4r + 2, m(1/2 - 2\epsilon)\}$. Theorem 14.2 improves this to q_r by using endpoint interpolation instead of literal substitution. It is this specific construction, not invariance of polynomial hierarchies under arbitrary nonlinear reformulations, that removes the degree loss.

15 Relative tolerance, tensor blocks, and escape cuts

A fixed relative-gap obstruction for increasing concave costs follows by using many independent blocks of fixed size. The node oracle is still global: its square multipliers can couple variables from every block. The independence is a feature of the optimization problem, and also gives a short alternative certificate outside the single-cover model.

Fix integers $r \geq 1$, $t \geq 2r - 1$, and $G \geq 1$. There are G blocks, each with $\ell = 3t$ original coordinates and its own balance:

$$\begin{aligned} n &= 3tG, & K &= t + \frac{1}{2}, & \sum_{i=1}^{3t} x_{b,i} &= K \quad (b = 1, \dots, G), & 0 \leq x_{b,i} \leq 1, \\ C(x) &= \sum_{b=1}^G \sum_{i=1}^{3t} [x_{b,i}(2 - x_{b,i}) + d_{b,i}x_{b,i}]. \end{aligned} \tag{15.1}$$

Assume $d_{b,1} < \dots < d_{b,3t}$ are positive and $\sum_i d_{b,i} \leq \eta$ for every block, with $\eta > 0$. Every coordinate cost is strictly increasing and strictly concave, by its derivatives $2 - 2x + d_{b,i} > 0$ and -2 . Corollary 13.4 applied blockwise gives the exact optimum

$$C^* = G(K + \frac{1}{4}) + \sum_{b=1}^G \left(\sum_{i \leq t} d_{b,i} + \frac{1}{2}d_{b,t+1} \right) \geq G(K + \frac{1}{4}). \tag{15.2}$$

On an arbitrary coordinate product domain, grant the order- r oracle (14.1)–(14.2) with all G balance equalities and every polynomial multiplier through *total* degree $2r$. Squares may involve all n variables. This includes the box preordering and, for boxes at $r = 1$, full global SDP–RLT with every balance product. Let $\text{LB}_r(S)$ be the infimum of $L[C]$. A relative certificate at tolerance $0 < \theta < 1$ is a finite cover of the feasible set by regions that are infeasible or satisfy

$$\text{LB}_r(S) \geq T_* = (1 - \theta)C^*. \tag{15.3}$$

Pruning at $(1 - \theta)U$ for any feasible incumbent $U \geq C^*$ implies this condition. The dimension in all ensuing bounds is original $n = 3tG$.

Lemma 15.1 (Tensor positivity with a total-degree truncation). *For each block b , suppose L_b is a normalized functional on block polynomials through degree $2r$, satisfying the block balance products, local equalities, and every block preordering inequality. Define L on global monomials of degree at most $2r$ by*

$$L \left[\prod_b x_b^{\alpha_b} \right] = \prod_b L_b[x_b^{\alpha_b}], \tag{15.4}$$

and extend linearly. Then L satisfies the corresponding global oracle, including every globally coupled square multiplier. The same assertion holds when a block contains coordinatewise graph auxiliaries and degree is measured in the lifted variables.

Proof. Each block degree is at most the global degree, so the definition is well-defined and normalized. To test a balance equality from block b times a global monomial, factor the latter across blocks. Its expectation is the block- b balance product, of degree at most $2r$, times moments of the other blocks, so it vanishes. Linearity treats every multiplier. The argument for local equalities is identical and also covers products involving several equalities within the total degree bound.

For a preordering expression let $g = \prod_b g_b$ group all local inequality factors according to their block, and let $d = \deg P$ with $\sum_b \deg g_b + 2d \leq 2r$. For every b , form the matrix

$$M_b(\alpha, \beta) = L_b[g_b x_b^\alpha x_b^\beta], \quad |\alpha|, |\beta| \leq d.$$

It is defined and positive semidefinite: the degree bound $\deg g_b + 2d \leq 2r$ allows the block preordering inequality for every polynomial in this index span. Their tensor product is positive semidefinite, as follows for example by tensoring Gram representations. Keep only tuples of indices whose total degree across blocks is at most d . The corresponding principal submatrix has entries exactly $L[gx^\alpha x^\beta]$ by (15.4); each such entry has available global degree at most $2r$. Its quadratic form on the coefficient vector of P is $L[gP^2] \geq 0$. The unused entries of the larger tensor matrix need not represent globally available moments: it is only a Gram construction whose required principal submatrix has the stated total-degree interpretation. Nothing in this proof uses that the coordinates within a block were unlifted. $\square$

Theorem 15.2 (A fixed relative-gap cover bound). *For (15.1) and the global product-domain oracle above, set*

$$q_0 = t - 2r + 2 \geq 1, \quad \tau = \frac{1}{4} - \theta(K + \frac{1}{4}) - \eta. \quad (15.5)$$

If $\tau > 0$, every certificate at target (15.3) has at least

$$(3/2)^{q_0 G \tau} = (3/2)^{q_0 \tau n / (3t)} \quad (15.6)$$

regions. For fixed r, t, θ, η satisfying these conditions, this is exponential in n . If $\tau \leq 0$, only the trivial bound one is asserted. The same conclusions hold for coordinatewise graph lifts satisfying the local oracle and full-graph objective-agreement assumptions of Theorem 14.2, retaining all original balances; there is no auxiliary-degree loss in r .

Proof. Choose independently in each block a uniform ordered partition (H_b, M_b, Z_b) into three sets of size t , and set $w_b = \mathbf{1}_{H_b} + p\mathbf{1}_{M_b}$ with $p = 1/(2t)$. Every combined witness is feasible. Fix a product domain containing one and define A_b, D_b, R_b by endpoint exclusion within block b . If $|R_b| < q_0$, the construction of Theorem 14.1 gives a feasible block functional whose penalty is at most

$$|R_b|p(1-p) \leq \frac{|R_b|}{2t} \leq \frac{|R_b|}{2q_0},$$

since $1 \leq q_0 \leq t$. No absolute pruning target is needed to construct this functional: only the endpoint restriction count is used. If $|R_b| \geq q_0$, instead use exact evaluation at the actual block witness. Its penalty is $tp(1-p) = 1/2 - 1/(4t) \leq 1/2 \leq |R_b|/(2q_0)$. Thus in either case the block functional is feasible through degree $2r$, its first moments lie in $[0, 1]$, and its penalty has the same upper bound. Lemma 15.1 gives a feasible global functional of objective at most

$$GK + \frac{1}{2q_0} \sum_b |R_b| + G\eta. \quad (15.7)$$

A certified region containing this witness must therefore satisfy

$$GK + \frac{1}{2q_0} \sum_b |R_b| + G\eta \geq \text{LB}_r(S) \geq (1 - \theta)C^* \geq (1 - \theta)G(K + \frac{1}{4}).$$

It follows that $\sum_b |R_b| \geq 2q_0G\tau$.

For a fixed domain, witness membership factors across blocks. In block b , endpoint avoidance bounds its witness fraction by

$$\min\{(2/3)^{|A_b|}, (2/3)^{|D_b|}\} \leq (2/3)^{|R_b|/2}.$$

Multiplying over independent blocks bounds the full witness fraction in a certified region by $(2/3)^{q_0G\tau}$. The union bound over a cover proves (15.6). The restriction counts belong to the region, not to the selected witness, so this is a uniform bound for every certified region with at least one witness.

For lifted regions work with the exact original-coordinate preimages. In a lightly restricted block use the affine endpoint substitution (14.6); in a heavily restricted block use evaluation at its true graph witness. These block functionals satisfy all lifted localizers and equality products by the proof of Theorem 14.2, with the original order r . Lemma 15.1 proves every global lifted localizer, including squares coupling auxiliaries from all blocks. It also proves all balance and local equality products.

For completeness, a lifted polynomial objective agreeing with C on the full product graph need not be written as a sum of block polynomials. After the block substitutions, it agrees with the substituted C at every Boolean assignment of all remaining formal variables; the heavily restricted blocks are actual fixed graph tuples. Affine substitution preserves its total degree at most $2r$, and uniqueness of Boolean reduction shows the two polynomials have the same value under the tensor functional. That functional annihilates Boolean identities in every remaining coordinate because each factor does. Hence (15.7) still holds, and the preimage witness count is identical. This proves the lifted assertion without assuming blockwise separability of the written lifted objective. $\square$

For the explicit parameters

$$r = 1, \quad t = 2, \quad \theta = 1/32, \quad \eta = 1/32, \quad K = 5/2, \quad q_0 = 2, \quad \tau = 17/128, \quad (15.8)$$

the lower bound is $(3/2)^{17n/384}$, even for full global SDP–RLT. For any other fixed order choose fixed $t \geq 2r - 1$, $0 < \eta < 1/4$, and $0 < \theta < (1/4 - \eta)/(K + 1/4)$. There is no assertion of a common positive relative tolerance while r grows without bound in this fixed-block construction.

Proposition 15.3 (Uniqueness and symmetry modulo the balances). *Problem (15.1) has a unique optimizer: in every block its first t coordinates equal one, coordinate $t + 1$ equals $1/2$, and the rest equal zero. For the explicit choice*

$$d_{b,i} = \frac{\eta((b-1)3t+i)^2}{3tn^2}, \quad b \in [G], \quad i \in [3t], \quad (15.9)$$

no nonidentity permutation of the original coordinates preserves both the feasible set and the objective restricted to that set.

Proof. Each independent block has the unique optimizer from Corollary 13.4; their concatenation is therefore the unique global optimizer. The coefficients in (15.9) are positive, globally distinct, and increasing in each block. Each is at most $\eta/(3t)$, so its block sum is at most η as required.

The point with every coordinate $K/(3t)$ is strictly between zero and one. Thus the affine hull of the feasible set is exactly the displayed balance space. Its normal row space consists of vectors constant on each block. A feasible-set-preserving coordinate permutation preserves this row space. The image of a block indicator is a zero-one vector in it with precisely $3t$ ones, so it is the indicator of one entire block. Hence such a permutation maps whole blocks to whole blocks.

The unperturbed quadratic and linear parts are invariant under every coordinate permutation. If the perturbed objective is preserved on the feasible set, its coefficient difference is constant within each block; this necessary condition follows by varying coordinates within their balance hyperplane. Consequently the coefficient multiset of each block must equal the coefficient multiset of its image block up to translation. Put $c = \eta/(3tn^2)$ and $\ell = 3t$. The sorted successive differences in block b are

$$c(2((b-1)\ell + i) + 1), \quad i = 1, \dots, \ell - 1.$$

Their first entries distinguish all blocks. Translation preserves these differences, so every block is fixed. Within a fixed block, comparison of the minimum shows that a finite coefficient set cannot equal a nonzero translate of itself. The translation is zero, and its distinct entries force the identity permutation. $\square$

Distinctness alone would not prove the statement modulo several balances: two blocks with coefficient lists differing by additive offsets could be exchanged without changing the objective on their equal-demand slices. The squared-index choice explicitly excludes this possibility. The symmetry assertion is about the original optimization problem; redundant auxiliary variables can of course introduce symmetries of their own.

15.1 A short alternative certificate and the scope of the obstruction

The blocks have fixed dimension. Solving each separately and summing its exact certified value gives a certificate of size proportional to G , if reuse and addition of component certificates is allowed. Even spatial certification of each separate block by Proposition 12.3 has fixed cost for fixed t . Theorem 15.2 counts regions in one global cover with the stated node oracle. It does not constrain a method that retains separate component certificates and adds their bounds. In this family the explicit sorted optimizer is an even simpler way to solve each component.

There is also a classical coupled cut that closes every root under study. Its low written degree does not mean that its nonnegativity follows from a low-degree local preordering.

Proposition 15.4 (A Boolean-quadric clique cut closes the root). *For integers $1 \leq k \leq n-1$, the inequality*

$$2 \sum_{i < j} X_{ij} - 2k \sum_i \mu_i + k(k+1) \geq 0 \tag{15.10}$$

is valid on the continuous quadratic graph $(\mu, X) = (x, xx^T)$ for $x \in [0, 1]^n$. Adding it to the root equality-product relaxation yields the exact lower bound $1/4$ for F , and the exact root bound for F_d and C_d . One such inequality per block makes the root bound for (15.1) exact.

Proof. This is Padberg's clique inequality [34, Lemma 2, equation (17)], with the whole coordinate set and parameter k , after multiplying by two and changing sides. Its continuous validity also has a direct proof. After substituting $X_{ij} = x_i x_j$, the left side is multiaffine. Successively choosing endpoints can only decrease its value, so its minimum on the cube is attained at a Boolean vertex. At a vertex with s ones its value is $(s-k)(s-k-1) \geq 0$, since $s-k$ is an integer.

The balance and its linearized products imply $\sum_{i,j} X_{ij} = K \sum_i \mu_i = K^2$. Equation (15.10) therefore gives

$$\sum_i X_{ii} \leq K^2 - 2kK + k(k+1) = k + \frac{1}{4}, \quad \sum_i (\mu_i - X_{ii}) \geq \frac{1}{4}.$$

For the perturbed objective, first moments meet the unit bounds and balance, and their linear cost is at least $\sum_{i \leq k} d_i + d_{k+1}/2$, the linear-program minimum proved in Corollary 13.4. Adding the two bounds gives F_d^* . The actual optimizer supplies an attaining exact moment matrix, so the root bound is exact. Adding the conserved K treats C_d , and applying the argument separately to all block balances treats C . The deduction does not use PSD. For the optional boundary case $k = 0$, the same polynomial is $2 \sum_{i < j} x_i x_j \geq 0$ and the algebra still applies directly. $\square$

Thus these easy-to-optimize families exhibit limits of particular spatial certificate systems, including full node preorderings at the specified orders, while known global cuts avoid the obstruction. In a penalty lift the coupled affine inequality $\sum_i y_i \geq 1/4$ would also close the unperturbed root; its validity is precisely the global conclusion, so it is not supplied as a local graph inequality. These escape certificates motivate seeking different families when stronger coupled cuts are granted. In particular, a later signed cubic XOR construction serves a different spatial purpose and must not be identified with the positive multilinear gap examples of the earlier sections.

16 Signed cubic certificates with exact quadratic moment information

The clique inequality in Proposition 15.4 closes the fractional-cardinality example. A signed cubic objective permits a different obstruction: all its first and second moments can be genuine moments while its three-variable correlations still give a false objective value. The Boolean moment construction used below is classical [36]. We prove its transfer to continuous coordinate regions, including the exact quadratic moment hull at each region.

Let E be a multiset of $m > 0$ clauses on $[n]$, each containing three *distinct* coordinates, with signs $b_e \in \{-1, 1\}$. Occurrences count multiplicity, and $\Delta = \max_i |\{e : i \in e\}| \geq 1$. Set

$$c_e(x) = \frac{1 - b_e \prod_{i \in e} x_i}{2}, \quad F(x) = \frac{1}{m} \sum_{e \in E} c_e(x), \quad x \in [-1, 1]^n. \quad (16.1)$$

Every c_e lies in $[0, 1]$ on the cube. Successively minimizing in each coordinate moves to a Boolean vertex without increasing F , so

$$\text{OPT} = \min_{[-1, 1]^n} F = \min_{w \in \{-1, 1\}^n} F(w). \quad (16.2)$$

Here clauses define the objective; the continuous problem does not impose perfect clause satisfaction as a feasibility constraint.

16.1 The node oracle and the general transfer

A Boolean pseudoexpectation through degree d is a linear functional $\mathcal{E}$ on $\mathbb{R}[x]_{\leq d}$ with $\mathcal{E}[1] = 1$, respecting $x_i^2 = 1$ through that degree and positive on P^2 whenever $2 \deg P \leq d$. Boolean reduction replaces a monomial by the character $\chi_A(x) = \prod_{i \in A} x_i$, where A consists of its odd exponents; $\chi_\emptyset = 1$. Respecting the Boolean identities means that a polynomial and its reduction have the same functional value.

For a nonempty coordinate product $B = \prod_i S_i \subseteq [-1, 1]^n$, define

$$\mathcal{Q}(B) = \text{conv}\{(x, xx^T) : x \in B\}. \quad (16.3)$$

Our order- r node functional L is defined through degree $2r$, normalized, and satisfies

$$L \left[\left(\prod_{j=1}^a g_j \right) P^2 \right] \geq 0, \quad \sum_j \deg g_j + 2 \deg P \leq 2r, \quad (16.4)$$

$$L[qv] = 0, \quad \deg q + \deg v \leq 2r, \quad (16.5)$$

$$((L[x_i])_i, (L[x_i x_j])_{ij}) \in \mathcal{Q}(B). \quad (16.6)$$

In (16.4), each g_j is any univariate polynomial nonnegative on its coordinate set. Products may repeat factors, and the empty product and globally coupled P are allowed. In (16.5), q is any univariate polynomial vanishing on its coordinate set, and v is arbitrary through degree. For boxes, the weaker oracle using just the two bound slacks per coordinate, with arbitrary repeated products, is also covered by all ensuing lower bounds. Let $\text{LB}_r(B) = \inf L[F]$, where $r \geq 2$ makes the cubic objective available. An empty feasible set of functionals has bound $+\infty$.

For compact boxes, (16.6) is equivalent to adding *linearly on moments* every quadratic polynomial inequality valid on the box: separation applies to the compact convex hull in (16.3). Products or square localizers of arbitrary coupled quadratic inequalities are not part of this grant. For nonclosed coordinate sets we impose actual hull membership, which can be stronger than all closed halfspace inequalities. No closure is taken.

Theorem 16.1 (Quadratic-hull cover transfer). *Let $r \geq 2$ be an integer. Suppose a Boolean pseudoexpectation $\mathcal{E}$ through degree $4r$ satisfies*

$$\mathcal{E}[\chi_e] = b_e \quad (e \in E), \quad (\mathcal{E}[x], \mathcal{E}[xx^T]) \in \text{conv}\{(w, ww^T) : w \in \{-1, 1\}^n\}. \quad (16.7)$$

Every finite cover of $[-1, 1]^n$ by coordinate products certified by (16.4)–(16.6) at a target $T_ > 0$ has at least*

$$2^{mT_*/\Delta} \quad (16.8)$$

regions. The coordinate sets may be disconnected, nonclosed, or have no finite polynomial description.

Proof. Use all 2^n Boolean vertices as equally weighted witnesses. For a domain B containing a witness w , define

$$R = \{i : \{-1, 1\} \not\subseteq S_i\}, \quad U = [n] \setminus R, \quad L[p] = \mathcal{E}[p(w_R, x_U)].$$

This fixes coordinates deterministically and then takes a marginal of $\mathcal{E}$. It is not conditioning on an event under $\mathcal{E}$ and requires no positive mass for the chosen signs. Substitution does not increase degree, and $L[1] = 1$.

First consider a box-slack product gP^2 . Slacks in R become nonnegative constants. In U , the interval contains both endpoints and therefore equals $[-1, 1]$. Each remaining slack is $1 + x_i$ or $1 - x_i$. Modulo Boolean identities, their product is zero if opposite signs occur on a coordinate, and otherwise is a nonnegative multiple of an assignment indicator I . Repeated factors change only the nonnegative multiplier. If $a = \deg g$ and $d = \deg P$, the indicator has degree at most a . Writing $\hat{P} = P(w_R, x_U)$, Boolean idempotence gives

$$I\hat{P}^2 = (I\hat{P})^2 \pmod{x_i^2 - 1}, \quad \deg(I\hat{P}) \leq a + d \leq 2r. \quad (16.9)$$

Both sides are within degree $4r$, so Boolean reduction and square positivity of $\mathcal{E}$ prove the inequality.

For the stronger local oracle, a factor on R is again a nonnegative constant. On U its Boolean reduction is

$$g(x_i) = g(1)\frac{1+x_i}{2} + g(-1)\frac{1-x_i}{2} \pmod{x_i^2-1}, \quad g(1), g(-1) \geq 0. \quad (16.10)$$

Remove constant factors and expand the product. Every summand is a nonnegative multiple of a product of endpoint indicators. The number of nonconstant factors is at most the original sum of generator degrees; hence its indicator degree is at most that sum. Applying (16.9) proves every allowed localizer. This argument includes arbitrary repeated factors. A local equality is zero at the fixed witness on R , or has both endpoint values zero on U ; its Boolean reduction, also after any allowed multiplier, vanishes.

Let μ be a Boolean distribution realizing the original moments through degree two. Marginalize μ on U and set its other coordinates to w_R . The resulting finite distribution is supported in B . Its means, diagonal moments, and cross moments are exactly those of L , proving (16.6) even for nonclosed sets.

An untouched clause keeps cost zero. At most $\Delta|R|$ clauses touch R , including multiplicity. After substitution each cost has the form $(1 - \sigma\chi_A)/2$ with $|A| \leq 3$. Since $\chi_A^2 = 1$,

$$0 \leq \mathcal{E}[(1 \pm \chi_A)^2] = 2(1 \pm \mathcal{E}[\chi_A])$$

proves its cost lies in $[0, 1]$; these squares have degree at most six, within $4r$. Thus

$$\text{LB}_r(B) \leq L[F] \leq \frac{\Delta|R|}{m}. \quad (16.11)$$

Certification forces $|R| \geq mT_*/\Delta$. Every coordinate in R permits exactly one Boolean endpoint when the domain contains a witness, so the witness fraction in B is $2^{-|R|}$ and is at most $2^{-mT_*/\Delta}$. Domains containing no witnesses have fraction zero. The union bound for any cover proves (16.8). $\square$

This result counts certified regions in the sense of Section 2.3. It permits arbitrary real coordinate split points and variable choices. A complete split tree has at least this many leaves; regions discarded by objective propagation must also be certified and charged. The theorem does not bound leaves remaining after free deletion of feasible regions.

16.2 Classical signed moments and an occurrence-controlled family

Lemma 16.2 (Realizing signed quadratic moments). *Let an augmented moment matrix indexed by $1, z_1, \dots, z_N$ be positive semidefinite, with all diagonal entries one and all entries in $\{0, -1, 1\}$. There is a probability distribution on $\{-1, 1\}^N$ with exactly its prescribed first and second moments.*

Proof. Represent the matrix as a Gram matrix of unit vectors $v_0, v_1, \dots, v_N$. Inner product 1 or -1 forces equality or opposition. Partition the vectors into classes up to sign and choose a representative for each. Distinct representatives are orthogonal because their inner products belong to $\{0, -1, 1\}$ and cannot have absolute value one. Choose v_0 as representative of its own class. Give this class the deterministic random value one; give every other class an independent unbiased sign. If $v_j = \sigma_j v_{C(j)}$, set $Z_j = \sigma_j \xi_{C(j)}$. A coordinate in the constant class has mean σ_j , and other coordinates have mean zero. Two coordinates in one class have product mean $\sigma_i \sigma_j$; different classes give product mean zero, including when one is the constant class. These are exactly all Gram entries, and the distribution has finite support. $\square$

We spell out the width convention of the source because two degree losses will occur in the spatial argument. The following is the signed-character construction in the proof of Schoenebeck's Lemma 13 [36, Theorems 11–12 and Lemma 13], written as a polynomial functional.

Lemma 16.3 (Width and character degree). *Suppose a signed XOR formula has no width- w XOR-resolution contradiction, where $w \geq 3$ is an integer and all initial clause widths are at most w . There is a normalized Boolean functional through degree w with every character moment in $\{0, -1, 1\}$, satisfying every clause in expectation and positive on squares of degree at most $\lfloor w/2 \rfloor$.*

Proof. In multiplicative notation, the resolution rule derives $\chi_{A \Delta B} = st$ from $\chi_A = s$ and $\chi_B = t$ when $|A \Delta B| \leq w$. Let $\mathcal{C}_w$ be its closure, including $\chi_\emptyset = 1$. Absence of contradiction means that a support can have at most one derived sign: opposite signs would derive $\chi_\emptyset = -1$. Define y_A for $|A| \leq w$ to be that sign when derived, and zero otherwise. In particular $y_\emptyset = 1$. Define $\mathcal{E}$ by Boolean reduction and $\mathcal{E}[\chi_A] = y_A$. This proves normalization, signed moments, and the clause means.

Index characters by supports of size at most $\ell = \lfloor w/2 \rfloor$. Declare $I \sim J$ if a sign for $I \Delta J$ is derived. This is reflexive and symmetric. For transitivity, multiply the relations for $I \Delta J$ and $J \Delta K$; the resulting support $I \Delta K$ has size at most $2\ell \leq w$. Thus the result is in the closure, with the product sign. Choose a representative I_0 per class and orient I by the derived sign s_I of $I \Delta I_0$. Multiplication of two such relations shows that $s_I s_J = y_{I \Delta J}$ within a class. Give distinct classes orthogonal unit vectors e_C and put $v_I = s_I e_{C(I)}$. Then for all $|I|, |J| \leq \ell$,

$$\langle v_I, v_J \rangle = y_{I \Delta J}. \quad (16.12)$$

For any polynomial P of degree at most ℓ , write its Boolean reduction as $\sum_I p_I \chi_I$. Equation (16.12) gives

$$\mathcal{E}[P^2] = \sum_{I,J} p_I p_J y_{I \Delta J} = \left\| \sum_I p_I v_I \right\|^2 \geq 0.$$

This proves positivity for arbitrary polynomial squares, including sums whose terms jointly involve more than ℓ variables. The constant character has a unit Gram vector, as normalization requires. $\square$

At density eight, Theorem 11 of Schoenebeck [36] gives a constant $a > 0$ such that width at least an is available with probability $1 - o(1)$, after reducing a to absorb integer rounding. Take the source parameters $k = 3$, $d = 8$, $\delta = \gamma = 1/4$, and its density exponent $\varepsilon_{\text{src}} = 0$. Theorem 12's density condition is $8 \geq 1 + 8 \log 2$, and the denominator $k/2 - \gamma - 1 = 1/4$ is positive. Theorem 11 supplies the positive width constant; Lemma 13 converts width w to level $\lfloor w/2 \rfloor$. Lemma 16.3 verifies directly that this means moments through degree w and square positivity through degree $\lfloor w/2 \rfloor$, not moments only through half the width. We need $4r \leq w$ below and $4rD \leq w$ for the lifts. This choice avoids the boundary $\gamma = 1/2$ in the source's illustrative three-variable specialization.

Theorem 16.4 (A bounded-occurrence cubic family). *There are absolute constants $a > 0$ and n_0 such that for every integer $n \geq n_0$ there is an objective (16.1) with*

$$7n \leq m \leq 8n, \quad \Delta \leq 64, \quad \text{OPT} \geq 1/8, \quad (16.13)$$

and a Boolean functional of available degree at least an as in Lemma 16.3. For every integer $r \geq 2$ with $4r \leq an$, any certificate at absolute tolerance $1/16$, or relative tolerance $1/2$, under the quadratic-hull node oracle has at least

$$2^{7n/1024} \quad (16.14)$$

regions. The same instance works simultaneously for all these orders.

Proof. Sample $m_0 = 8n$ independent clauses, each with a uniform three-element support and an independent uniform sign. This is the random-XOR model in Schoenebeck's definition: ordered distinct variables and independent literal negations induce the same support and parity-sign distribution. For a fixed Boolean assignment, its violation count V is $\text{Bin}(8n, 1/2)$, regardless of overlap among supports. For $t > 0$, the generating function and $\cosh u \leq e^{u^2/2}$ give

$$\mathbb{P}\{V \leq 2n\} \leq e^{-2nt} \mathbb{E} e^{-t(V-4n)} = e^{-2nt} \cosh(t/2)^{8n} \leq e^{-2nt+nt^2}.$$

The inequality for $\cosh$ follows by integrating $(\log \cosh u)' = \tanh u \leq u$ for $u \geq 0$ and using symmetry. Taking $t = 1$ and a union bound over all assignments bounds the probability of any assignment with at most $2n$ violations by $2^n e^{-n} = e^{-(1-\log 2)n} = o(1)$. Independently of needing this elementary bound, the source gives the signed functional with linear available degree with probability $1 - o(1)$.

Let D_i be the original number of clauses containing coordinate i ; $D_i \sim \text{Bin}(8n, 3/n)$. Delete every clause incident to a coordinate with $D_i > 64$. If Z is the number deleted, then $Z \leq \sum_i D_i \mathbf{1}_{D_i > 64}$. Differentiating the binomial generating polynomial yields

$$\begin{aligned} \mathbb{E}[D_i \mathbf{1}_{D_i > 64}] &\leq 2^{-64} \mathbb{E}[D_i 2^{D_i}] = \frac{48(1 + 3/n)^{8n-1}}{2^{64}} \\ &\leq \frac{48e^{24}}{2^{64}} < \frac{1}{8}. \end{aligned} \tag{16.15}$$

For example $e < 3$ makes the last inequality follow from $384 \cdot 3^{24} < 2^{64}$, an integer inequality. Hence $\mathbb{P}\{Z > n\} \leq \mathbb{E}Z/n < 1/8$ by Markov's inequality. No independence between different D_i is used. For every sufficiently large n , the sum of this failure bound and the two preceding $o(1)$ failure probabilities is less than one. Fix an instance in their intersection.

At least $7n$ clauses remain, all remaining occurrences are at most 64, and every assignment still violates at least n clauses. Equation (16.2) gives $\text{OPT} \geq n/m \geq 1/8$. Deleting clauses preserves the original functional, its positivity, all character moments, and the means of retained clauses. Its degree-two moments have an actual Boolean realization by Lemma 16.2. Apply Theorem 16.1 with any $4r \leq an$. At the best incumbent the absolute target is $\text{OPT} - 1/16 \geq 1/16$ and the relative target is $\text{OPT}/2 \geq 1/16$. Thus $mT_*/\Delta \geq 7n/(16 \cdot 64)$, proving the count. Restricting one source functional to lower degrees proves simultaneous validity. $\square$

Proposition 16.5 (Sensitivity and curvature). *Throughout $[-1, 1]^n$, the objective (16.1) satisfies*

$$|\partial_i F| \leq \frac{\Delta}{2m}, \quad \sum_{j \neq i} |\partial_i \partial_j F| \leq \frac{\Delta}{m}, \quad \|\nabla^2 F\|_2 \leq \frac{\Delta}{m}. \tag{16.16}$$

In the family of Theorem 16.4, the last bound is at most $64/(7n)$.

Proof. An incident clause contributes at most $1/(2m)$ to a first derivative and at most two mixed second derivatives of that magnitude. The diagonal second derivatives vanish. Summing absolute contributions gives the first two bounds, and the symmetric Hessian's spectral norm is at most its maximum absolute row sum. $\square$

These estimates record the normalization of this family. They do not assert a general implication from small curvature to large certificates.

17 Certificates after bounded monomial reformulation

The preceding transfer survives coordinate branching on bounded products. The number of auxiliary coordinates can be arbitrary; the count depends on the original variables represented by independent parity restrictions.

Consider a map $h : [-1, 1]^n \rightarrow [-1, 1]^N$ of the form

$$h_j(x) = \sigma_j \chi_{A_j}(x), \quad \sigma_j \in \{-1, 1\}, \quad 1 \leq |A_j| \leq D, \quad j = 1, \dots, N, \quad (17.1)$$

where $D \geq 1$ is an integer. Include every original coordinate as an identity coordinate. The nonempty supports are squarefree; repeated or overlapping supports and either sign are allowed, and N is finite but otherwise unrestricted. The continuous feasible set is the full graph $\Gamma_h = \{h(x) : x \in [-1, 1]^n\}$. Retaining originals makes its Boolean witnesses $h(w)$ distinct. Let $\Phi \in \mathbb{R}[z]_{\leq 2r}$ satisfy the exact polynomial identity

$$\Phi(h(x)) = F(x). \quad (17.2)$$

Agreement merely on Boolean points is not the hypothesis here.

For $B = \prod_{j=1}^N S'_j \subseteq [-1, 1]^N$, each S'_j nonempty, use the local oracle (16.4)–(16.6) in the z coordinates, with objective Φ . In addition grant

$$L[qv] = 0 \quad \text{if } q(h(x)) \equiv 0, \quad \deg q + \deg v \leq 2r. \quad (17.3)$$

Thus every polynomial graph identity through available degree, with every allowed multiplier, can be imposed. One may use just a particular generating set instead; the stronger version makes the lower bound more informative. A polynomial vanishing on the entire continuous graph has identically zero pullback because the original cube has nonempty interior. The quadratic requirement is

$$(L[z], L[zz^T]) \in \text{conv}\{(z, zz^T) : z \in B\}, \quad (17.4)$$

the hull of the entire coordinate domain. It is not the hull of $\Gamma_h \cap B$. The latter would already optimize a linear or quadratic objective exactly. These quadratic inequalities enter linearly on moments; the oracle does not close them under arbitrary products or localizers.

Theorem 17.1 (Bounded monomial transfer, including order one). *Let $r, D \geq 1$ be integers with $rD \geq 2$. Suppose a Boolean pseudoexpectation $\mathcal{E}$ is available through degree $4rD$, is positive on squares through degree $2rD$, satisfies $\mathcal{E}[\chi_e] = b_e$ for every clause, and has every character moment through its available degree in $\{0, -1, 1\}$. For any map (17.1) and objective (17.2), every finite cover of Γ_h by coordinate products certified by the preceding order- r oracle at $T_* > 0$ contains at least*

$$2^{mT_*/(\Delta D)} \quad (17.5)$$

regions. Arbitrary nonclosed coordinate sets and arbitrary numbers of repeated or overlapping supports are allowed.

Proof. Let B contain $h(w)$ for $w \in \{-1, 1\}^n$. Put

$$R = \{j : \{-1, 1\} \not\subseteq S'_j\}, \quad C = \bigcup_{j \in R} A_j, \quad U = [n] \setminus C, \quad \tilde{h}(x_U) = h(w_C, x_U). \quad (17.6)$$

Each coordinate in R is now fixed to its witnessed sign. An unrestricted coordinate becomes a signed character on U , possibly constant, and both of its endpoints belong to its coordinate domain. Set

$$L[p(z)] = \mathcal{E}[p(\tilde{h}(x_U))]. \quad (17.7)$$

Pullback increases degree by at most D , so L is defined through lifted degree $2r$ and is normalized. This again uses deterministic substitution followed by marginalization, with no conditioning step.

We verify all localizers, keeping the degree budget explicit. Suppose $\sum_j \deg g_j + 2 \deg P \leq 2r$, where every g_j is a locally valid univariate polynomial in a lifted coordinate. A factor on a restricted coordinate becomes a nonnegative constant. For any other coordinate, Boolean reduction of g_j after pullback equals

$$g_j(1) \frac{1 + \tilde{h}_{k_j}}{2} + g_j(-1) \frac{1 - \tilde{h}_{k_j}}{2}, \quad g_j(1), g_j(-1) \geq 0,$$

where k_j is the coordinate used by factor g_j ; these indices may repeat. A constant factor needs no indicator and is removed first. Expanding the nonconstant factors expresses the product as a sum of nonnegative coefficients times products I of parity indicators. Overlapping supports cause no difficulty: all these factors commute and are idempotent in the Boolean quotient, so $I^2 = I$, even when $I = 0$. With $a = \sum_j \deg g_j$ and $d = \deg P$,

$$\deg I \leq Da, \quad \deg \hat{P} \leq Dd, \quad \hat{P} = P \circ \tilde{h}, \quad \deg(I\hat{P}) \leq D(a + d) \leq 2rD. \quad (17.8)$$

Consequently $I\hat{P}^2 = (I\hat{P})^2$ modulo Boolean equations, within degree $4rD$. Original square positivity proves the localizer. In particular, this checks every repeated bound-slack product, not just individual bound multipliers. If a univariate equality vanishes on its coordinate set, it vanishes at the fixed value or at both permitted endpoints; its pulled-back Boolean reduction times any allowed multiplier is zero. All graph identities in (17.3) vanish as literal polynomials after pullback and substitution. Their multiplied pullbacks have degree at most $2rD$, so every stated equality is respected.

To verify (17.4), form the augmented first/second moment matrix of L . A lifted linear polynomial pulls back to degree at most $D \leq 2rD$, so this matrix is positive semidefinite. Every diagonal entry is one because $\tilde{h}_j^2 = 1$ after Boolean reduction. Every other entry is an original character moment, multiplied by a fixed sign, and therefore belongs to $\{0, -1, 1\}$. Lemma 16.2 provides a finite Boolean distribution on the lifted coordinates with exactly this matrix. For $j \in R$, its coordinate mean equals the fixed sign $h_j(w)$. A Boolean variable with mean 1 or -1 equals that sign almost surely. All other coordinates have both endpoints in their sets. Thus the realizing distribution is supported in B , including when the sets are nonclosed. It may leave Γ_h : the graph moment identities have already been proved for L , and no common graph-supported distribution realizing its higher moments is asserted.

Finally, (17.2) gives $L[\Phi] = \mathcal{E}[F(w_C, x_U)]$. Only clauses meeting C change their costs. Their number is at most $\Delta|C|$. As in the unlifted proof, each substituted clause is $(1 - \sigma\chi_A)/2$ with $|A| \leq 3$. The squares $(1 \pm \chi_A)^2$ have degree at most six; they are available because $2rD \geq 4 \geq 3$. Therefore all such pseudo-costs lie in $[0, 1]$, and

$$\text{LB}_r(B) \leq L[\Phi] \leq \frac{\Delta|C|}{m}. \quad (17.9)$$

Certification forces $|C| \geq mT_*/\Delta$.

For witness counting, write each original Boolean sign as $w_i = (-1)^{t_i}$ with $t_i \in \mathbb{F}_2$. Each restricted coordinate imposes one equation with row $\mathbf{1}_{A_j}$ and a right-hand side determined by σ_j and its permitted endpoint. Let A_R be this binary row matrix and $q = \text{rank}_{\mathbb{F}_2} A_R$. The system is consistent, since it contains w , and has exactly 2^{n-q} solutions. Every such solution meets all endpoint restrictions and all unrestricted sets contain both endpoints, so the witness fraction of B is exactly 2^{-q} .

Select q basis rows from the original rows of A_R . If a column is zero in each basis row, it is zero in their entire span, hence zero in every row. The union of all supports is therefore the union of the

basis-row supports. Each basis support has at most D elements, giving

$$|C| \leq Dq, \quad 2^{-q} \leq 2^{-|C|/D} \leq 2^{-mT_*/(\Delta D)}. \quad (17.10)$$

The union bound over the graph witnesses proves the theorem. Counting restricted lifted coordinates alone would not prove this bound: repeated parities can create arbitrarily many restrictions with rank one. $\square$

The proof needs $r \geq 1$ and $rD \geq 2$, together with availability of Φ through degree $2r$. It uses no separate lower bound $r \geq 2$. The unlifted cubic objective still requires $r \geq 2$ because it must be evaluated by the node functional. The all-character assumption is stronger than quadratic realizability in Theorem 16.1; a lifted pair of coordinates can represent a character on as many as $2D$ original variables. Lemma 16.3 supplies precisely the required signed moments whenever $4rD \leq w$.

Corollary 17.2 (A quadratic formulation with a linear objective). *For every sufficiently large original dimension n , the family of Theorem 16.4 has an equivalent continuous formulation with $N = n + 2m \leq 17n$ coordinates, $2m$ quadratic equalities, and a linear objective. At every integer order $r \geq 1$ with $12r \leq an$, certificates at absolute tolerance $1/16$ or relative tolerance $1/2$ require at least*

$$2^{7n/3072} \geq 2^{7N/52224} \quad (17.11)$$

coordinate regions under the oracle of Theorem 17.1. This includes branching on all pair and clause coordinates at order one.

Proof. Order the three distinct variables of each clause as (i, j, k) and introduce

$$u_e = x_i x_j, \quad v_e = u_e x_k, \quad \Phi(x, u, v) = \frac{1}{m} \sum_e \frac{1 - b_e v_e}{2}. \quad (17.12)$$

All coordinates have bounds $[-1, 1]$. For every original x , these equalities determine exactly $u_e = x_i x_j$ and $v_e = x_i x_j x_k$, which lie in the stated bounds. Conversely every feasible tuple has those values. Thus the feasible set is the continuous monomial graph, the objectives agree as polynomials after substitution, and the degree parameter is $D = 3$. Repeated clauses can have separate auxiliaries; neither support repetitions nor auxiliary occurrence counts alter the theorem. The source condition is $4rD = 12r \leq an$, which allows $r = 1$ for all sufficiently large n . Using $T_* \geq 1/16$, $m \geq 7n$, and $\Delta \leq 64$ in (17.5) gives the first bound. The identity $N = n + 2m \leq 17n$ gives the second. $\square$

Corollary 17.3 (Support degree versus region count). *For the same family, whenever $r \geq 1$, $rD \geq 2$, and $4rD \leq an$, the fixed-tolerance lower bound for degree- D monomial coordinates is*

$$2^{7n/(1024D)}. \quad (17.13)$$

Within this degree regime, a number of regions polynomial in original dimension n requires $D = \Omega(n/\log n)$. For fixed D , the lower bound is exponential in n even with arbitrarily many auxiliaries and node order up to a sufficiently small constant multiple of n .

Proof. Apply Theorem 17.1 with the family parameters. If the number of regions is at most Cn^k for constants C, k , taking base-two logarithms in (17.13) gives $7n/(1024D) \leq \log_2 C + k \log_2 n$. This is the asserted necessary condition. It is not a sufficient construction and makes no assertion outside $4rD \leq an$. $\square$

For an arbitrarily inflated N , the exponent is in the original dimension n , not automatically in N ; likewise “polynomial” in this corollary means polynomial in n . The endpoint interpolation of Theorem 14.2 concerned functions of one original coordinate at a time. Its absence of auxiliary-degree loss does not extend by that proof to these multivariable monomials. Here literal pullback and the parity-rank argument supply a different, explicit $4rD$ degree budget.

18 Finite upper certificates and the boundary of affine branching

The fixed-tolerance lower bounds have finite upper certificates of the same exponential order in dimension. We give two certificates because they use different information. The Bernstein proof uses the order-two preordering and polynomial graph identities. A separate order-one proof for the quadratic formulation uses the exact quadratic hull of the box.

Proposition 18.1 (Order-two Bernstein certificates). *For every objective (16.1) and $\epsilon > 0$, at most $\lceil 3/\epsilon \rceil^n$ boxes certify the target $\text{OPT} - \epsilon$ at order two. This holds both in the original variables and in the quadratic formulation (17.12). In the latter, only original coordinates need subdivision. The quadratic moment hull is unnecessary for this upper bound.*

Proof. Set $M = \lceil 3/\epsilon \rceil$ and partition each original interval into M equal intervals, of width $h = 2/M \leq 2\epsilon/3$. Fix a resulting box $B = \prod_i [a_i, a_i + h]$. Every clause cost has derivatives of magnitude at most $1/2$ in each of its three coordinates. Moving between points of its three-coordinate box one coordinate at a time changes its value by at most $3h/2 \leq \epsilon$. Let β_e be its minimum on that box. In particular

$$\frac{1}{m} \sum_e \beta_e \geq F(a) - 3h/2 \geq \text{OPT} - \epsilon. \quad (18.1)$$

For $i \in e$, write $\lambda_i^1 = (x_i - a_i)/h$ and $\lambda_i^0 = (a_i + h - x_i)/h$. Multiaffine interpolation is the exact polynomial identity

$$c_e(x) - \beta_e = \sum_{s \in \{0,1\}^e} (c_e(a_e + hs) - \beta_e) \prod_{i \in e} \lambda_i^{s_i}(x_i). \quad (18.2)$$

Each coefficient is nonnegative, and each basis product is a positive multiple of three coordinate-bound slacks. The minimum β_e is attained at a corner by multiaffinity. Thus the node preordering through degree four certifies $L[c_e] \geq \beta_e$. Averaging and (18.1) give the original upper certificate.

For the lifted formulation, keep every auxiliary interval $[-1, 1]$. The exact identity

$$v_e - x_i x_j x_k = (v_e - u_e x_k) + x_k(u_e - x_i x_j) \quad (18.3)$$

has graph multipliers of total degree at most three. Order two therefore implies $L[v_e] = L[x_i x_j x_k]$. The same Bernstein certificate in the original bound slacks proves $L[\Phi] \geq \text{OPT} - \epsilon$. There are M^n lifted boxes, and together they cover the full feasible graph. $\square$

Proposition 18.2 (Order-one certificates from the quadratic box hull). *The quadratic formulation (17.12) admits an order-one certificate at target $\text{OPT} - \epsilon$ with at most $\lceil 3/\epsilon \rceil^n$ lifted boxes under the exact quadratic-box-hull oracle, for every $\epsilon > 0$. This certificate uses only the original coordinate subdivisions and the two graph equations per clause imposed on moments.*

Proof. Use the same grid, with original lower corner a and width $h = 2/M$, and leave auxiliaries in $[-1, 1]$. Any feasible order-one functional has its first and second moments realized by an actual distribution μ on this entire lifted box. The graph moment equations give

$$L[u_e] = L[x_i x_j], \quad L[v_e] = L[u_e x_k]. \quad (18.4)$$

We use these equalities only in expectation; μ need not satisfy the graph equations pointwise. On the full box,

$$|x_i x_j - a_i a_j| \leq |x_i - a_i| |x_j| + |a_i| |x_j - a_j| \leq 2h, \quad |u_e(x_k - a_k)| \leq h.$$

Taking expectations, all of which are degree at most two, yields

$$\begin{aligned} |L[v_e] - a_i a_j a_k| &\leq |L[u_e x_k] - a_k L[u_e]| + |a_k| |L[u_e] - a_i a_j| \\ &\leq h + 2h = 3h. \end{aligned} \tag{18.5}$$

For either clause sign this gives $L[(1-b_e v_e)/2] \geq c_e(a) - 3h/2$. Averaging proves $L[\Phi] \geq F(a) - 3h/2 \geq \text{OPT} - \epsilon$. The inequality holds for every feasible node functional and hence for its infimum. The M^n boxes cover the graph.

There is also an explicit quadratic inequality certifying each step. Set

$$q_e = \frac{3h}{2} - \frac{b_e}{2} [u_e(x_k - a_k) + a_k(x_i x_j - a_i a_j)]. \tag{18.6}$$

The preceding pointwise bounds give $q_e \geq 0$ on the *full* lifted box. Moreover,

$$\frac{1 - b_e v_e}{2} - c_e(a) + \frac{3h}{2} = q_e - \frac{b_e}{2} [(v_e - u_e x_k) + a_k(u_e - x_i x_j)]. \tag{18.7}$$

All polynomials in this identity have lifted degree at most two. The quadratic hull gives $L[q_e] \geq 0$, and the graph moment equations kill the two remaining terms. Thus the proof is a certificate using a valid quadratic box inequality linearly, without any higher-degree localizer or graph-supported realizing distribution. $\square$

At $\epsilon = 1/16$, both propositions give 48^n regions. For the family with $\text{OPT} \geq 1/8$, these also certify relative tolerance $1/2$ with the best incumbent: $\text{OPT} - 1/16 \geq \text{OPT}/2$. Consequently the quadratic formulation has both an order-one lower bound $2^{7n/3072}$ and an order-one upper bound 48^n , for all sufficiently large n . These counts measure regions, with no claim that the exact quadratic-box-hull oracle is computationally tractable. For the original cubic objective the distinct order-two Bernstein certificate remains the applicable statement.

18.1 Two obstructions to the same proof under affine branching

The parity-rank count depends on restricted coordinates being signed products with bounded support. A balanced affine halfspace behaves differently even under an actual probability distribution.

Proposition 18.3 (A halfspace rejects the preserved quadratic moments). *For every $n \geq 1$, let $\mathcal{E}$ be uniform expectation on $\{-1, 1\}^n$ and put $S(x) = \sum_i x_i$. The first and second moments of $\mathcal{E}$ have no actual realization on*

$$H = \{x \in [-1, 1]^n : S(x) \geq 0\},$$

although H contains at least half the Boolean witnesses. Equivalently, adding the affine coordinate $z = S/n$ and restricting it to $[0, 1]$ rejects the preserved moments by the valid quadratic inequality $z^2 \leq z$.

Proof. The uniform moments give $\mathcal{E}[S] = 0$ and $\mathcal{E}[S^2] = n$. A distribution supported where $S \geq 0$ with mean $S = 0$ must have $S = 0$ almost surely, contradicting that second moment. In the affine coordinate the same contradiction is $\mathcal{E}[z] = 0$ and $\mathcal{E}[z^2] = 1/n > 0$. The involution $w \mapsto -w$ proves $\mathbb{P}\{S(w) \geq 0\} \geq 1/2$; for even n the boundary can add mass. $\square$

Proposition 18.4 (Substitution cost for a halfspace localizer). *Start from the same uniform Boolean expectation, fix an arbitrary coordinate set C to arbitrary signs w_C , and leave the others independently uniform. If the resulting functional L satisfies*

$$L[SP^2] \geq 0 \quad (1 + 2 \deg P \leq 2r)$$

for an integer $r \geq 1$, then necessarily

$$|C| \geq \min\{r - 1, \lceil n/2 \rceil\}. \quad (18.8)$$

Proof. Write $t = |C|$, $a = \sum_{i \in C} w_i$, and $s = n - t$. Suppose both $t < r - 1$ and $t < \lceil n/2 \rceil$. Then $t \leq r - 2$ and $s \geq t + 1$. If $a < 0$, $P = 1$ already violates the localizer. Otherwise a is an integer in $[0, t]$. Select $a + 1$ unrestricted coordinates, which is possible because $a + 1 \leq t + 1 \leq s$, and set

$$P(x) = \prod_{i \text{ selected}} \frac{1 - x_i}{2}.$$

Its degree is $a + 1 \leq r - 1$, so the localizer has permitted total degree. On its indicator event the selected variables sum to $-(a + 1)$, all other unrestricted variables have conditional mean zero, and the fixed ones sum to a . Boolean idempotence therefore gives the exact value

$$L[SP^2] = (a - (a + 1)) 2^{-(a+1)} = -2^{-(a+1)} < 0.$$

This is the contradiction. For $r = 1$ the conclusion is the stated trivial bound $t \geq 0$. $\square$

At order proportional to n , this halfspace requires linearly many substitutions in the displayed construction while retaining at least half the witnesses. Thus large substitution cost cannot itself imply exponentially small witness mass for general affine inequalities. These propositions obstruct this proof method. They do not prove that a different node functional is impossible, or that the constant-gap XOR objective has a short affine-branch certificate.

18.2 Comparison with other disjunctive certificate models

Disjunctive sum-of-squares certificates can keep polynomial degree fixed while increasing the number of regions. Ahmadi, Dash, Hua, and Stellato [2, Sections 2–4 and 6] develop such certificates and a spatial framework on simplicial cones intersected with the sphere, with a corresponding simplex method for matrix copositivity. Their Section 6 states termination of the algorithms at positive prescribed tolerance. A convergence result of that kind is compatible with exponentially many regions. The coordinate-product bounds here do not apply to all simplicial or algebraic disjunctions in that framework.

Stabbing Planes instead branches on an integer linear inequality $ax \geq b$ and its integer negation $ax \leq b - 1$. It may discard the intervening slab because the slab has no integer points [8, Section 1]. Continuous subdivision must account for those points and therefore has a different covering requirement. Beame et al. give quasipolynomial-size Stabbing Planes refutations of Tseitin formulas [8, Theorem 1]. Fleming et al. give quasipolynomial-size Cutting Planes refutations for CNF encodings of inconsistent finite-field linear systems, and size lower bounds for a coefficient-restricted Stabbing Planes subsystem [21, Theorems 1.3–1.4]. These source statements do not transfer a parity SOS obstruction directly to unrestricted affine branching, nor do they give a constant-gap certificate for our objective. A refutation of perfect Boolean satisfiability proves only that some clause fails. Its direct normalized objective consequence is $F \geq 1/m$ on Boolean vertices, and hence on the cube by multiaffinity; it does not establish $F \geq 1/16$ when m grows. Proving or disproving comparable constant-gap region bounds for a stronger affine-branching model requires an additional argument.

19 Representation and other computational resources

The preceding lower bounds specify what a node can certify. Other forms of difficulty require different statements. Global lift size counts cone factors or matrix order; integer precision counts discrete coordinates; bound propagation studies a limit or a sequence of local updates. None of these counts can be substituted for certified spatial regions.

Several short geometric comparisons also show why the representation must be fixed. Appendix F derives the four established point-packing RLT/SDP values, including symmetry-adjusted secants and attaining covariances. This benchmark illustrates the gain from moment consistency and the care needed when the underlying box changes. Appendix G gives exact scale and multiplicity hulls by classical disaggregation, with a retained common intensive variable. It separates this hull identity from cost separation, which requires additional structure. Appendix H gives a retained-domain counterexample to the universal nonexactness assertion in the published P-split theorem, a sufficient directional repair, and exact translated-ball projection and distance formulas. A rational change of coordinates can turn a growing gap into a bounded one; convexifying the exact auxiliary image can then be exact in aligned coordinates. These claims concern the stated epigraph model and retained domain.

Appendix I exhibits a stable correlation-polytope face inside the unit-capacity rank-one hull. Established correlation lower bounds transfer to exact conic lifts, while the explicit stability estimate transfers fine entrywise- ℓ_1 approximations at order m^{-2} . The fixed-block exponential bound and the unrestricted PSD superpolynomial bound are separate conclusions. Appendix J transfers least monotone fixed points to fair forward FBBT, proves PosSLP-hard constant-error approximation of the limiting lower bound, and gives a doubly exponential primitive-update example valid under every fair ordering. Its primitive contractor restriction is essential. Finally, Appendix K contrasts scalar and simultaneous-bilinear approximation accuracy with spatial certification. It records short midpoint and cover proofs, without extending the present paper to the broader theory of mixed-integer representability.

20 Consequences and open questions

The central distinction is whether locally admissible representations can be made simultaneous. For envelope comparisons this is a probability law with all the prescribed means. Positive coefficients permit common upper attainment and reduce the comparison to lower-envelope rounding; signs instead lead to cut ranges. Structural restrictions improve the available couplings, while positive physical lower bounds change the coefficients and may change the relevant incidence structure. In spatial certification, a pseudoexpectation can satisfy a node's prescribed moments and polynomial constraints without supplying a feasible graph distribution. Counting regions that preserve it requires both a degree calculation and a witness cover argument.

Several bounded refinements sharpen this picture. The fixed-ambient balanced orientation improves the positive-box upper bound by replacing two with $\beta_N < 2$, and the coefficient-regularity cardinality argument uses the same law under its stated coefficient ratios. The general-radix cutoff construction gives exact attainment beyond the simpler large-radix regime. Coordinatewise graph lifts admit endpoint interpolation without polynomial-degree loss, including everywhere-finite nonpolynomial coordinate functions. Bounded signed-monomial lifts admit the lower-bound transfer at order $r \geq 1$ when $rD \geq 2$, and the quadratic formulation has a separate order-one finite upper certificate. The unlifted cubic still requires $r \geq 2$. These completions retain their domain, oracle and full-graph agreement hypotheses.

The remaining questions are quantitative or concern stronger methods. The exact cubic constant lies between the certified lower value $1610000/743033$ and $31/12$; optimality of the three-law mixture does not determine it. The optimized harmonic upper asymptotic has no matching second-order lower term. For a fixed physical aspect ratio the interval $\max\{2, \rho_B\} \leq C_{\text{box}}(\rho_B) \leq \rho_B + 2$ is not closed by the large-aspect leading constant. Exact structural constants for incidence treewidth at least three remain open; the width-two coloring proof does not extend merely by increasing the number of colors. On unequal-aspect positive boxes even bipartite frequency-two exactness fails, and the bipartite supremum remains between $3/2$ and two.

For spatial methods, arbitrary affine branching, additional coupled-cut localizers, and convex hulls of feasible graphs exceed the proved oracle scope. The balanced-halfspace calculation identifies an obstruction to the preserved-moment proof under affine branching; it provides neither a fast affine tree nor a lower bound for all such trees. The examples therefore suggest precise questions about stronger local information, rather than a conclusion about every global optimization algorithm.

A Exhaustive verification of the finite signing constants

This appendix gives the finite computational proof used in Proposition 3.6. Label vertices $0, \dots, n-1$. For vertex signs t_i , switching replaces a_{ij} by $a'_{ij} = a_{ij}t_it_j$. The identity $Q_{a'}(s) = Q_a(t \cdot s)$ shows that switching preserves both extrema of Q , and hence R_V . Taking $t_0 = 1$ and $t_j = a_{0j}$ makes every coefficient incident to zero positive. This normalized representative is unique within each switching class: a switching that keeps those coefficients positive has $t_j = t_0$ for all j , so changes no edge. Thus it suffices to enumerate the $2^{\binom{n-1}{2}}$ choices of negative edges among vertices $1, \dots, n-1$. There are respectively 2, 8, 64, 1024, 32768 representatives for $n = 3, \dots, 7$.

Encode a cut by the subset on side one, always leaving vertex zero on side zero. Complementation changes no cut, so the 2^{n-1} such subsets include every cut. If a cut has c crossing edges, of which h are negative, its weight is exactly $c - 2h$. The following complete Python 3.10+ program computes the first row of Proposition 3.6 using unbounded integers. The operation `bit_count()` counts set bits in an integer; each free-edge mask records which edges cross the cut. No floating-point arithmetic or solver is used.

```
from itertools import combinations
def min_range(n):
    edges = list(combinations(range(n), 2))
    free = [(i, j) for i, j in edges if i != 0]
    cuts = []
    for side in range(1 << (n - 1)):
        side <= 1 # Vertex zero stays on side zero.
        cross = lambda i, j: ((side >> i) ^ (side >> j)) & 1
        size = sum(cross(i, j) for i, j in edges)
        mask = sum(cross(i, j) << k
                    for k, (i, j) in enumerate(free))
        cuts.append((size, mask))
    best = len(edges) + 1
    for negative in range(1 << len(free)):
        weights = [size - 2 * (negative & mask).bit_count()
                    for size, mask in cuts]
        best = min(best, max(weights) - min(weights))
    return best
result = [min_range(n) for n in range(2, 8)]
assert result == [1, 2, 4, 4, 5, 8]
```

```
print(result)
```

Every signed cut weight lies between $-\binom{n}{2}$ and $\binom{n}{2}$; its range is at most $\binom{n}{2}$, because two cut weights differ by a sum of at most that many coefficients in $\{-1, 1\}$. The initial value `len(edges) + 1` therefore exceeds every candidate range. The two loops cover all normalized signings and all cuts, so their minimum is exactly $\min_a R_V(a)$, rather than merely an upper bound supplied by the best witness found. This is an exhaustive finite proof, not an analytic formula for general n .

n	Negative edges; all other edges have coefficient +1	$\min Q$	$\max Q$
3	$\emptyset$	-1	3
4	$\emptyset$	-2	6
5	12, 14, 23	-4	4
6	14, 15, 23, 25, 34	-5	5
7	12, 13, 16, 23, 25, 34	-7	9

Table 4: Attaining full-center signings, with vertices numbered from zero. An entry ij denotes the unordered pair $\{i, j\}$.

The extrema in Table 4 can be checked by direct evaluation of Q on the 2^{n-1} vertex-sign vectors with $s_0 = 1$. They give $R_V = (\max Q - \min Q)/2$ equal to the computed minimum in each case. Extending the listed K_6 witness by vertex six and positive new edges gives $\min Q = -9$, $\max Q = 11$ at the full center, hence ratio 21/10; its original six-vertex face retains ratio three.

The accompanying script `verification/check_complete_signings.py` implements the same integer enumeration, records a histogram of all candidate ranges, and checks the displayed witnesses by direct evaluation of Q on every induced face. Running it with Python 3.10+ writes `verification/complete_signings.json`, including the witnesses, counts, exact rational ratios, and the extended six-vertex example.

B Distinct multiscale arguments and finite refinements

B.1 Arbitrary partitions and the dense predecessor

The exact dyadic formula in Theorem 4.2 relies on a simultaneous prefix-hitting construction. A separate logarithmic estimate works for arbitrary partitions in (4.5).

Proposition B.1 (Partition-independent estimate). *For the polynomial and means in (4.5), with any choice of equal-size level partitions,*

$$0 < H \leq 5 + \log_2(1 + (L - 1) \log 2), \quad T = L.$$

Proof. Use the quantile surrogate from (4.6), writing $r(t)$ for the nonincreasing failure-count quantile and $\delta = 2^{-L}$. The interval $0 < t \leq \delta$ contributes at most $\delta \sum_{j=1}^L 2^j < 2$, and $t > 1/2$ contributes zero. For $\delta < t \leq 1/2$, let l be the largest integer with $t \leq 2^{-l}$. Write $\log_2^+ v = \max\{0, \log_2 v\}$ for $v > 0$. Then $2^l \leq 1/t$, and for $r > 0$,

$$\sum_{j=1}^l \min(2^j, r) \leq r \left(3 + \log_2^+ \frac{1}{tr} \right). \quad (\text{B.1})$$

If $r \geq 2^l$, the sum is less than $2r$. If $r < 2$, it is lr and $l \leq 1 + \log_2(2^l/r)$. Otherwise, with $h = \lfloor \log_2 r \rfloor$, the first h capped terms sum to less than $2r$ and the others sum to $(l-h)r$; use $h > \log_2 r - 1$. This proves (B.1), interpreted as zero at $r = 0$.

Put $D = (\delta, 1/2)$ and $M_0 = \int_D r(t) dt \leq 1$. As $\log_2^+ v \leq \log(1+v)/\log 2$, the bound becomes

$$H \leq 2 + 3M_0 + \frac{1}{\log 2} \int_D r(t) \log \left(1 + \frac{1}{tr(t)} \right) dt.$$

For $M_0 > 0$, Jensen's inequality under the probability measure $r(t) dt/M_0$ gives

$$\int_D r(t) \log \left(1 + \frac{1}{tr(t)} \right) dt \leq M_0 \log \left(1 + \frac{(L-1) \log 2}{M_0} \right) \leq \log(1 + (L-1) \log 2).$$

The final step uses that $M \log(1+B/M)$ increases for $M > 0$, since $\log(1+s) \geq s/(1+s)$. At $M_0 = 0$ the integral is zero. This proves the upper bound. Independent rounding and common-threshold rounding have distinct expected polynomial values at these interior means, so $H > 0$. $\square$

A dense symmetric construction gives a different exact reduction. With $m = 2^L$, $u_j = 2^{-j}$, $k_j = mu_j$, define

$$\tilde{f}_L(a, z) = \sum_{j=1}^L \frac{1}{u_j \binom{m}{k_j}} \sum_{|K|=k_j} a_j \prod_{i \in K} z_i, \quad a_j = u_j, \quad z_i = 1 - 1/m. \quad (\text{B.2})$$

Again $T = \text{cav } \tilde{f}_L = L$. Conditional on r failed leaves, the fraction of size- k_j subsets hit is

$$g_j(r) = 1 - \frac{\binom{m-r}{k_j}}{\binom{m}{k_j}} \leq \min(1, u_j r).$$

The inequality is the union bound: a uniform size- k_j subset contains each specified failed leaf with probability $k_j/m = u_j$. Thus the deficiency is $\sum_j A_j g_j(R)/u_j$, bounded by the same surrogate $\sum_j A_j \min(2^j, R)$ and consequently by Proposition B.1.

In this dense model the hull gap itself is exactly the optimum of the finite rational program

$$\begin{aligned} \max \quad & \sum_{j=1}^L \sum_{r=0}^m z_{jr} g_j(r) / u_j \\ \text{subject to} \quad & p_r \geq 0, \quad 0 \leq z_{jr} \leq p_r, \\ & \sum_{r=0}^m p_r = 1, \quad \sum_{r=0}^m r p_r = 1, \quad \sum_{r=0}^m z_{jr} = u_j \quad (j = 1, \dots, L). \end{aligned} \quad (\text{B.3})$$

Every feasible binary law supplies $p_r = \mathbb{P}(R = r)$ and $z_{jr} = \mathbb{P}(A_j = 1, R = r)$. Conversely draw R with probabilities p_r , choose a uniform r -subset of failed leaves, and, conditional on $R = r$, choose anchors independently with probabilities z_{jr}/p_r when $p_r > 0$. These choices realize all means and the objective; zero-mass states can be ignored. Thus (B.3) is an exact reduction, not a formula for the sparse model. No floating-point values from this program are needed for the divergent bound.

B.2 Dyadic conditional-failure rounding and its tail lemma

The following proof uses finitely many dyadic scales rather than a harmonic kernel. Though its $O(\log d)$ bound is weaker, its tail argument is distinct.

Lemma B.2 (Tail integral). *For a finite collection $p_j \geq 0$, let $N(s) = |\{j : p_j \geq s\}|$ and $S = \sum_j p_j$. For $u > 0$,*

$$\int_0^u \min\{N(s), u/s\} ds \geq \min(u, S). \quad (\text{B.4})$$

Proof. If $S \leq u$, then every $p_j \leq u$ and $\int_0^u N(s) ds = S$. Monotonicity gives $sN(s) \leq \int_0^s N(t) dt \leq S \leq u$, so the integrand equals $N(s)$ throughout. If $S > u$, then $\int_0^u N(s) ds \geq u$: either a $p_j \geq u$, or that integral equals S . By continuity choose $v \leq u$ with $\int_0^v N(s) ds = u$. For $s \leq v$ the same monotonicity bound gives $sN(s) \leq u$. Integrating on $(0, v)$ therefore gives u . $\square$

Proposition B.3 (Dyadic degree bound). *Let $K = 1 + \lfloor \log_2(d-1) \rfloor$ for $d \geq 2$. Every positive multilinear polynomial of maximum degree at most d on a nonnegative box satisfies $T \leq 2(K+1)H/c_0$.*

Proof. For each scale $B \in \{1, 2, \dots, 2^{K-1}\}$ restricted to powers of two, draw a common uniform U . Low coordinates succeed when $U \leq x_i$. A high coordinate with failure mean $p > 0$ has $h = \min(Bp, 1)$ and, conditional on U , fails independently with probability p/h when $U \leq h$, and never otherwise. Coordinates with $p = 0$ never fail. The marginal failure is $h(p/h) = p$ and the probabilities are valid because $B \geq 1$.

For a one-low term with anchor mean $u > 0$, let $p_1, \dots, p_r$ be its high failure means. At $U = t \leq u$, each $p_j \geq t/B$ is active and has conditional failure probability at least $1/B$. Conditional independence and $1 - e^{-z} \geq c_0 \min(1, z)$ give deficiency

$$G_B \geq c_0 \int_0^u \min(1, N(t/B)/B) dt = c_0 \int_0^{u/B} \min(B, N(s)) ds.$$

For $0 < s \leq u$ with $N(s) > 0$, choose the largest power of two $B \leq \min(N(s), u/s)$. It is in the allowed set, since $N(s) \leq r \leq d-1$, and exceeds half this minimum. Therefore

$$\sum_B \mathbf{1}\{s \leq u/B\} \min(B, N(s)) \geq \frac{1}{2} \min(N(s), u/s).$$

Integrating and invoking Lemma B.2 yields $\sum_B G_B \geq c_0 \min(u, S)/2 = c_0 t_e/2$. Terms with $u = 0$ have zero gap. Independence supplies the same guarantee for terms with zero or at least two low coordinates by Lemma 2.3. The uniform mixture of the K scale laws and independence thus guarantees $c_0 t_e/(2(K+1))$ for every term. Sum and apply Proposition 2.4. $\square$

B.3 The simplest harmonic cutoff and its reciprocal correction

The general finite theorem permits $M = d-1$, so $\eta = 1$ and $\Lambda = 1 + \log(d-1)$. Let z_Λ solve $J_{\Lambda,1}(z_\Lambda) = z_\Lambda$ and put $\alpha = \Lambda z_\Lambda$, $\ell = \log \Lambda$. For $\ell \geq 4$,

$$z_\Lambda \geq \frac{\ell - \log \ell - 2}{\Lambda}. \quad (\text{B.5})$$

Indeed $A = \ell - \log \ell - 2 \geq 1/2$ and $A \leq \ell < \Lambda$. Equation (4.15) with $\eta = 1$ gives $\Lambda J(A/\Lambda) \geq \ell - \log \ell - 1 - 1/(2A) \geq A$. Monotonicity of $J(z) - z$ proves the finite bound.

The integrated linear upper bound and quadratic lower bound imply

$$\ell - \log \alpha - 1 - \frac{1}{2\alpha} \leq \alpha \leq \ell - \log \alpha - 1 + \frac{\alpha}{\Lambda}.$$

Together with (B.5), these show $\alpha/\ell \rightarrow 1$: once $\alpha \geq 1$, the upper bound gives $\alpha \leq \ell/(1 - 1/\Lambda)$, while the finite lower bound is $\ell - \log \ell - 2$. Substitution yields

$$\Lambda z_\Lambda = \log \Lambda - \log \log \Lambda - 1 + o(1). \quad (\text{B.6})$$

The constant -1 belongs to this fixed cutoff, whereas the tuned cutoff in (4.14) removes that constant after expressing the bound with numerator N .

There is also a reciprocal refinement. If $we^w = \Lambda/e$, then

$$\alpha = w - \frac{1}{2w} + O(w^{-2}) \quad (\Lambda \rightarrow \infty). \quad (\text{B.7})$$

To prove it, integrate the cubic Taylor bound for $1 - e^{-x}$ with $x = (1 - v)/(\Lambda v)$. After multiplication by Λ , the error beyond the quadratic term is at most

$$\Lambda \int_{\alpha/\Lambda}^1 \frac{x^3}{6} dv \leq \frac{1}{12\alpha^2}.$$

The exact quadratic integral is $\int_z^1 (1/v - 1)^2 dv = 1/z + 2 \log z - z$. Its omitted terms, and the linear endpoint term, are $O(\log \Lambda/\Lambda) = o(\alpha^{-2})$ because $\alpha \sim \ell$. Consequently

$$\alpha + \log \alpha = \log \Lambda - 1 - \frac{1}{2\alpha} + O(\alpha^{-2}).$$

Compare this with $w + \log w = \log \Lambda - 1$. Since $\alpha \sim w$, the mean value theorem for $t + \log t$ first gives $\alpha - w = O(1/w)$ and, after substitution, gives (B.7). This refines the scalar certificate only; it is not a second-order lower bound on R_d .

B.4 A second rational two-level parameter choice

A related finite variant illustrates that the two-level mechanism is not tied to the particular coefficients in Theorem 5.6. For real $a, c \in [0, 1]$, direct expansion gives

$$\begin{aligned} ac^2 + \frac{24}{25}a^2 - \left(\frac{41}{25}a + \frac{4}{5}c - \frac{68}{75} \right) &= \frac{24}{25} \left(a - \frac{41 - 25c^2}{48} \right)^2 \\ &+ \frac{(1 - c)(5c - 3)^2(5c + 11)}{480}. \end{aligned} \quad (\text{B.8})$$

Both terms on the right are nonnegative. The equality atoms $(1/3, 1)$ and $(2/3, 3/5)$, each with probability $1/2$, have mean $(1/2, 4/5)$ and scalar envelope value $83/150$. For $25 \mid m$, use groups of m variables with these two means and $f_m = AE_2(W) + (24m/25)E_2(U)$. At binary normalized counts,

$$\frac{2f_m}{m^3} = ac^2 + \frac{24}{25}a^2 - \frac{ac + (24/25)a}{m}.$$

The expected correction is at most $49/(50m)$ because $ac \leq a$. The scaled concave envelope and termwise gap are, respectively, $(49/50)(1 - 1/m)$ and $(22/25)(1 - 1/m)$; cubic terms have lower envelope $1/10$, while quadratic terms have lower envelope zero. The affine minorant gives scaled hull gap at most $49/50 - 83/150 = 32/75$. Conditional independent sampling from the equality atoms, as in Theorem 5.6, gives scaled hull gap at least $(32/75)(1 - 1/m)$. Thus

$$\frac{33}{16}(1 - 1/m) \leq \frac{T(f_m)}{H(f_m)} \leq \frac{33}{16}.$$

At $m = 50$ the lower certificate is $1617/800 > 2$. This variant has a smaller limiting ratio than the two-level family in the main text.

C Exact finite cubic certificates

The following certificates use success counts, so their verification does not enumerate all 2^{192} binary vectors. Within each group, every binary vector with the same count has the same polynomial value. The finite enumeration therefore covers every vertex, including every possible dependence pattern relevant to the envelope.

For the six-term form (5.9), let $\Phi_m(A, B, C)$ be its binary count polynomial as defined in the proof of Proposition 5.5. Table 5 gives integers D, b_A, b_B, b_C, b_0 such that

$$D\Phi_m(A, B, C) \geq b_AA + b_BB + b_CC + b_0 \quad (0 \leq A, B, C \leq m \text{ integers}). \quad (\text{C.1})$$

Table 6 gives attaining laws. Each listed weight is nonnegative, the weights sum to one, and their mean count vector is $(m/4, m/2, 3m/4)$. Uniform success subsets conditional on these counts therefore give the required singleton marginals. The checker below verifies the inequality at every integer count triple and equality of primal and dual expectations in rational arithmetic.

m	D	b_A	b_B	b_C	b_0
6	13	962	778	842	−4816
8	7	793	770	798	−5882
64	105	871710	900446	899046	−51743768

Table 5: Integer affine minorants for the three-group witnesses.

m	Count states (A, B, C)	Corresponding probabilities
6	$(1, 4, 4), (2, 1, 6), (6, 2, 1)$	$17/26, 4/13, 1/26$
8	$(1, 2, 8), (1, 5, 6), (8, 4, 2)$	$2/7, 4/7, 1/7$
64	$(4, 19, 64), (5, 19, 64), (16, 40, 43), (64, 31, 17)$	$241/735, 12/735, 419/735, 63/735$

Table 6: Primal laws attaining the affine bounds.

For $m = 8$, minimizing the residual in (C.1) over A, B for each $C = 0, \dots, 8$ gives

$$168, 42, 0, 30, 22, 0, 0, 16, 0.$$

The three primal states have values 405, 507, 734 respectively, with expected value $3572/7$. In all three examples the orbit sizes, upper values per unweighted monomial, and lower values per unweighted monomial are respectively

$$s = \left(\binom{m}{3}, m \binom{m}{2}, \binom{m}{2}, m^2, m^2, \binom{m}{2} \right),$$

$$u = (3/4, 1/2, 1/2, 1/4, 1/4, 1/4), \quad l = (1/4, 0, 0, 0, 0, 0).$$

Thus $\text{cav } f = \sum_i c_i s_i u_i$ and $\sum_e \text{vex } f_e = \sum_i c_i s_i l_i$. These formulas prove the remaining columns of Table 3. The exact hull gaps for dimensions 18, 24, 192 are $10411/52, 3225/7, 27562048/105$, respectively.

For the two-level $m = 16$ certificate (5.17), the integer residual is

$$2 \left[A \binom{C}{2} + 20 \binom{A}{2} \right] - 428A - 177C + 3372.$$

Its minimum over $C = 0, \dots, 16$ for successive $A = 0, \dots, 16$ is

$$540, 352, 204, 96, 28, 0, 9, 7, 0, \\ 0, 19, 63, 132, 235, 369, 540, 742.$$

At counts $(8, 12)$ the value and its affine bound both equal 1088.

For the smaller members of (5.13), write $\Phi_m(A, C) = A \binom{C}{2} + (5m/4) \binom{A}{2}$. Table 7 gives affine minorants valid at every integer pair $0 \leq A, C \leq m$. At the listed counts each minorant equals Φ_m . Independently choosing uniform success subsets of those fixed sizes in the two groups gives every singleton mean $1/2, 3/4$ and attains the bound. Taking expectations of the minorant under any feasible law therefore proves the stated exact convex-envelope value. Every term has lower envelope zero and upper envelope $1/2$, so $\text{cav } f_m = T(f_m) = (9m/8) \binom{m}{2}$; subtraction gives the exact ratios. The checker verifies all $5^2 + 9^2 + 13^2$ minorant inequalities and these attainment and envelope equalities in rational arithmetic.

m	Affine minorant	Counts (A, C)	$\text{vex } f_m$	$\text{cav } f_m = T$	T/H
4	$9A + 4C - 19$	$(2, 3)$	11	27	$27/16$
8	$47A + 20C - 188$	$(4, 6)$	120	252	$21/11$
12	$(231/2)A + 48C - 684$	$(6, 9)$	441	891	$99/50$

Table 7: Exact smaller two-level certificates and attaining counts.

Here is a complete Python 3 checker using only the standard library. It enumerates $7^3 + 9^3 + 65^3$ three-group states and $5^2 + 9^2 + 13^2 + 17^2$ two-group states. The data are exactly those in the manuscript tables; no numerical optimization or proprietary solver is involved. The same executable text is supplied as `verification/check_stage02_finite.py`. Concatenate the two code blocks in order to obtain that file.

```
from fractions import Fraction as Q
from itertools import product
from math import comb

# m, coefficients, denominator, integer affine coefficients,
# probability denominator, (state, probability numerator), ratio
cases = [
    (6, (2,3,9,10,7,7), 13, (962,778,842,-4816), 26,
     [((1,4,4),17), ((2,1,6),8), ((6,2,1),1)],
     Q(20891,10411)),
    (8, (2,3,13,12,8,7), 7, (793,770,798,-5882), 7,
     [((1,2,8),2), ((1,5,6),4), ((8,4,2),1)],
     Q(6601,3225)),
    (64, (2,3,120,105,70,63), 105,
     (871710,900446,899046,-51743768), 735,
     [((4,19,64),241), ((5,19,64),12),
      ((16,40,43),419), ((64,31,17),63)],
     Q(7443345,3445256))]

def phi(a, b, c, coef):
    values = (comb(c,3), b*comb(c,2), comb(b,2),
              a*c, a*b, comb(a,2))
    return sum(v*w for v,w in zip(values,coef))
```

```

for m,coef,D,dual,P,atoms,ratio in cases:
    for a,b,c in product(range(m+1), repeat=3):
        rhs = sum(v*w for v,w in zip((a,b,c,1),dual))
        assert D*phi(a,b,c,coef) >= rhs
        assert all(p >= 0 for _,p in atoms)
        assert sum(p for _,p in atoms) == P
        means = [sum(Q(p,P)*state[j] for state,p in atoms)
                  for j in range(3)]
        assert means == [Q(m,4), Q(m,2), Q(3*m,4)]
        primal = sum(Q(p,P)*phi(*state,coef) for state,p in atoms)
        lower = sum(Q(v,D)*w for v,w in zip(dual,means+[1]))
        assert primal == lower
        sizes = (comb(m,3), m*comb(m,2), comb(m,2),
                  m*m, m*m, comb(m,2))
        upper = (Q(3,4), Q(1,2), Q(1,2), Q(1,4), Q(1,4), Q(1,4))
        cav = sum(c*s*u for c,s,u in zip(coef,sizes,upper))
        term_lower = Q(coef[0]*comb(m,3),4)
        assert cav > primal
        assert (cav-term_lower)/(cav-primal) == ratio
        assert ratio > 2
        print(3*m, cav, term_lower, primal, ratio)

# m, affine coefficients (A,C,1), attaining value, cav=T, ratio
small = [
    (4, (9,4,-19), 11, 27, Q(27,16)),
    (8, (47,20,-188), 120, 252, Q(21,11)),
    (12, (Q(231,2),48,-684), 441, 891, Q(99,50))]
for m,dual,vex,cav,ratio in small:
    def value(a,c):
        return a*comb(c,2)+Q(5*m,4)*comb(a,2)
    slacks = [value(a,c)-dual[0]*a-dual[1]*c-dual[2]
              for a,c in product(range(m+1), repeat=2)]
    assert min(slacks) == 0
    a,c = m//2,3*m//4
    assert Q(a,m) == Q(1,2) and Q(c,m) == Q(3,4)
    assert value(a,c) == dual[0]*a+dual[1]*c+dual[2] == vex
    assert (m+Q(5*m,4))*comb(m,2)/2 == cav
    assert cav > vex and Q(cav,cav-vex) == ratio < 2
    print(2*m, cav, vex, ratio)

mins = []
for a in range(17):
    residuals = [2*(a*comb(c,2)+20*comb(a,2))
                 -428*a-177*c+3372 for c in range(17)]
    mins.append(min(residuals))
assert mins == [540,352,204,96,28,0,9,7,0,0,19,63,
                132,235,369,540,742]
assert 8*comb(12,2)+20*comb(8,2) == 1088
assert 214*8+Q(177,2)*12-1686 == 1088
assert Q(2160,2160-1088) == Q(135,67)
assert Q(135,67) > 2
assert Q(243,115)*(1-Q(1,20)) == Q(4617,2300) > 2
print("All finite cubic certificates passed.")

```

D Auxiliary structural reductions

The following arguments are useful independently of the width-two proof. They concern the stated local laws or reductions and make no claim about arbitrary larger-width extensions.

Lemma D.1 (Canonical pair laws on a forest). *On a forest of binary variables of prescribed means p_i , let Q_{ij} be the midpoint of the two Fréchet endpoint pair laws on each edge:*

$$Q_{ij}(1, 1) = \frac{\min(p_i, p_j) + \max(0, p_i + p_j - 1)}{2}.$$

For any target set S there is a global law with these pair marginals and target union probability at least $[\min(1, \sum_{i \in S} p_i) + \max_{i \in S} p_i]/2$, with both expressions zero for $S = \emptyset$. For any nonnegative target payoff, such a law also attains at least half its unconstrained singleton-feasible maximum. The resulting law may depend on the target.

Proof. Every pair cell is affine in its $(1, 1)$ probability and nonnegative on the Fréchet interval. Its midpoint value is therefore at least half every feasible cell value. Choose a singleton-feasible law P maximizing the target union, using consecutive circular event intervals to attain $c = \min(1, \sum_S p_i)$. The edge measures $R_{ij} = 2Q_{ij} - P_{ij}$ are nonnegative probability measures with the prescribed singleton means. They glue on the forest by rooting each component and sampling each child conditionally on its parent. Null conditioning states are irrelevant. Thus a law R with these edge marginals exists. The mixture $(P + R)/2$ has exactly the desired Q marginals, and its target union is at least $(c + b)/2$ since every law has union at least $b = \max_S p_i$. For a nonnegative payoff use its maximizing law P and $\mathbb{E}_{Rg} \geq 0$ in the same construction. $\square$

This residual construction cannot simply be applied on a triangle: for three fair bits, take P equally on the two constant states. Each canonical pair is uniform, so $2Q_{ij} - P_{ij}$ requires perfect anticorrelation. Three bits cannot be pairwise anticorrelated. This is a failure of that residual-gluing step, not a counterexample to a positive-monomial width-two bound.

Lemma D.2 (Identical-neighborhood compression). *In a positive unit-cube monomial factorization, suppose a nonempty variable block B occurs in every incident factor in its entirety. Replace it by one bit W of mean $w = \max(0, \sum_{i \in B} x_i - |B| + 1)$ in those factors. The full lower envelope and sum of local lower envelopes equal those of the reduced instance. For some $\Delta \geq 0$,*

$$T_{\text{original}} = T_{\text{reduced}} + \Delta, \quad H_{\text{original}} = H_{\text{reduced}} + \Delta.$$

Thus every reduced gap inequality with constant at least one transfers.

Proof. For any original binary law, the product $V = \prod_{i \in B} X_i$ has mean $q \geq w$. If $q > 0$, independently retain its ones with probability w/q , producing $W \leq V$ of mean w ; if $q = 0$ use $W = 0$. The reduced objective is nondecreasing in W , so this operation cannot increase it. Conversely choose a local law on B attaining product mean w with all the original means. Attach it to a reduced law by conditioning on its product value W . This lifts the law with the same objective and original means. These two operations prove full lower equality, including the endpoint cases. For a factor with scope $B \cup C$, its original local lower value is

$$\max(0, \sum_B x_i + \sum_C x_i - |B| - |C| + 1) = \max(0, w + \sum_C x_i - |C|),$$

where if $\sum_B x_i - |B| + 1 < 0$ both sides vanish. The upper value can only decrease under compression because $w \leq \min_B x_i$; this also holds when C is empty and the reduced factor is affine. Common upper attainment permits summing these decreases into one $\Delta \geq 0$. The two gap identities follow, and $T_{\text{reduced}} + \Delta \leq c(H_{\text{reduced}} + \Delta)$ for $c \geq 1$ proves the transfer without dividing by a gap. $\square$

Proposition D.3 (Signed forest gluing on arbitrary boxes). *For any multiaffine local factors on finite boxes whose incidence graph is a forest, both upper and lower envelopes add. In particular this holds for frequency-two factors whose dual multigraph is a forest.*

Proof. Choose an optimizing vertex law for each factor and normalize its coordinate endpoints. Root each incidence-tree component at a factor and traverse it, sampling each new factor conditionally on its single already sampled variable. Consistent singleton means ensure consistency at that separator; every other variable is new, or there would be an incidence cycle. Null separator states have zero probability. This simultaneously preserves all chosen optimizing laws. Use minimizing laws for the lower identity and maximizing laws for the upper identity. Different components and unused variables can be sampled independently. Fixed coordinates are substituted first. Signs do not enter the proof. $\square$

A factor partition into q incidence forests would give a q bound by mixing their exact laws for positive monomials, but treewidth does not bound such a partition number. In $K_{2,m}$, $m \geq 2$, with the two-vertex side interpreted as variables, any two factors create a four-cycle; all m factors need distinct forest colors, although treewidth is two. Adding one private variable to each factor makes their monomial scopes distinct and preserves the obstruction. This explains why the stronger matrix argument is useful.

For the radix construction, the general cutoff following Theorem 7.3 also records a fixed-radix limitation. For integers $b \geq 2, L \geq 2$ its profile capacities satisfy

$$M_s = \frac{(L-s)(b-1)+b}{b^s}, \quad \sum_{q=s+1}^L M_q = \frac{L-s}{b^s}.$$

The sum identity follows by telescoping $M_q = (L-q+1)b^{1-q} - (L-q)b^{-q}$. If s brackets one between M_s and M_{s+1} , the exact hull is $s + (L-s)b^{-s}$, as proved by its two attaining profiles. For fixed b , $s = \log_b L + O_b(1)$: the inequality $M_s \geq 1$ gives $s \leq \log_b((b-1)L+b)$, hence $L-s \sim L$; the reverse bracket then gives $b^{s+1} \geq (b-1)(L-s-1)+b$. The remainder term is $O_b(1)$, so the hull is $\log_b L + O_b(1)$. This is a limitation of this fixed-radix construction, not a sharp second-order lower theorem for all positive polynomials.

E Distinct positive-box comparisons

These proofs are superseded as upper bounds by Theorem 10.2, but use different estimates. Throughout this appendix $\rho > 1$, $t = \rho - 1$, $\alpha = 1/\rho$, and $\eta = 1 - \alpha$. Work first on $[1, \rho]^n$. The law I is independent endpoint rounding and O is independently fair endpoint orientation. For one product write C, V, P, O for its upper, lower, independent, and orientation values, and $D_I = C - P, D_O = C - O$.

E.1 The symmetric low/high argument

Proposition E.1 (A coarse finite constant). *For positive monomials on a box whose aspect ratios are at most ρ ,*

$$T \leq K(\rho)H, \quad K(\rho) = 2(1 + 1/\rho) + \max\{4(1 + \rho), 8\eta^{-3}\}.$$

In particular, with $R = \max\{4, \max_i u_i/\ell_i\}$, $T \leq (4R + 6 + 2/R)H$, which is $45/2$ for aspect ratios at most four. Direct use of $\rho = 2$ instead gives the older constant 67.

Proof. Classify a normalized mean as low when it is at most $1/2$. Put $q_i = p_i$ for lows and $q_i = 1 - p_i$ for highs, so $q_i \leq 1/2$. Divide the product's values by ρ^h , where h is the number of high coordinates. Let $L(s), H(s)$ count the respective deviations at least s , and let Q_L, Q_H be their sums. Integrals in this paragraph are over $[0, 1/2]$. Define

$$\begin{aligned} C_L &= 1 + \int (\rho^{L(s)} - 1) ds, & P_L &= \prod_{\text{low}} (1 + tq_i), \\ C_H &= 1 + \int (\alpha^{H(s)} - 1) ds, & P_H &= \prod_{\text{high}} (1 - \eta q_i). \end{aligned}$$

Common-threshold low successes and high failures occupy opposite ends of the interval. Thus $C = C_L + C_H - 1$, $P = P_L P_H$, and

$$D_I = (C_L - P_L) + (C_H - P_H) + J_I, \quad J_I = (P_L - 1)(1 - P_H) \geq 0. \quad (\text{E.1})$$

All three terms are nonnegative. Set $A = C_L - 1 - tQ_L$, $B = C_H - 1 + \eta Q_H$, and let q_L, q_H be the respective largest deviations, zero for empty groups. Integer-count inequalities give

$$A \geq t^2(Q_L - q_L), \quad B \geq \eta^2(Q_H - q_H). \quad (\text{E.2})$$

For the first, use $(1+t)^l - 1 - tl \geq t^2(l-1)_+$; for the second, $\alpha^l - 1 + \eta l$ has increments $\eta(1 - \alpha^l) \geq \eta^2$ for $l \geq 1$. Integrating proves both. Conditional orientation expectations at either end of the interval are $b_L = (1 + t/2)^{L(s)}$ and $b_H = (1 - \eta/2)^{H(s)}$, so

$$O = 1 + 2 \int (b_L b_H - 1), \quad D_O = D_{OL} + D_{OH} + J_O, \quad J_O = 2 \int (b_L - 1)(1 - b_H).$$

Here $D_{OL}, D_{OH} \geq 0$ are the separate-group deficiencies. Expansion in powers of t gives $(1+t)^l - 2(1+t/2)^l + 1 \geq [(1+t)^l - 1 - tl]/2$, since each order- $j \geq 2$ coefficient is multiplied by $1 - 2^{1-j} \geq 1/2$. Therefore

$$D_O \geq A/2 + J_O, \quad J_O \geq (t\eta/2) \min(q_L, q_H). \quad (\text{E.3})$$

The high separate-group deficiency is also nonnegative directly from convexity of the integer power at the midpoint of α and one.

Write $D_{IH} = C_H - P_H$. If $Q_H \leq 1$, put $q = q_H$ and $R' = Q_H - q$. Bonferroni's bound gives

$$D_{IH} \geq B - \eta^2 \sum_{i < j} q_i q_j \geq B - \eta^2 (qR' + R'^2/2) \geq (1 - q - R'/2)B \geq B/4.$$

Here $B \geq \eta^2 R'$, $q \leq 1/2$, and $q + R' \leq 1$; no division by R' is needed. If $Q_H \geq 1$, order deviations decreasingly. At a fixed sum, the linear common-threshold high value is minimized by filling the largest-weight coordinates to their cap $1/2$ and at most one remainder. Convex interpolation then gives

$$C_H \geq \frac{1}{2} + \frac{1}{2} \alpha^{2Q_H}, \quad P_H \leq e^{-\eta Q_H}.$$

The difference $g(Q) = 1/2 + \alpha^{2Q}/2 - e^{-\eta Q}$ is nondecreasing for $Q \geq 1$. Indeed $-\log \alpha \leq \eta/\alpha$ and $\alpha^{2Q-1} \leq e^{-\eta Q}$ imply $g'(Q) \geq \eta[e^{-\eta Q} - \alpha^{2Q-1}] \geq 0$. The fourth-order exponential remainder yields

$$D_{IH} \geq g(1) = 1 - \eta + \eta^2/2 - e^{-\eta} \geq \eta^3/6 - \eta^4/24 \geq \eta^3/8.$$

Let ϕ_ρ interpolate ρ^k between every consecutive integer, including negative integers. The exact normalized lower envelope is $V = \phi_\rho(Q_L - Q_H)$, because the original count sum is shifted by integer h . Supporting slopes at zero are η and t , so

$$C - V \leq A + B + t\eta \min(Q_L, Q_H). \quad (\text{E.4})$$

For $Q_H \leq 1$, use $\min(Q_L, Q_H) \leq \min(q_L, q_H) + A/t^2 + B/\eta^2$ to obtain

$$T \leq (1 + 1/\rho)A + (1 + \rho)B + 2J_O \leq 2(1 + 1/\rho)D_O + 4(1 + \rho)D_I.$$

For $Q_H \geq 1$, $T \leq C_L = 1 + A + tQ_L$ and $J_I \geq tQ_L(1 - e^{-\eta}) \geq t\eta Q_L/2$. The preceding bounds therefore imply

$$T \leq 2D_O + 8\eta^{-3}D_{IH} + 2\eta^{-1}J_I \leq 2D_O + 8\eta^{-3}D_I.$$

Combining the cases proves the per-product inequality with the two coefficients in $K(\rho)$. Mix the global laws proportionally, undo the product's normalization, and sum to prove the common-aspect bound. The positive affine substitution of Theorem 10.2 transfers it to unequal aspects. For $R \geq 4$, $8(1 - 1/R)^{-3} \leq 512/27 < 20 \leq 4(1 + R)$, giving $K(R) = 4R + 6 + 2/R$. For $\rho = 2$ the two coefficients are 3 and 64. $\square$

The high-mass argument in fact permits replacing $8\eta^{-3}$ by $1/g(1)$. Indeed $g(1) \leq 1 - e^{-\eta}$, since their difference is $\eta - \eta^2/2 \geq 0$; thus the same coefficient controls both the constant and the J_I term in C_L . This gives the distinct sharper finite expression $2(1 + 1/\rho) + \max\{4(1 + \rho), 1/g(1)\}$.

A tempting all-high simplification $T \leq 2D_I$ is false. On $[1, 2]^3$ with failure probabilities $(1/2, 49/100, 1/100)$, divide values by 2^3 . Exact values are

$$C = 501/800, \quad V = 1/2, \quad P = 90147/160000, \quad T = 20200/160000 > 2D_I = 20106/160000.$$

The boundary mean $1/2$ is allowed in this all-high calculation. It does not invalidate the mixed-law inequality.

E.2 The asymmetric split

Proposition E.2 (An earlier leading-order bound). *For $\rho \geq 64$, positive products on boxes of aspect ratios at most ρ satisfy*

$$T \leq \left(2 + \frac{\rho + 1}{1 - 3/\sqrt{\rho}}\right) H.$$

Proof. Put $\delta = \rho^{-1/2} \leq 1/8$ and $b = (\rho + 1)/(1 - 3\delta)$. Now classify normalized means as low if $p_i \leq 1 - \delta$, and high otherwise. High deviations are $q_i = 1 - p_i < \delta$. Normalize by ρ^h again. Reset the counts to $L(s) = \#\{i \text{ low} : p_i \geq s\}$ and $H(s) = \#\{i \text{ high} : q_i \geq s\}$, with $Q_L = \sum_{\text{low}} p_i$ and $Q_H = \sum_{\text{high}} q_i$. The separate groups' common-threshold upper values and independent values are now

$$\begin{aligned} C_L &= 1 + \int_0^{1-\delta} (\rho^{L(s)} - 1) ds, & P_L &= \prod_{\text{low}} (1 + tp_i), \\ C_H &= 1 + \int_0^\delta (\alpha^{H(s)} - 1) ds, & P_H &= \prod_{\text{high}} (1 - \eta q_i). \end{aligned}$$

Low successes and high failures under common-threshold rounding are disjoint, so (E.1), the definitions $A = C_L - 1 - tQ_L$, $B = C_H - 1 + \eta Q_H$, and (E.4) remain valid. The dispersion bounds (E.2) hold with maxima a_L, a_H of low means and high failures; they did not require low means at most $1/2$. A different estimate now replaces $D_{OL} \geq A/2$:

$$D_{IL} = C_L - P_L \geq \delta A.$$

Expand the low physical product only for this calculation. Every subset of degree at least two has independent product at most $(1 - \delta)$ times its minimum mean, and the sum of its common upper values is exactly A . Constant and affine terms cancel.

For $s \in [0, \delta]$ and $U = s$ or $1 - s$, conditional low expectations are $1 + (t/2)\mathbf{1}\{s \leq p_i\}$, while high normalized expectations are $1 - (\eta/2)\mathbf{1}\{s \leq q_i\}$. Outside these two end strips the conditional high

product is one. Conditional independence of the orientation coins gives $D_O = D_{OL} + D_{OH} + J_O \geq J_O$, where

$$J_O = 2 \int_0^\delta \left(\prod_{\text{low}} [1 + (t/2) \mathbf{1}\{s \leq p_i\}] - 1 \right) \left(1 - \prod_{\text{high}} [1 - (\eta/2) \mathbf{1}\{s \leq q_i\}] \right) ds \geq (t\eta/2) \min(a_L, a_H).$$

Combining this with dispersion and the target inequality yields $T \leq (1 + 1/\rho)A + (1 + \rho)B + 2J_O$. If $Q_H \leq 4\delta$, the same Bonferroni argument with $q = a_H \leq \delta$ and $R' = Q_H - q \leq 4\delta$ gives $D_{IH} \geq (1 - q - R'/2)B \geq (1 - 3\delta)B$. Since $(1 + 1/\rho)/\delta \leq b$, it follows that $T \leq bD_I + 2D_O$.

If $Q_H \geq 4\delta$, the common high failure set has measure at most δ , so $C_H \geq 1 - \delta$. Also $P_H \leq e^{-4\eta\delta} \leq 1 - 2\delta$, by $e^{-z} \leq 1 - z + z^2/2$ and $4\eta - 8\eta^2\delta \geq 3 - 1 = 2$. Hence $D_{IH} \geq \delta$ and $J_I \geq \delta t Q_L$, giving $D_I \geq \delta(A + 1 + tQ_L) = \delta C_L$. As $T \leq C_L$, again $T \leq bD_I + 2D_O$. Mix the two global laws with weights proportional to $b, 2$, sum over products, and use positive affine transfer for unequal aspects. $\square$

Its expansion is $\rho + 3\sqrt{\rho} + O(1)$; the coefficient theorem improves the additive error to two.

E.3 Optimality within a fixed two-law mixture

Proposition E.3 (A limitation of fixed independent/fair-orientation mixtures). *Fix $\rho > 1$ and an orientation weight $w \in [0, 1]$. For mixtures of I and independently fair O , the best uniform per-physical-monomial captured fraction over all dimensions is at most $\min\{w/2, (1-w)/\rho\}$. Optimizing the weight gives exactly $1/(\rho + 2)$, attained at $w = 2/(\rho + 2)$.*

Proof. For a bilinear product at normalized means $(\epsilon, 1 - \epsilon)$, $0 < \epsilon < 1/2$, deleting affine parts gives $D_I/T = \epsilon$ and $D_O/T = 1/2$. Letting $\epsilon \downarrow 0$ bounds the mixture fraction by $w/2$. For the other obstruction use means ϵ and k copies of $1 - \epsilon/k$, where $k \geq 1$. Divide physical values by ρ^{k+1} , and write $\alpha = 1/\rho$, $\eta = 1 - \alpha$, $\gamma = (1 + \alpha)/2$. The sum of success means is k , giving $V = \alpha$. Common-threshold and fair-orientation integration, split at ϵ/k and ϵ , and direct independent multiplication give

$$\begin{aligned} C &= \alpha + \epsilon[\eta - \alpha(1 - \alpha^k)/k], \\ P &= \alpha[1 + (\rho - 1)\epsilon][1 - \eta\epsilon/k]^k, \\ O &= \alpha + \epsilon[\eta - 2\gamma(1 - \gamma^k)/k]. \end{aligned}$$

For clarity, in the orientation law the two end intervals of length ϵ/k have normalized expected product γ^{k+1} , the two next intervals of total length $2(\epsilon - \epsilon/k)$ have value γ , and the middle interval of length $1 - 2\epsilon$ has value α ; these yield the stated O . For each fixed k , divide deficiencies by ϵ and let $\epsilon \downarrow 0$. Then

$$\begin{aligned} T/\epsilon &= \eta - \alpha(1 - \alpha^k)/k, \\ D_I/\epsilon &\longrightarrow \alpha\eta - \alpha(1 - \alpha^k)/k, \\ D_O/\epsilon &= [2\gamma(1 - \gamma^k) - \alpha(1 - \alpha^k)]/k. \end{aligned}$$

The denominator coefficient is at least $\eta^2 > 0$ because $1 - \alpha^k \leq k\eta$. Now let $k \rightarrow \infty$ to obtain $D_I/T \rightarrow 1/\rho$ and $D_O/T \rightarrow 0$, bounding the mixture by $(1 - w)/\rho$. The maximum of the minimum of these two bounds is $1/(\rho + 2)$ at their intersection. Lemma 10.1 with fair orientations supplies the matching guarantee. $\square$

This is only an optimality statement for a fixed two-law mixture’s per-monomial guarantee over unrestricted dimensions. Other laws, coefficient-dependent mixtures, and the true optimum joint law for a sum of monomials are outside it. In particular it does not prove $C_{\text{box}}(\rho) = \rho + 2$, nor conflict with the finite-ambient balanced-orientation refinement.

F A classical point-packing benchmark

The following four values were conjectured by Anstreicher [4, Section 4] and proved by Khajavirad [28, Propositions 1(i–ii) and 3]. We include a variance proof and explicit feasible covariances to illustrate how an exact relaxation value differs from a spatial certificate bound. For $n \geq 2$ points $(x_i, y_i) \in [0, 1]^2$, maximize γ subject to $(x_i - x_j)^2 + (y_i - y_j)^2 \geq \gamma$ for $i < j$. Replace the two sets of products by symmetric matrices X, Y and write

$$D_{ij} = X_{ii} - 2X_{ij} + X_{jj} + Y_{ii} - 2Y_{ij} + Y_{jj} \geq \gamma.$$

RLT imposes all four products of coordinate-bound slacks, including repeated indices, as in (12.10). SDP instead imposes $X \succeq xx^T$, $Y \succeq yy^T$ and the diagonal secants $X_{ii} \leq (\ell_i + u_i)x_i - \ell_i u_i$, and likewise for Y . The symmetry bounds, denoted SYM, restrict $x_i \in [1/2, 1]$ for $i \leq n_x = \lceil n/2 \rceil$ and $y_i \in [1/2, 1]$ for $i \leq k = \lceil n/4 \rceil$. Reflection gives the lower-half convention in Khajavirad’s paper. Both RLT and SDP secants are recomputed on these intervals; adding only first-moment bounds would define different models.

Proposition F.1 (Four exact values). *The optimal values are*

<i>model</i>	<i>value</i>	<i>range</i>
RLT	2	$n \geq 2$
SDP (with or without RLT)	$n/(n-1)$	$n \geq 2$
RLT + SYM	1/2	$n \geq 5$
SDP + SYM (with or without RLT)	$k/[4(k-1)]$	$n \geq 5$.

Here $k-1 = \lfloor (n-1)/4 \rfloor$.

Proof. For two coordinates in a common interval $[\ell, \ell + L]$, the RLT maximum of $X_{ii} - 2X_{ij} + X_{jj}$ at fixed means is

$$L \min\{x_i + x_j - 2\ell, 2\ell + 2L - x_i - x_j\} \leq L^2.$$

Take both diagonals at their secants and the off-diagonal at the larger of its two lower McCormick planes. This is feasible for all pairs at once: the diagonal secant dominates both diagonal lower planes, and each pair’s lower planes lie below its upper planes. Thus the bound two holds without symmetry and is attained at $x = y = 1/2$, with diagonal 1/2 and off-diagonal zero in each matrix. With symmetry, two of the first $k \geq 2$ points give the bound 1/2. Put the three groups $Q = [k]$, $H = [n_x] \setminus Q$, and $F = [n] \setminus [n_x]$ at $(3/4, 3/4)$, $(3/4, 1/2)$, and $(1/2, 1/2)$. Set every diagonal to its interval secant and each off-diagonal to its lower RLT bound. In group-pair order QQ, QH, QF, HH, HF, FF , the resulting squared distances are

$$1/2, \quad 7/8, \quad 5/4, \quad 5/4, \quad 13/8, \quad 2.$$

Pairs from groups with fewer than two members impose no extra requirement. This proves both RLT values.

For SDP, restrict to any $k \geq 2$ points sharing $[\ell, \ell + L]$ in one coordinate, and let $e = \mathbf{1} \in \mathbb{R}^k$. With $s = \sum_i (x_i - \ell)$, PSD and the secants give

$$\begin{aligned} \sum_{i < j} (X_{ii} - 2X_{ij} + X_{jj}) &= k \operatorname{tr} X - e^T X e \\ &\leq k[(2\ell + L) \sum_i x_i - k\ell(\ell + L)] - (\sum_i x_i)^2 \\ &= kLs - s^2 \leq k^2 L^2 / 4. \end{aligned}$$

The pair average per coordinate is at most $kL^2/[2(k-1)]$. Add the two coordinates, first using $k = n, L = 1$ and then the symmetry subset with $L = 1/2$, to obtain the two asserted SDP upper bounds.

Without symmetry, set $x = y = e/2$ and

$$X = Y = ee^T/4 + Z, \quad Z = \frac{n}{4(n-1)}(I - ee^T/n) \succeq 0. \quad (\text{F.1})$$

Its diagonals equal $1/2$ and its off-diagonals equal $1/4 - 1/[4(n-1)]$. Every pair has total distance $n/(n-1)$. These entries also satisfy all RLT bounds.

For symmetry use the same three group means as above. In each coordinate set the moment matrix to mean outer product plus a covariance that is block diagonal in Q, H, F . On Q use in both coordinates

$$Z_Q = \frac{k}{16(k-1)}(I - ee^T/k).$$

On H use $I/16$ in x and $I/4$ in y ; on F use $I/4$ in both. These matrices are PSD because $I - ee^T/k$ is an orthogonal projection. Every diagonal equals its tightened secant. The group-pair distances are

$$\frac{k}{4(k-1)}, \quad \frac{1}{2}, \quad \frac{3}{4}, \quad \frac{5}{8}, \quad \frac{7}{8}, \quad 1.$$

They are at least the first value for every $k \geq 2$. For RLT feasibility, cross-group and distinct H, F entries are products of their means and hence obey the four planes. On Q an off-diagonal is $9/16 - 1/[16(k-1)] \in [1/2, 5/8]$, exactly within the tightened RLT range at mean $3/4$. Diagonal secants satisfy their diagonal lower planes as above. Thus adding RLT cannot change either SDP optimum. $\square$

In dimension d , the same one-coordinate average and the same covariance in each coordinate give RLT value d and SDP value $dn/[2(n-1)]$ on $[0, 1]^d$, for $n \geq 2$. A subset of $k \geq 2$ points in a common side- L cube forces an SDP bound at most $dL^2k/[2(k-1)]$. This includes $L = 0$. These are dimension-scaled averaging statements, not claims about all symmetry or ordering models. Numerical symmetry values at $n < 5$ and ordering formulations are not needed here. The benchmark explains weakness of these particular root relaxations; it establishes no lower bound on every possible strengthening or spatial algorithm.

G Scale disjunctions and shared operating variables

Scaling gives an example where a representation can remove an artificial gap. The arguments are affine interpolation and classical disjunctive convexification [5, Section 2, Theorem 2.1]; we give

them explicitly to identify the role of shared intensive variables. Let $E \subset \mathbb{R}^d \times \mathbb{R}^p$ be nonempty compact, with per-unit extensive coordinates ξ and intensive coordinates T . Define

$$\lambda \odot E = \{(\lambda\xi, T) : (\xi, T) \in E\}, \quad F(\Lambda, E) = \{(\lambda, \lambda\xi, T) : \lambda \in \Lambda, (\xi, T) \in E\}.$$

The nonnegative scale set Λ is nonempty compact, with extremes λ_-, λ_+ . At zero scale, $X = 0$ and $T \in \text{proj}_T E$; a different off-state domain must be modeled explicitly.

Proposition G.1 (Endpoint hull and multiplicities). *With $C = \text{conv } E$, the hull of $F(\Lambda, E)$ is the hull of just its two extreme-scale slices, with E replaced by C . It is exactly the projection of*

$$\begin{aligned} 0 \leq \theta \leq 1, \quad \lambda = (1 - \theta)\lambda_- + \theta\lambda_+, \quad X = X^- + X^+, \quad T = T^- + T^+, \\ (X^-, T^-) \in (1 - \theta)(\lambda_- \odot C), \quad (X^+, T^+) \in \theta(\lambda_+ \odot C), \end{aligned} \quad (\text{G.1})$$

where $0K = \{0\}$ for nonempty K . For $N \geq 1$ identical units sharing T , allow integer $0 \leq n \leq N$, $X = \sum_{i=1}^n \xi_i$, and $(\xi_i, T) \in E$. Their hull is the same as $\text{conv } F(\{0, \dots, N\}, E)$. Writing $P_T = \text{proj}_T C$, it is

$$0 \leq n \leq N, \quad \exists T^+ : (X, T^+) \in \frac{n}{N}(N \odot C), \quad T - T^+ \in (1 - n/N)P_T. \quad (\text{G.2})$$

Proof. For fixed (ξ, T) , the point $(\lambda, \lambda\xi, T)$ depends affinely on λ . If the extremes differ, interpolate with $\theta = (\lambda - \lambda_-)/(\lambda_+ - \lambda_-)$; if they coincide, there is only one slice. Every original point is therefore a combination of endpoint points. A fixed-scale map is linear in (ξ, T) , so its hull is that slice of C . Combining two compact convex slices gives (G.1), including zero weights. For an integer $n > 0$ and fixed T , a sum of n points of E_T lies in $n \text{conv } E_T$, and identical copies include nE_T . Equivalently, a multiplicity point is the average of its n same-scale points $(n, n\xi_i, T)$, while every same-scale point is a feasible multiplicity point. Their hulls therefore agree at fixed n after convexification. Taking the hull over n and applying the endpoint result proves (G.2). At $n = 0$ both models use the stated projection off-state; $N = 0$ consists only of that slice. $\square$

If $C = \{(\xi, T) : A\xi + BT \leq b\}$ and $P_T = \{T : DT \leq e\}$ are polytopes, (G.2) becomes

$$AX + NBT^+ \leq bn, \quad D(T - T^+) \leq e(1 - n/N), \quad 0 \leq n \leq N.$$

Compact conic-representable slices can also be homogenized at zero weight. Indeed, if $K_0 = \{z : \exists y, Az + By + b \in \mathcal{K}\}$ is nonempty compact, then at weight $t > 0$ division in $Az + Bw + tb \in \mathcal{K}$ gives tK_0 . At $t = 0$ a feasible (z, w) is a recession direction of the lifted set: add any nonnegative multiple to a feasible lift. Its projected direction must be zero by compactness of K_0 . Conversely $z = w = 0$ is feasible. This checks zero weights even when the auxiliary lift is unbounded. The aggregation mechanism also appears in Wu et al. [41, Section 4.1, Lemma 3 and Theorem 3, under Assumption 1].

For example, let per-unit load $\xi \in [\ell, u]$, $0 \leq \ell < u$, common $T \in [a, b]$, $a < b$, and per-unit extensive output $\omega = \xi T$. Then C is the McCormick hull of this bilinear graph. Formula (G.2) supplies its exact local aggregate hull with one disaggregated intensive coordinate. Its endpoint vertices are the two off-state points and the four corners at scale N . Besides $n\ell \leq X \leq nu$, $0 \leq n \leq N$, $a \leq T \leq b$ and the global McCormick inequalities for $W = XT$ on $[0, Nu] \times [a, b]$, this hull has

$$W \leq bX + \ell[NT - Na - n(b - a)], \quad W \geq aX + \ell[NT - Nb + n(b - a)]. \quad (\text{G.3})$$

To check sufficiency without a facet calculation, introduce $S = NT^+$. The scaled per-unit McCormick system is equivalent to

$$\begin{aligned} an \leq S \leq bn, \quad NT - b(N - n) \leq S \leq NT - a(N - n), \\ \ell S + aX - a\ell n \leq W \leq \ell S + bX - b\ell n, \quad uS + bX - bun \leq W \leq uS + aX - aun. \end{aligned}$$

Eliminating S by comparing each lower and upper bound gives precisely the displayed system (redundant comparisons follow from the stated bounds). When $\ell = 0$, use these inequalities directly without dividing by ℓ . Alternatively (G.2) itself is a complete formulation at that boundary.

G.1 When scale costs do and do not separate

Example G.2 (A scale-cost gap on a convex per-unit set). Take $E = \{(\xi, T) : \xi = T \in [0, 1]\}$, $\Lambda = \{0, 1, 2\}$, and $f(0) = f(2) = 1$, $f(1) = 0$. The point $(\lambda, X, T) = (1, 1, 1/2)$ is the average of $(0, 0, 0)$ and $(2, 2, 1)$. Its true minimum convexified cost is one although the separate lower scale-cost envelope has $\check{f}(1) = 0$. In any representation let p_i be the mass at scale i . Mean scale one gives $p_0 = p_2$. On scales zero, one, and two, respectively, $X - T$ is at most zero, zero, and one. Thus the target $X - T = 1/2$ forces $p_2 \geq 1/2$, and the cost is at least $p_0 + p_2 \geq 1$. The endpoint representation attains it. Scaling f scales this error arbitrarily.

An operating cost must also be lifted before convexifying a nonconvex per-unit set. With $E = \{-1, 1\}$, scale one and cost $|\xi|$, its convexified cost at $X = 0$ is one, not $|0|$. Further, $\lambda c(X/\lambda, T)$ need not be convex with an unscaled intensive coordinate: the convex cost $c(\xi, T) = T^2$ gives λT^2 .

Proposition G.3 (Extensive-only perspective). *Let $E \subset \mathbb{R}^d$ be nonempty compact, $c : E \rightarrow \mathbb{R}$ continuous, and $\Lambda \subset [0, \infty)$ finite and nonempty. Let h be the lower boundary of $\text{conv}\{(\xi, c(\xi)) : \xi \in E\}$ and $\check{f}$ the lower convex envelope of the finite scale-cost data. For $\lambda > 0$, the exact hull of $Z \geq \lambda c(\xi) + f(\lambda)$, $X = \lambda \xi$, is*

$$\lambda \in \text{conv } \Lambda, \quad X \in \lambda \text{conv } E, \quad Z \geq \lambda h(X/\lambda) + \check{f}(\lambda).$$

If $0 \in \Lambda$, its zero-scale slice is $X = 0$, $Z \geq f(0)$.

Proof. For a representation with weights μ_j and positive mean scale λ , the size-biased weights $\mu_j \lambda_j / \lambda$ sum to one. They give per-unit cost at least $h(X/\lambda)$, while the original weights give scale cost at least $\check{f}(\lambda)$. Conversely choose attaining per-unit atoms with weights ν_i and attaining scale weights μ_j . Their product weights $\mu_j \nu_i$ attain both bounds and the required X . Compactness gives finite attaining representations; adding vertical slack gives every larger Z . At zero mean scale, nonnegativity forces all positively weighted scales to be zero. $\square$

This is the usual perspective argument after the cost has been included in the per-unit graph. The multiplicity statement follows by treating that cost as another additive extensive coordinate.

Proposition G.4 (Rectangularity criterion). *Let $E \subset \mathbb{R}^d \times \mathbb{R}^p$ be nonempty compact and let the finite catalogue $\Lambda \subset [0, M]$ satisfy $\{0, s, M\} \subseteq \Lambda$ and $0 < s < M = \max \Lambda$. Put $C = \text{conv } E$, $Q = \text{proj}_\xi C$, and $P_T = \text{proj}_T C$. Every scale-only cost separates, in the precise sense*

$$\begin{aligned} \text{conv}\{(\lambda, X, T, Z) : (\lambda, X, T) \in F(\Lambda, E), Z \geq f(\lambda)\} \\ = \{(\lambda, X, T, Z) : (\lambda, X, T) \in \text{conv } F(\Lambda, E), Z \geq \check{f}(\lambda)\}, \end{aligned} \tag{G.4}$$

if and only if $C = Q \times P_T$. It suffices to test the single cost $f(s) = 0$, $f(\lambda) = 1$ at every other catalogue scale.

Proof. The epigraph hull equals the compact hull of the cost graph plus the nonnegative vertical ray. Hence all required minimum-cost representations are attained; no closure operation is hidden. If the diagnostic cost separates, every point of the scale- s base-hull slice has a zero-cost representation. All its constituents must have scale s , so this slice is exactly $s \odot C$. Put $\alpha = s/M \in (0, 1)$. For $(\xi, T_1) \in C$ and $T_0 \in P_T$, mixing the extreme slices gives $(s, s\xi, \alpha T_1 + (1 - \alpha)T_0)$ in that slice. Thus the fibre of C at ξ contains $\alpha T_1 + (1 - \alpha)T_0$. Iteration gives $\alpha^j T_1 + (1 - \alpha^j)T_0$ in the fibre for every j . Closedness puts T_0 in it, proving rectangularity. Conversely, under rectangularity the base hull is $\{(\lambda, X, T) : 0 \leq \lambda \leq M, X \in \lambda Q, T \in P_T\}$. For $\lambda > 0$ choose $\xi = X/\lambda$ and use any attaining scale-cost weights with the same $(\xi, T) \in C$ in every scale slice. Decompose that point into E within each slice; the resulting representation attains $\check{f}(\lambda)$. At zero choose any $\xi \in Q$ and use the zero slice. Every representation has cost at least $\check{f}$ by convexity, proving equality. Universal separation implies the diagnostic case, completing the equivalence. $\square$

Without an interior scale, two endpoint weights are already fixed by λ , so all scale-only costs separate even for nonrectangular C . Affine scale costs always separate. The criterion concerns all scale-only costs with the stated off-state domain, not arbitrary operating costs.

G.2 Integral counts and aggregate rounding

For nonempty compact extensive-only unit types $E_j \subset \mathbb{R}^d$ and an integral polytope $P \subset \mathbb{R}_+^k$ of allowable counts, the exact aggregate hull is

$$\{(n, X) : n \in P, X \in \sum_j n_j \text{conv } E_j\}. \quad (\text{G.5})$$

Indeed it is convex and contains every integer multiplicity point. For the reverse inclusion write $X = \sum_j n_j \xi_j$ with $\xi_j \in \text{conv } E_j$, decompose n into integral vertices of P , and keep these ξ_j fixed. At each integral vertex the sum of scaled convex hulls is the hull of the corresponding finite Minkowski sum. This proves (G.5); no integer-decomposition property beyond integrality of P is needed.

For n identical compact extensive sets in $\mathbb{R}^d$, the classical Shapley–Folkman lemma [38, Appendix 2, Lemma 2 and corollary] represents any $X \in n \text{conv } E$ as a sum with at most $\min(d, n)$ constituents outside E . Replacing each exceptional constituent by any point of E changes the sum by at most $\min(d, n) \text{diam}(E)$ in a chosen norm. Thus the same bound holds for the Hausdorff distance between the two aggregate sets. An L -Lipschitz objective has gap at most $L \min(d, n) \text{diam}(E)$ when minimized over those sets alone. A relative $O(d/n)$ bound additionally requires a positive normalization of order n and uniformly bounded L and diameter.

All these identities concern the stated local or aggregate sets. Added balance constraints can exclude the rounded point, and $\text{conv}(F \cap H)$ can be strictly smaller than $\text{conv } F \cap H$. Endpoint hull equality also does not license removing intermediate catalogue choices from the feasible model. If the per-unit set depends on scale, even the hull equality can fail: $\xi = \lambda$ at scales 1, 2, 3 gives $(\lambda, X) = (2, 4)$ outside the segment joining $(1, 1)$ and $(3, 9)$.

H Retained domains and coordinate dependence in P-split

Kronqvist, Misener, and Tsay’s P-split construction splits convex additive constraints by groups of variables, introduces epigraph auxiliaries for the groups, and convexifies the resulting bounded auxiliary disjunction [29, Sections 2–4]. The original domain $x \in X$ remains a global constraint. The bounds are computed over this retained compact convex X , and identical or scaled group functions share an auxiliary under the paper’s minimal-sharing Assumption 3 and Remark 1. In the

following comparisons, “basic” means these epigraph links, exact global interval bounds, and the exact auxiliary disjunction hull. Extra original–auxiliary links change this model.

H.1 A domain counterexample and a local nonexactness criterion

The published Definition 4 calls disjuncts fully disjoint when they are pairwise disjoint. Theorem 6 states universal nonexactness when the bounds are additive and the constraint functions strictly convex. Its universal statement fails with the retained-domain convention; this concerns that theorem’s scope, not validity of the P-split formulations.

Example H.1 (An exact retained-box relaxation). On $X = [0, 3] \times [-1, 1]$, use

$$D_0 = \{(t, w) \in X : t^2 + w^2 \leq 1\}, \quad D_3 = \{(t, w) \in X : (t - 3)^2 + w^2 \leq 1\}.$$

The disjuncts are nonempty and disjoint, and are defined by strictly convex quadratics, with additive component bounds on the box. Together they contain all four vertices of X . Hence $\text{conv}(D_0 \cup D_3) = X$, and every convex valid relaxation retaining X , including every P-split partition, is exact. For a direct full-split lift use shared w^2 and set

$$(\alpha_0, \alpha_3, \alpha_w) = (3t, 9 - 3t, 1).$$

This lies in the auxiliary hull as the combination of $(0, 9, 1)$ and $(9, 0, 1)$ with weights $1 - t/3, t/3$. The links follow from $t^2 \leq 3t$, $(t - 3)^2 \leq 9 - 3t$, and $w^2 \leq 1$. In dimension $n \geq 2$ replace w^2 by $(n - 1)^{-1} \sum_{i=2}^n x_i^2$ on $[0, 3] \times [-1, 1]^{n-1}$. The same box-vertex argument applies, with rational positive definite quadratics and one constraint per disjunct.

The source explicitly calls its “few constraints” Assumption 4 technically unnecessary, and the last construction also satisfies its intended large-dimension setting. In the proof of its Theorem 6, strict Jensen slack is claimed in every component. Strict convexity supplies that slack only when the two arguments differ. Along the top hull segment in the example, w is fixed at one. Further, a neighborhood allowed by epigraph slack need not remain in X : an outward step through its top facet is forbidden.

Proposition H.2 (A sufficient directional repair). *Let X and Q be convex, A finite, and F_a continuous convex. Define $R = \{x \in X : \exists \alpha \in Q, F_a(x) \leq \alpha_a \ (a \in A)\}$. Assume R contains the entire true feasible set. Suppose true feasible points x^0, x^1 have lifts $F(x^0), F(x^1) \in Q$. At $x^\theta = (1 - \theta)x^0 + \theta x^1$, $0 < \theta < 1$, let $\Delta_a = (1 - \theta)F_a(x^0) + \theta F_a(x^1) - F_a(x^\theta)$. If a hull-supporting inequality $c^T x \leq \beta$ is tight at x^θ and a direction v has $c^T v > 0$, $x^\theta + \epsilon v \in X$ for all small $\epsilon \geq 0$, and $F_a(x^\theta + \epsilon v) \leq F_a(x^\theta)$ for every zero-slack link and all small $\epsilon \geq 0$, then R strictly contains the true hull.*

Proof. Keep $\alpha = (1 - \theta)F(x^0) + \theta F(x^1) \in Q$. Every positive slack persists for a sufficiently small step by continuity. There are finitely many links, so one common positive radius suffices. The assumptions preserve the remaining links and domain, while the support violation is $\epsilon c^T v > 0$. If varying links have directional Lipschitz constants L_a on a common radius, imposing $\epsilon L_a \leq \Delta_a$ for every positive-slack link gives an explicit allowable step within that radius. $\square$

H.2 Exact basic relaxation of two translated balls

Proposition H.3 (Projection and Hausdorff error). *For $n \geq 2$, radius $r > 0$ and separation $d > 2r$, let $x = (t, w)$ and*

$$D_0 = \{t^2 + \|w\|_2^2 \leq r^2\}, \quad D_d = \{(t - d)^2 + \|w\|_2^2 \leq r^2\}, \quad X = [-r, d + r] \times [-r, r]^{n-1}.$$

Their hull is $C = [0, d]e_1 + r\mathbb{B}_2^n$. The basic full coordinate split, sharing the transverse square auxiliaries, projects exactly to

$$R = \{(t, w) : \|w\|_2 \leq r, \quad 2(t - d/2)^2 + \|w\|_2^2 \leq 2(d/2 + r)^2\}. \quad (\text{H.1})$$

The two-group partition $\{t\}, \{w_1, \dots, w_{n-1}\}$ gives the same set. With $e_* = [(d/2 + r)^2 - r^2/2]^{1/2} - d/2 > 0$,

$$d_H(R, C) = \sqrt{r^2 + e_*^2} - r, \quad d_H(R, C)/r \longrightarrow \sqrt{2} - 1 \quad (d/r \rightarrow \infty). \quad (\text{H.2})$$

Proof. Put $U = (d + r)^2$. The auxiliaries are $a \geq t^2$, $b \geq (t - d)^2$, $c_i \geq w_i^2$, with $0 \leq a, b \leq U$, $0 \leq c_i \leq r^2$. The auxiliary disjunction is $a + \sum_i c_i \leq r^2$ or $b + \sum_i c_i \leq r^2$. Its hull is exactly

$$0 \leq a, b \leq U, \quad c_i \geq 0, \quad \sum_i c_i \leq r^2, \quad a + b + \sum_i c_i \leq U + r^2. \quad (\text{H.3})$$

Necessity holds in each disjunct. For sufficiency fix c and put $h = r^2 - \sum_i c_i \in [0, r^2]$. At this fixed c , the union in (a, b) is $[0, h] \times [0, U] \cup [0, U] \times [0, h]$. Its hull is the square cut by $a + b \leq U + h$: every vertex of that truncated square lies in the union. This proves sufficiency with the same c in each constituent. The set (H.3) is downward closed, so its epigraph projection follows by substituting the actual squares. Completing the square gives (H.1); its cylinder and ellipsoid already imply all the retained box bounds. In the two-group partition replace $\sum_i c_i$ by $c \geq \|w\|^2$. Its global upper bound $(n - 1)r^2$ is reduced by either disjunct to r^2 , giving the same proof. The additive-bound refinement property [29, Theorem 4 and Corollary 3] puts this full-split set inside every basic coarser coordinate partition for this box.

Between the centers every point of the cylinder is in C . By reflection, consider only $t \leq 0$ and put $\rho = \|w\| \in [0, r]$. The leftmost allowed coordinate is $t = -e(\rho)$ with $e(\rho) = \sqrt{(d/2 + r)^2 - \rho^2/2} - d/2 > 0$. The squared distance to the nearer center is

$$e(\rho)^2 + \rho^2 = (d/2 + r)^2 + d^2/4 + \rho^2/2 - d\sqrt{(d/2 + r)^2 - \rho^2/2}.$$

Its derivative with respect to ρ^2 is $1/2 + d/[4\sqrt{(d/2 + r)^2 - \rho^2/2}] > 0$. The maximum occurs at $\rho = r$, where the distance outside the capsule is $\sqrt{r^2 + e_*^2} - r$. At $\rho = 0$ that distance is zero, so no positive part was lost. Since $C \subset R$, directed and ordinary Hausdorff distances coincide. Finally $e_*/r \rightarrow 1$, proving the limit. $\square$

H.3 An unbounded coordinate effect with rational data

First take unit balls centered at 0 and (D, D) , $D \geq 5$, retaining $[-1, D + 1]^2$. The basic split introduces four separate squares $a_i \geq x_i^2$, $b_i \geq (x_i - D)^2$ with upper bound $U = (D + 1)^2$, and convexifies $a_1 + a_2 \leq 1$ or $b_1 + b_2 \leq 1$. At $p = (2D/3, D/3)$ its true square vectors a, b have $2a_i, 2b_i \leq 8D^2/9 \leq U$. Their average representation $(a, b) = \frac{1}{2}(0, 2b) + \frac{1}{2}(2a, 0)$ proves feasibility in the auxiliary hull. Projection onto the center segment is $(D/2, D/2)$, so the Hausdorff error is at least $D/(3\sqrt{2}) - 1$. Under the orthogonal change $t = (x_1 + x_2)/\sqrt{2}$, $w = (x_1 - x_2)/\sqrt{2}$, the retained domain is exactly

$$-\sqrt{2} \leq t + w, t - w \leq \sqrt{2}(D + 1),$$

a diamond, and the centers are $0, \sqrt{2}De_t$. The shared w^2 auxiliary is at most one throughout either disjunct, hence $|w| \leq 1$ in the relaxation. For $t \leq 0$, the domain implies $-t + |w| \leq \sqrt{2}$, giving $t^2 + w^2 \leq (\sqrt{2} - |w|)^2 + w^2 \leq 2$; the convex quadratic in $|w| \in [0, 1]$ has its maximum at an endpoint. The right end is identical. Between centers the cylinder lies in the capsule. The new error is therefore at most $\sqrt{2} - 1$. No additive-bound hierarchy is asserted on the transformed diamond.

Proposition H.4 (Rational coordinate comparison). *For rational $D \geq 2$, unit balls with centers $0, v = (3D, 4D)$ in $X = [-1, 3D + 1] \times [-1, 4D + 1]$ have basic original-coordinate Hausdorff error at least $4D/5 - 1$. Under the fixed rational orthogonal map*

$$t = (3x_1 + 4x_2)/5, \quad w = (4x_1 - 3x_2)/5, \quad (\text{H.4})$$

using the exact transformed domain and tight recomputed bounds, the error is at most $\sqrt{2} - 1$.

Proof. The witness $p = (2D, 4D/3)$ has each coordinate at one or two thirds of v_i . Twice each original square is at most $8v_i^2/9 \leq (v_i + 1)^2$, so the preceding half-weight lift is feasible. Its transformed coordinates are $t(p) = 34D/15 \in [0, 5D]$ and $w(p) = 4D/5$, proving the lower bound. In aligned coordinates the common transverse auxiliary again forces $|w| \leq 1$. For $t \leq 0$, the inverse map gives $x_1 \leq 4/5$ and $x_2 \leq 3/5$, while the retained domain gives $x_1, x_2 \geq -1$. Thus $t^2 + w^2 = x_1^2 + x_2^2 \leq 2$. For $t \geq 5D$, apply the same argument to $x - v$: its two components lie in $[-4/5, 1]$ and $[-3/5, 1]$. The error outside either end is at most $\sqrt{2} - 1$, and between centers it is zero. The map has determinant -1 ; changing the sign of w gives a proper rotation with identical conclusions. All bounds and domain inequalities remain rational for rational D . $\square$

H.4 The exact auxiliary image is still different from the graph

Proposition H.5 (Strongest auxiliary-only comparison). *In either original-coordinate example, let $F(x) = (x_i^2, (x_i - v_i)^2)_{i=1}^2$. Every convex auxiliary set Q containing $F(0)$ and $F(v)$ preserves the original witness in*

$$R(Q) = \{x \in X : \exists \alpha \in Q, F(x) \leq \alpha\}.$$

In particular the lower error survives $Q = \text{conv } F(D_0 \cup D_v)$. In aligned coordinates, however, the exact feasible auxiliary image hull for $F'(t, w) = (t^2, (t - d)^2, w^2)$ gives exactly the true capsule, for $d > 2$. This holds with either exactly transformed retained domain above.

Proof. Convexity forces $[F(0) + F(v)]/2 \in Q$, whose entries are $v_i^2/2$. At the witness, both coordinate squares are at most $4v_i^2/9$ and hence satisfy all epigraph links to that point. This argument uses only the two feasible center images, so every valid convex auxiliary-only strengthening of the same epigraph map retains the witness.

For alignment write $(a, b, c) = F'(t, w)$ and set

$$f(c) = d^2 + 1 - c + 2d\sqrt{1 - c}, \quad 0 \leq c \leq 1.$$

At an actual feasible point, $t \in [-q, q] \cup [d - q, d + q]$ with $q = \sqrt{1 - c}$, so $a, b \leq (d + q)^2 = f(c)$. The function f is concave and decreasing. Its two hypographs therefore contain the whole convex auxiliary image hull. If the epigraph links are feasible there, then $|w| \leq 1$ and, by monotonicity,

$$|t| \leq d + \sqrt{1 - w^2}, \quad |t - d| \leq d + \sqrt{1 - w^2}.$$

Their intersection is precisely $-\sqrt{1 - w^2} \leq t \leq d + \sqrt{1 - w^2}$, the capsule. Conversely the epigraph relaxation is convex and contains both balls, so it contains their hull. The retained domain contains that hull, which proves equality. In fact, just $0 \leq c \leq 1$ and $a, b \leq f(c)$ suffice: each upper inequality has a conic representation with $s \geq 0$, $s^2 + c \leq 1$ and, respectively, $a + c - d^2 - 1 \leq 2ds$ or $b + c - d^2 - 1 \leq 2ds$. $\square$

The exact auxiliary image hull retains only values of the chosen functions; it is not the hull of their full feasible graph in (x, α) . Original-space cuts, constraints coupling original and auxiliary variables, or identities replacing the epigraph links can remove the witness. These calculations establish explicit coordinate effects in this family, without an algorithm for choosing good coordinates in general and without an assertion about every strengthened P-split formulation.

I A stable correlation face and global lift size

This appendix compares global formulation size with the local gap and spatial resources in the main text. All matrix ℓ_1 norms here are *entrywise* sums, and $\|H\|_\infty = \max_{ij} |H_{ij}|$ for a cost matrix. Scalar inequalities count as nonnegative cone factors.

I.1 The face and inherited exact lower bounds

For $n \geq 1$ define the compact sets

$$\mathcal{R}_n = \{W \geq 0 : \text{rank } W \leq 1, We \leq e, W^T e \leq e\}, \quad K_n = \text{conv } \mathcal{R}_n.$$

For a nonzero generator, its row and column marginals r, c and total S satisfy $W = rc^T/S$, $0 \leq r, c \leq e$, and $e^T r = e^T c = S \leq n$. Indeed, writing $W = uv^T$ and computing its marginals proves this identity. Then $\text{tr } W = r^T c/S \leq 1$. Equality forces $r_i(1 - c_i) = c_i(1 - r_i) = 0$ for every i , so $r = c = 1_A$ for a nonempty set A . Thus the trace-one face is the hull of $1_A 1_A^T/|A|$.

Proposition I.1 (Explicit correlation face). *In K_{2m} put*

$$T(W) = e^T W e, \quad D(W) = 1 - \text{tr } W, \quad B(W) = \sum_{i=1}^m W_{i,m+i},$$

$$F = \{W \in K_{2m} : D = B = 0, T = m\}.$$

Then F is a face affinely isomorphic to $\text{COR}(m) = \text{conv}\{xx^T : x \in \{0, 1\}^m\}$. The forward map is $X = mW_{[m],[m]}$, and, with $x = \text{diag } X$, its inverse is

$$W = \frac{1}{m} \begin{pmatrix} X & xe^T - X \\ ex^T - X & ee^T - xe^T - ex^T + X \end{pmatrix}. \quad (\text{I.1})$$

Moreover, the affine functional

$$g(W) = m - T(W) + (2m + 1)B(W) + (4m + 1)D(W)$$

satisfies $g \geq B + D \geq 0$ on K_{2m} and exposes exactly F .

Proof. Within the trace face, $B = 0$ selects sets A with at most one element from each pair $\{i, m+i\}$. Their totals are $|A| \leq m$; maximizing the total selects exactly one element per pair. These successive faces give atoms $(x, e - x)(x, e - x)^T/m$. Expanding their blocks gives (I.1), which extends affinely to their hull. For exposure, on a nonzero generator the inequality $|a - b| \leq a(1 - b) + b(1 - a)$ gives $\|r - c\|_1 \leq 2SD$. Also $r_i + r_{m+i} - r_i r_{m+i} \leq 1$, so

$$S \leq m + \sum_i r_i r_{m+i} \leq m + SB + \|r - c\|_1 \leq m + 2mB + 4mD.$$

The zero generator satisfies the same affine inequality, hence so does K_{2m} . Substitution proves $g \geq B + D$ and the asserted zero set. $\square$

Adding affine equations to a lift and composing its output map leaves its cone unchanged. Thus every lift of K_{2m} gives one of $\text{COR}(m)$ over the same cone, without a strict-feasibility assumption. The established correlation-polytope bounds therefore imply:

- For fixed d and $m \geq d$, a lift over $(\mathbb{S}_+^d)^q$ has

$$q \geq (3^d - 1)^{-(1-1/d)}(1 - 3^{-d})^{-m/d}.$$

For $d = 2$ the stronger bound is $q \geq 7^{-1/2}(9/7)^{m/2}$ [20, Theorem 1].

- The total dimension of an exact second-order-cone lift is at least $7^{-1/2}(9/7)^{m/2}$. A Lorentz cone $\{(t, z) : \|z\|_2 \leq t, z \in \mathbb{R}^k\}$ decomposes into $k - 1$ three-dimensional Lorentz cones for $k \geq 2$, each isomorphic to $\mathbb{S}_+^2$; one-dimensional nonnegative factors and the two-dimensional Lorentz cone require at most their dimension in such blocks. Apply the preceding bound.
- An unrestricted PSD lift has matrix order at least $2^{\alpha m^{2/13}}$ for an absolute $\alpha > 0$, for sufficiently large m [30, Theorem 5.4]. Several PSD blocks count by total order.

These are inherited conic lower bounds through the displayed face. They assert neither nonrepresentability by finitely many second-order cones nor an exponential number of Lorentz cones of unrestricted dimension. For comparison, K_2 is already nonpolyhedral: the face where its first row and column sums equal one is the hull of $W(t) = (1, t)^T(1, t)/(1 + t)$, $0 \leq t \leq 1$. With $q = W_{11}$ its lower-right entry is $q + 1/q - 2$, a strictly convex function on $[1/2, 1]$. Its infinitely many graph points are extreme. Positive-error LP bounds below consequently carry information that exact finite LP nonrepresentability alone does not.

I.2 A linear error bound and a quantitative section transfer

Proposition I.2 (Stability). *For every $W \in K_{2m}$,*

$$\text{dist}_1(W, F) \leq (136m + 10)g(W). \quad (\text{I.2})$$

If a convex set Q satisfies $K_{2m} \subset Q \subset K_{2m} + \epsilon U_1$, and $H_F = \{D = B = 0, T = m\}$, then each $Y \in Q \cap H_F$ is within $(184m + 6)\epsilon$ of F . In particular, with $A_m = m(184m + 6)$,

$$\text{COR}(m) \subset \{mY_{[m],[m]} : Y \in Q \cap H_F\} \subset \text{COR}(m) + A_m \epsilon U_1. \quad (\text{I.3})$$

Proof. For a nonzero generator $W = rc^T/S$, round r to a binary vector a . Since $\min(t, 1 - t) \leq 2t(1 - t)$ and $\|r - c\|_1 \leq 2SD$,

$$u := \|r - a\|_1 \leq 6SD, \quad v := \|c - a\|_1 \leq 8SD.$$

Indeed $\sum r_i(1 - r_i) \leq \sum r_i(1 - c_i) + \|r - c\|_1 \leq 3SD$. Write $k = \sum a_i$, and let h, z count pairs with two and zero ones. Then $k = m + h - z$ and the product difference inequality on $[0, 1]^2$ gives $h \leq SB + \|r - c\|_1 + u \leq SB + 8SD$. Change one coordinate in each full or empty pair to obtain b with exactly one one per pair. Thus

$$\|a - b\|_1 = h + z = 2h + m - k \leq 2SB + 22SD + |m - S|.$$

Set $V = bb^T/m \in F$ and $\tilde{b} = (S/m)b$. Outer-product norm factorization, using that $r, c, \tilde{b}$ have total S , gives

$$\begin{aligned} \|W - V\|_1 &\leq \|r - b\|_1 + \|c - b\|_1 + 3|S - m| \\ &\leq 4SB + 58SD + 5|S - m|. \end{aligned}$$

Here the first two terms bound the comparison with $\widetilde{bb}^T/S$, and its distance from V is $|S - m|$. Since $S \leq 2m$ and $|S - m| \leq g + (2m + 1)B + (4m + 1)D$, the last bound is at most $(18m + 5)B + (136m + 5)D + 5g \leq (136m + 10)g$. For $W = 0$ the distance to any paired atom is m , while $g(0) = 5m + 1$. Taking finite convex combinations proves (I.2).

For the stronger transfer retain the bound $8mB + 116mD + 5|S - m|$ on each generator, including zero. Exposure gives $(S - m)_+ \leq 2mB + 4mD$. In a convex decomposition of W , therefore,

$$\sum_{\ell} \lambda_{\ell} |S_{\ell} - m| \leq 4mB(W) + 8mD(W) + m - T(W).$$

The corresponding combination $V \in F$ consequently obeys

$$\|W - V\|_1 \leq 28mB(W) + 156mD(W) + 5[m - T(W)].$$

For $Y \in Q \cap H_F$ choose $W \in K_{2m}$ with $\|Y - W\|_1 \leq \epsilon$. The coefficient norms of B, D, T give $0 \leq B(W), D(W) \leq \epsilon$ and $m - T(W) \leq \epsilon$. The triangle inequality yields $(184m + 6)\epsilon$. The block projection has norm at most m , proving the last assertion. $\square$

An immediate exact-penalty consequence is that for $\lambda > (136m + 10)\|H\|_{\infty}$, minimizing $\langle H, W \rangle + \lambda g(W)$ over K_{2m} has all minimizers in F and value $\min_F \langle H, V \rangle$. Compare W with the point V provided by (I.2); the excess is at least $[\lambda - (136m + 10)\|H\|_{\infty}]g(W)$.

I.3 Fine LP and SDP approximations

Proposition I.3 (Explicit accuracy scales). *Let $K_{2m} \subset Q \subset K_{2m} + \epsilon U_1$ with Q convex. If $\epsilon \leq 1/A_m$, any LP lift of Q has $2^{\Omega(m)}$ inequalities. If $\epsilon \leq 1/(2A_m)$, any PSD lift of Q has matrix order at least $\exp(c(m/\log m)^{2/13})$ for an absolute $c > 0$ and all sufficiently large m .*

Proof. For the LP assertion apply (I.3). The inequalities $\langle 2 \operatorname{diag}(a) - aa^T, X \rangle \leq 1$, $a \in \{0, 1\}^m$, have coefficient norm at most one and slack $(1 - a^T x)^2$ at xx^T . The projected set is therefore between $\operatorname{COR}(m)$ and the polyhedron defined by these inequalities with right side two. The fixed-dilation lower bound of Braun et al. [11, Theorem 6(i), author version p. 18] gives the result. This particular sandwich has a small SDP relaxation and does not imply the next assertion.

For that assertion use the sign correlation polytope $\operatorname{SCOR}(m) = \operatorname{conv}\{yy^T : y \in \{-1, 1\}^m\}$. We first establish its fixed-error consequence of the pseudo-density bounds in Lee et al. [30, Theorem 3.8, equation (3.11), and Theorem 5.3]. For odd $k \geq 3$ their construction supplies a degree- k pseudo-density D on $\{0, 1\}^k$ with

$$\mathbb{E}D = 1, \quad \|D\|_{\infty} \leq k^{3/2}, \quad \mathbb{E}Df_k = -1/(4k^2), \quad f_k(x) = \frac{(\sum x_i - k/2)^2 - 1/4}{k^2}.$$

Here degree- k means $\mathbb{E}Dp^2 \geq 0$ for $\deg p \leq k/2$. The cited quantitative theorem states that if $f : \{0, 1\}^k \rightarrow [0, 1]$ and $\mathbb{E}Df < -\delta$, then the matrix $M(S, x) = f(x_S)$, indexed by k -subsets of $[m]$ and binary x , has, for $m \geq 2k$, PSD rank at least

$$\left[\frac{c_0 \delta m}{k^3 \|D\|_{\infty} \log m} \right]^{k/4} \left[\frac{\delta}{\|D\|_{\infty}} \right]^{3/2} \sqrt{\mathbb{E}f}. \quad (\text{I.4})$$

These pseudo-density and rank statements are established inputs.

Suppose $\operatorname{SCOR}(m) \subset R \subset \operatorname{SCOR}(m) + \eta U_1$ with $\eta \leq 1/2$. For every k -subset S the affine function

$$L_S(Y) = \frac{\sum_{i,j \in S} Y_{ij} - 1}{4k^2}$$

is nonnegative on the sign polytope. Its coefficient norm is $1/(4k^2)$, so $L_S + \theta \geq 0$ on R , where $\theta = \eta/(4k^2)$. At $y = 2x - e$ its slack is $f_k(x_S) + \theta$. This shifted function is in $[0, 1]$, has mean at least $(k - 1)/(4k^2) \geq 1/(6k)$, and has pseudo-expectation at most $-1/(8k^2)$. Taking $\delta = 1/(16k^2)$ in (I.4) gives

$$\text{rank}_{\text{psd}} M \geq c_1 k^{-23/4} \left[\frac{c_2 m}{k^{13/2} \log m} \right]^{k/4}. \quad (\text{I.5})$$

If R has a PSD lift of order q , these valid affine slacks have a PSD factorization of order at most $q + 1$. To justify this without a proper-lift assumption, restrict the lift to the smallest PSD face containing its feasible set. A finite combination of feasible matrices spanning all feasible ranges is positive definite on this face. Conic duality for the minimization of each nonnegative affine slack then writes that slack on the affine slice as $\langle B, Y \rangle + \tau$, $B \succeq 0$, $\tau \geq 0$. Appending the scalar factors τ and 1 supplies the asserted factorization. Choose odd k comparable to $a(m/\log m)^{2/13}$ with $a > 0$ small enough that the bracket in (I.5) is at least 16. This proves the claimed superpolynomial order for R .

Finally partition W into four m -by- m blocks and set $J(W) = m(W_{11} - W_{12} - W_{21} + W_{22})$. Its entrywise norm is at most m , and $J(F) = \text{SCOR}(m)$. The full-matrix transfer in Proposition I.2 shows that $R = J(Q \cap H_F)$ satisfies the preceding sandwich with $\eta = A_m \epsilon \leq 1/2$. Affine sections and images preserve the lift order, completing the proof. $\square$

The scale in both assertions is explicit order m^{-2} . For compact convex outer sets this norm guarantee is equivalent, by separation, to uniform additive objective error ϵ over all costs with $\|H\|_\infty \leq 1$. It does not cover only one objective, a restricted cost family, or a larger accuracy tolerance. No spatial region-count lower bound follows solely from these global lift-size bounds.

At the same SDP accuracy scale, a second-order-cone lift also needs superpolynomial total cone dimension: decompose its Lorentz cones into three-dimensional cones as above and embed them block-diagonally into a PSD cone of order at most twice the original total dimension. This transfers only the displayed superpolynomial approximate bound, whereas the exact second-order-cone bound is exponential in m .

J Limiting bounds and primitive FBBT updates

Feasibility-based bound tightening (FBBT) contracts a box by propagating individual constraints. Its limit, the number of primitive updates, and the finite tolerance used by an implementation are distinct objects. When the limiting box is nonempty, continuous linear FBBT limits can be obtained by an LP [9, Theorem 4.1], although primitive iteration need not terminate finitely. Here all variables start in $[0, 1]$. A *primitive update* computes the interval hull for one affine equality or one nonnegative product equality and intersects it with the current box. For the limiting-bound theorem we allow any additional sound contracting updates, provided every defining equation receives its forward natural interval-arithmetic update infinitely often. This is fairness, with no bounded-delay assumption. Exact real arithmetic specifies the mathematical limit; an approximation algorithm is measured in the ordinary binary Turing model.

Lemma J.1 (Least fixed point transfer). *Let P have nonnegative polynomial coefficients, and suppose $x = P(x)$ has a solution in the unit cube. Under sound contracting updates and fair forward propagation, its limiting lower vector is the least nonnegative fixed point q of P . This includes lifted arithmetic-node equations.*

Proof. The sequence $s^0 = 0$, $s^{j+1} = P(s^j)$ increases and is bounded by every nonnegative fixed point. Its limit q is a fixed point by continuity and is the least one. Soundness preserves q , so every

lower vector l satisfies $l \leq q$. The natural interval lower endpoint is exactly $P_i(l)$. Inductively, once $l \geq s^j$, fairness supplies a finite later time at which all coordinates have received a forward update and $l \geq s^{j+1}$. Hence the lower limit is at least every s^j and equals q . Apply the same argument to the augmented polynomial system for an expression graph. $\square$

Proposition J.2 (Restricted limiting-bound hardness). *Computing a designated limiting lower bound to additive error $1/4$ is PosSLP-hard, even for feasible affine and bilinear defining systems in the unit cube with at most three variable names per equation, nonnegative coefficients in $\{1/2, 1\}$, constants in $\{0, 1/2, 1\}$, and distinct names in every product. Every strongly connected component of the directed defining-equation dependency graph has at most four vertices. The feasible set has at most two points, all rational.*

Proof. PosSLP asks whether an integer arithmetic circuit using $0, 1, +, -, \times$ has positive output. Following the circuit normalization and amplification mechanism of Etessami and Yannakakis [19, Theorem 5.2], replace every gate by positive and negative parts. Multiplication uses $(uv)^+ = u^+v^+ + u^-v^-$ and $(uv)^- = u^+v^- + u^-v^+$; addition and subtraction use the corresponding sums. It suffices to compare two nonnegative integer outputs U, V .

Pad these fan-in-two circuits to common depth $L \geq 1$ with alternating addition and multiplication layers, using identities $v + 0, v \cdot 1$. This costs polynomial size. Define proof-only normalizers $M_0 = 1$, $M_j = 2M_{j-1}$ on addition layers and $M_j = M_{j-1}^2$ on multiplication layers. Encode each gate by its normalized value p and complement r . At inputs they are zero or one; the equations for subsequent gates are

$$p = (p_j + p_k)/2, \quad r = (r_j + r_k)/2$$

for addition and

$$p = p_j p_k, \quad v = p_j r_k, \quad r = r_j + v$$

for multiplication. These identities give $p = \text{value}/M_j$ and $r = 1 - p$, with unique values in $[0, 1]$. Introduce a copy variable whenever a product would repeat a name. Each equation has at most three names and all these dependencies are acyclic. With $M = M_L \leq 2^{2^L}$, put

$$c = (p_U + r_V)/2 = \frac{1}{2} + (U - V)/(2M), \quad d = (r_U + p_V)/2 = 1 - c.$$

The four-equation component

$$a' = a, \quad t = aa', \quad v = ct, \quad a = d + v$$

implements $a = d + ca^2$. Its least nonnegative root a_* equals one if $c \leq 1/2$ and d/c if $c > 1/2$. In the latter case, writing $\Delta = U - V \geq 1$, $1 - a_* = 2\Delta/(M + \Delta) \geq 1/M$.

Construct $b = 2^{-2^{L+2}}$ and its complement by $L + 2$ paired squarings starting from $1/2$. These numbers require polynomially many equations, not their full rational expansions as input constants. They satisfy $b \leq 1/(8M)$. Add $e = (1 - b)a$, represented using the complement variable, and the two-equation component $w = ez$, $z = b + w$. At the least detector root it has the unique solution

$$z_* = \frac{b}{1 - (1 - b)a_*} = 1 \quad (U \leq V), \quad z_* \leq bM \leq 1/8 \quad (U > V).$$

All displayed values lie in the cube and form a fixed point. The least fixed point of the full system has the unique acyclic values, the least detector root (bounded above by this displayed fixed point), and then this unique final value. Lemma J.1 identifies z_* with the FBBT limit. An additive- $1/4$ answer is at least $3/4$ in the first case and at most $3/8$ in the second; threshold $1/2$ decides PosSLP.

The largest directed component has the four detector variables, and the last has two. The detector has at most two rational roots, and each admissible root determines all other variables uniquely and rationally. $\square$

The directed graph has an edge from a defined variable to each argument on its right side; this is not an undirected treewidth restriction. PosSLP is not known to be NP-hard. The proposition does not assert hardness of executing a prescribed finite number of sweeps or of obtaining an approximately stationary box. A unique rational feasible point is possible if one further affine inequality is allowed: introduce $y = ca$ and impose $y \leq 1/2$. It retains the least detector root and excludes $a = 1$ when $c > 1/2$; soundness and the fair original forward updates retain the same limiting-bound proof.

Proposition J.3 (Every primitive schedule can be slow). *For each $n \geq 1$ a constant-coefficient affine-bilinear system with $O(n)$ variables and equations requires at least 2^{2^n-1} applications of one affine constraint before a designated lower bound reaches $1/2$, although its limit is one. This holds for every fair ordering of the bidirectional primitive interval-hull updates defined above. The only feedback component has two variables and is linear after its upstream values are fixed.*

Proof. Start with $b_0 = c_0 = 1/2$ and, for $i = 1, \dots, n$, impose

$$h_i = b_{i-1}, \quad b_i = b_{i-1}h_i, \quad v_i = b_{i-1}c_{i-1}, \quad c_i = c_{i-1} + v_i.$$

These acyclic equations give $b = b_n = 2^{-2^n}$, $c = c_n = 1 - b$. Add $w = cz$, $z = b + w$. The unique feasible values are $z = 1, w = c$. For a stronger initialization fix all acyclic variables exactly, leaving $z, w \in [0, 1]$. Primitive interval-hull contractors are monotone under box inclusion, so the original run with the same schedule has no larger lower bounds.

In the stronger run the invariant is $0 \leq l_w \leq cl_z$, $l_z \leq 1$. A product update sets l_w to cl_z at most; its inverse bound $l_z \geq l_w/c$ cannot improve l_z . An affine update sets

$$l_z^{\text{new}} = \max(l_z, b + l_w) \leq b + cl_z, \quad l_w^{\text{new}} = \max(l_w, l_z - b).$$

Since $l_z - b - cl_z = b(l_z - 1) \leq 0$, both terms defining the new lower bound on w are at most cl_z^{new} . This proves the invariant, including simultaneous hull updates. Upper endpoints cannot improve these formulas: feasibility forces $u_z = 1$, while u_w is either one or c . After K affine updates, induction and Bernoulli's inequality give

$$l_z \leq 1 - c^K = 1 - (1 - b)^K \leq Kb.$$

Thus $l_z \geq 1/2$ requires $K \geq 1/(2b) = 2^{2^n-1}$. $\square$

The precise bound is in the displayed parameter n ; encoding indexed variable names as binary strings gives ordinary input length $O(n \log n)$. At $l_z = l_w = 0$ with upstream variables fixed, every primitive update changes any lower endpoint by at most b , although l_z is distance one from its limit. Small local residual and small distance to the limiting box are therefore different guarantees.

Slow and nonfinite linear FBBT iteration is classical, as are repeated squaring and error amplification. The slow-Newton examples of Esparza et al. [18, Section 7, Theorem 7.1] and Stewart et al. [39, Section 4.1, equation (18)] also imply a doubly exponential first-bit bound for ordinary Kleene iteration. This is an inference from their displayed systems, not their stated Newton theorem. For Stewart et al.'s chain, $x_0 = (x_0^2 + 1)/2$ and $x_i = (x_i^2 + x_{i-1})/2$, the errors from the fixed point one satisfy $e_0(k+1) = e_0(k) - e_0(k)^2/2$ and $e_i(k)^2 \geq e_{i-1}(k)$, the latter because each iterate is below its next update. The first recurrence gives $e_0(k) \geq 1/(k+1)$ by induction (the map is increasing

on $[0, 1]$). Hence $e_n(k) \geq (k+1)^{-1/2^n} > 1/2$ when $k+1 < 2^{2^n}$. The specific scope proved here is every ordering of the stated bidirectional primitive contractors, with one feedback component that becomes linear. Symbolic substitution, equation aggregation, fixed-point acceleration, and global contractors are outside this iteration bound.

K A short comparison with integer precision

Here the resource is the number p of integer coordinates in a convex lift, or binaries in a linear lift. A relaxation contains the graph and has vertical error at most $\epsilon > 0$ at every projected point over its box. There is no bound on its continuous auxiliary dimension or on the ranges of general integer coordinates. The published parity midpoint argument [31, Lemma 4.1] groups graph points by the parity of a feasible integer lift. The midpoint of two lifts in one class has integer coordinates and remains feasible by convexity. Taking closures of these at most 2^p classes in the compact input box preserves every continuous midpoint-error inequality and avoids any measurability assumption on the original lift.

For x^2 on $[0, 1]$, the midpoint error is $(a-b)^2/4$. Each closed class therefore has diameter at most $2\sqrt{\epsilon}$. Covering the interval gives $1 \leq 2^{2^p}\sqrt{\epsilon}$, or

$$\epsilon \geq 2^{-2^{p-2}}. \quad (\text{K.1})$$

This is attained by a binary linear relaxation: divide the interval into 2^p equal intervals of length h . On each interval use its secant as upper bound and the maximum of the endpoint tangents as lower bound. Both vertical errors are at most $h^2/4$, with equality in the upper bound at the midpoint. The finite union of these bounded polyhedra has a linear encoding with p binary labels (unused auxiliary dimension and row count are unrestricted). Thus the minimum integer dimension and the minimum binary count both equal $\max\{0, \lceil (\log_2(1/\epsilon) - 2)/2 \rceil\}$. The more compact sawtooth construction has the same exact worst error [7, combined 2022 preprint, Section 5.1.1, PDF p. 21]; that inspected preprint has different numbering from the published parts.

For xy , the corresponding midpoint inequality is $|(a_1 - b_1)(a_2 - b_2)| \leq 4\epsilon$. A compact class S has area at most $16\epsilon \log 2$. Indeed choose points a, b attaining the extreme first coordinates, with width $X > 0$. At first coordinate $a_1 + t$ its section is contained in intervals centered at a_2, b_2 of lengths $8\epsilon/t, 8\epsilon/(X-t)$, respectively. Integrating their minimum gives $2 \int_0^{X/2} 8\epsilon/(X-t) dt = 16\epsilon \log 2$. The zero-width case has zero area. Hence

$$\frac{1}{16(\log 2)2^p} \leq \epsilon_{\text{best}}(p) \leq 2^{-p-2}. \quad (\text{K.2})$$

For the upper bound partition the first coordinate into 2^p strips and use the four McCormick inequalities in each. On a rectangle of side lengths h, k their maximum vertical error is $hk/4$, as follows by scaling the unit-square formulas. Thus both general integer dimension and binary count are $\log_2(1/\epsilon) + O(1)$ for this scalar product.

The simultaneous graph $w_{ij} = x_i x_j$, $ij \in E(G)$, has a different leading coefficient: the fractional vertex-cover number

$$\tau^*(G) = \min\{\sum_i a_i : a_i \geq 0, a_i + a_j \geq 1 \ (ij \in E)\}.$$

To see the quantitative comparison, for positive edge errors define

$$L_C = \min\left\{\sum_i s_i : s_i \geq 0, s_i + s_j \geq \max\{0, \log_2(1/(C\epsilon_{ij}))\} \ (ij \in E)\right\}.$$

Then the least general integer dimension and binary linear count obey

$$\lceil L_{20} \rceil \leq p_{\min} \leq p_{\text{bin}} \leq L_4 + |V|. \quad (\text{K.3})$$

For completeness, if a compact planar set satisfies $|\Delta x \Delta y| \leq 4\epsilon$, its coordinate widths satisfy $d_x d_y \leq 20\epsilon$: two extreme- x points differ in y by at most $4\epsilon/d_x$, and every other point is within $8\epsilon/d_x$ in y of one of them. Apply this to each edge projection of a closed parity class. If all its widths d_i are positive, $s_i = -\log_2 d_i$ is feasible for L_{20} , so its volume is at most $\prod_i d_i \leq 2^{-L_{20}}$; a zero width gives zero volume. The cover proves the lower bound.

For the upper bound take an optimum s of L_4 , put $p_i = \lceil s_i \rceil$, and share the binary expansion $x_i = \sum_{k=1}^{p_i} 2^{-k} \beta_{ik} + r_i$, $0 \leq r_i \leq 2^{-p_i}$, across every incident product. For each edge, expand first in x_i and then expand the residual term in x_j . Binary-continuous products have exact four-row linear descriptions; only $r_i r_j$ needs a McCormick relaxation. Its error is at most $2^{-p_i - p_j} / 4 \leq \epsilon_{ij}$. This proves (K.3) with at most $O(|V| + |E| + \sum_i \deg(i) p_i)$ continuous variables and rows. For fixed nonempty G and common error tending to zero, $L_C = \tau^*(G) \log_2(1/(C\epsilon))$. Consequently

$$p_{\min}(\epsilon) = p_{\text{bin}}(\epsilon) + O_G(1) = \tau^*(G) \log_2(1/\epsilon) + O_G(1).$$

These arguments count binary assignments or integer parity classes. Their feasible midpoint belongs to the mixed-integer projection; it need not be a region certified by any spatial bounding oracle.

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